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College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

BrewInsight BI: Revenue Pattern Analytics System for Kape Tubong

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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INTRODUCTION

1.1 Background of the Study

The Food and Beverage (F&B) industry has grown significantly over the past decade due to changing consumer lifestyles, urbanization, and the increasing demand for convenient dining and beverage options. Among the many businesses within this sector, coffee shops have become one of the fastest-growing industries because they serve not only as places to purchase beverages but also as spaces for meetings, studying, remote work, and social interaction. As customer expectations continue to evolve, coffee shop businesses are expected to maintain high standards in service quality, operational efficiency, and overall customer experience in order to remain competitive in the market. In addition, the increasing presence of local and international coffee brands has intensified market competition, requiring coffee shop businesses to adopt more efficient operational strategies and innovative management practices. As consumer preferences shift toward convenience, speed, and personalized service, businesses must continuously improve their operations to remain relevant and profitable.The modern coffee shop environment has evolved beyond simple beverage retailing into a highly dynamic service ecosystem requiring efficient coordination of multiple operational processes. Businesses in this industry must manage not only product quality but also customer experience, branch consistency, and rapid service delivery. The rise of digitally connected consumers has also increased expectations for faster transactions and personalized offerings. Social media and online customer feedback further amplify the need for businesses to maintain strong operational performance. In such a competitive environment, data-driven management becomes increasingly essential to sustain growth and profitability. Coffee shops that fail to adapt to evolving consumer expectations risk losing market share to more technologically advanced competitors. Operational efficiency has therefore become a strategic priority in the F&B industry. Businesses are expected to leverage modern technologies to streamline internal processes and improve service responsiveness. As competition intensifies, analytical decision-making tools are becoming critical for maintaining operational excellence. Thus, the F&B industry increasingly demands innovative and technology-supported management approaches.

As coffee shop businesses expand to multiple locations, managing operations becomes more challenging. Opening additional branches or kiosks increases the complexity of handling transactions, monitoring inventory, scheduling staff, tracking sales, and evaluating branch performance. Tasks that are manageable in a single-location business become more difficult when operations are spread across several outlets. Because of this, businesses require a centralized way of managing and monitoring operations to maintain consistency and efficiency across all locations. Without a standardized system, inconsistencies in branch operations may arise, leading to variations in service quality, reporting practices, and overall business performance. Maintaining operational control across multiple branches becomes increasingly difficult when management lacks centralized visibility into day-to-day branch activities.Multi-branch expansion introduces additional administrative and analytical challenges that cannot be effectively managed through informal monitoring methods. Branch managers may apply inconsistent operational procedures when centralized oversight mechanisms are absent. Variations in transaction recording, stock monitoring, and performance reporting can reduce organizational consistency. As the number of locations increases, communication gaps between branches and management may also widen. Delays in reporting from remote kiosks can hinder timely strategic interventions. Furthermore, evaluating comparative branch performance becomes increasingly difficult when data is fragmented across multiple systems. Branch-level accountability may weaken if performance metrics are not consistently monitored and standardized. Centralized operational visibility is therefore essential for maintaining business alignment across distributed locations. Businesses with multiple outlets require integrated systems to ensure consistent oversight and informed management. Effective branch expansion depends not only on market growth but also on scalable operational monitoring infrastructure.

Kape Tubong has shown continuous growth by expanding from a single coffee outlet into a business with multiple kiosks and a main branch. This expansion reflects increasing customer demand and positive market acceptance of the brand. However, as the business grows, the volume of sales and operational data generated from different locations also increases. Without an integrated information system, managing and monitoring this data becomes more difficult, resulting in fragmented records and inefficient reporting processes. The increasing volume of distributed data also creates challenges in maintaining data consistency, accuracy, and accessibility across all locations. As the organization expands further, reliance on disconnected data sources may hinder the company’s ability to scale efficiently and make timely strategic decisions. The expansion of Kape Tubong reflects positive organizational growth but also introduces new operational demands that require more sophisticated information management practices. Larger transaction volumes increase the complexity of monitoring revenue performance and operational efficiency across all branches. The accumulation of disconnected sales records may lead to inconsistent reporting and delayed consolidation of business information. Branch managers may also experience difficulty in maintaining accurate local records without centralized oversight mechanisms. As operational scale increases, manual coordination between branches and management becomes less sustainable. Fragmented data environments reduce management’s ability to obtain a holistic view of organizational performance. This may hinder strategic planning and delay responses to emerging operational concerns. Furthermore, business growth without corresponding digital infrastructure may create scalability limitations. Expanding enterprises require systems capable of supporting increased transaction volume and analytical complexity. Therefore, Kape Tubong’s continued growth creates a strong need for centralized digital business intelligence support.

Many small and medium-sized enterprises (SMEs), including coffee shop businesses, still rely on traditional or semi-manual methods of recording and managing operational data. These methods often include handwritten transaction logs, spreadsheet-based monitoring, and standalone Point of Sale (POS) systems that operate independently in each branch. While these approaches may work during the early stages of business operations, they become less effective as the organization expands. Separate systems create isolated data records that make information consolidation difficult and reduce management's visibility into overall business performance. Furthermore, manual and disconnected systems increase dependency on human intervention for data processing, which may compromise speed, reliability, and consistency of reporting. As operational complexity increases, these limitations can negatively impact organizational efficiency and management responsiveness.Traditional reporting practices often lack the automation necessary to support modern business analysis requirements. Manual encoding processes are vulnerable to human error, omissions, and inconsistencies in formatting. Spreadsheet-based reports may become difficult to manage as data volume increases over time. Standalone systems also limit the ability to generate consolidated enterprise-wide reports automatically. Businesses relying on disconnected systems often face challenges in reconciling conflicting data from multiple sources. These inefficiencies consume administrative resources that could otherwise be allocated to strategic functions. In addition, manual reporting processes delay the availability of performance information needed for decision-making. The lack of automated validation also increases the likelihood of inaccurate reporting outputs. SMEs that continue to rely on outdated reporting methods may struggle to compete with more digitally enabled organizations. Thus, modernizing operational data management has become increasingly necessary for growing enterprises.

One of the major issues caused by fragmented systems is the difficulty of obtaining real-time and unified sales data from all branches. Since each branch records transactions separately, managers must manually collect reports from each location before consolidating and analyzing them. This process consumes time, increases administrative workload, and creates opportunities for human error, delayed reporting, and data inconsistencies. As a result, management decisions may be based on incomplete or outdated information. Delays in obtaining accurate sales information may prevent management from responding quickly to sudden changes in customer demand, branch performance, or operational issues. The inability to access real-time data limits the organization's responsiveness and weakens its ability to make timely operational adjustments.Real-time information availability is increasingly recognized as a critical requirement in competitive business environments. Businesses operating without immediate access to updated data may fail to detect urgent operational issues promptly. Delayed reporting reduces the strategic value of business information because decisions are made after performance issues have already occurred. Fragmented reporting also prevents management from observing emerging trends across branches simultaneously. This weakens comparative analysis and branch-level benchmarking. Managers may be forced to rely on assumptions when immediate data is unavailable. Such reliance can result in less accurate and less strategic decision-making. Furthermore, inconsistent reporting intervals between branches may create additional visibility gaps. Unified real-time reporting therefore becomes essential for effective business control. Organizations lacking timely data access may experience reduced operational agility and strategic responsiveness.

Another challenge is the limited ability of traditional systems to provide meaningful analytical insights. Basic sales reports often present only raw figures without deeper analysis of trends or patterns. Without proper analytical tools, management may find it difficult to identify peak sales periods, determine customer purchasing behavior, evaluate branch performance, and monitor product demand. This limits the organization's ability to make informed and proactive business decisions. Without visualization and analytical reporting features, management may struggle to interpret large amounts of transactional data efficiently. Consequently, potentially valuable business insights may remain hidden within raw sales records and unused in strategic planning.
Raw numerical reports alone often fail to provide sufficient interpretive value for strategic decision-making. Managers may find it difficult to detect patterns when information is presented only in tabular or spreadsheet form. Analytical limitations reduce the practical usefulness of collected data despite high transaction volumes. Businesses may therefore possess large amounts of information without the capability to extract actionable insights. Trend analysis and performance visualization are necessary to transform transactional data into strategic knowledge. Without analytical tools, organizations may overlook important operational opportunities and risks. The inability to identify customer behavior trends can weaken promotional and inventory strategies. Similarly, lack of product-level analysis may prevent optimization of menu offerings and pricing decisions. Effective analysis requires both data collection and interpretive mechanisms. Thus, analytical system capabilities are essential for maximizing the strategic value of business data.

The lack of revenue pattern analysis also affects workforce scheduling and inventory planning. Without reliable information on peak operating hours and product demand, staffing schedules may rely on assumptions rather than actual business data. This may lead to overstaffing during slow periods or understaffing during peak hours, both of which can reduce efficiency and affect customer service quality. Similarly, inventory decisions made without accurate sales analysis may result in overstocking, stock shortages, increased waste, or lost sales opportunities. Inefficient workforce and inventory planning can directly affect profitability by increasing labor costs and reducing inventory turnover efficiency. Over time, these inefficiencies may significantly impact operational sustainability and customer satisfaction.Workforce scheduling based on assumptions rather than empirical demand data often leads to inefficient labor utilization. Overstaffing increases unnecessary payroll expenses while understaffing can reduce service quality and increase customer waiting times. Poor inventory planning may also cause avoidable spoilage, particularly in perishable F&B products. Understocking of high-demand items may lead to missed sales opportunities and dissatisfied customers. Effective resource allocation requires accurate understanding of historical and recurring demand patterns. Revenue pattern analysis can provide the necessary insight to align staffing and stock decisions with actual operational behavior. This improves both cost efficiency and customer service performance. Businesses that fail to optimize labor and inventory based on demand patterns may experience reduced profitability. Strategic planning in F&B operations therefore requires data-supported demand forecasting. Revenue analytics directly supports these optimization efforts.

To address these challenges, many organizations adopt Business Intelligence (BI) systems to improve operational monitoring and decision-making. Business Intelligence refers to the technologies and tools used to collect, organize, analyze, and present business data in a meaningful way. Through BI systems, raw sales data can be transformed into useful insights using dashboards, automated reports, trend analysis, and performance metrics. These tools enable businesses to make more strategic and data-driven decisions. BI systems also help organizations identify hidden opportunities, detect operational inefficiencies, and monitor business performance in real time. As data-driven management becomes increasingly essential in competitive industries, BI adoption has emerged as a valuable strategy for modern business operations.Business Intelligence platforms have become widely recognized as strategic tools for enhancing organizational awareness and performance visibility. These systems enable businesses to convert fragmented data into centralized analytical information environments. Through automation and visualization, BI tools reduce the effort required to interpret operational metrics. They also improve accessibility of business information for non-technical decision-makers. Organizations can use BI systems to monitor both real-time and historical performance indicators simultaneously. This supports immediate operational adjustments and long-term strategic planning. BI adoption also contributes to standardization of reporting practices across organizational units. Furthermore, centralized analytics platforms improve transparency and accountability in management processes. Businesses leveraging BI technologies often gain stronger analytical and competitive capabilities. Therefore, BI systems are increasingly essential in data-intensive business environments.

For multi-branch coffee shop operations, implementing a centralized BI platform provides significant advantages. By integrating sales data from all branches into a single database, management can access real-time business information and monitor performance across all locations. Dashboards can display important metrics such as total sales, peak sales hours, product demand, branch comparisons, and revenue trends. This allows management to make faster, more accurate, and more informed operational decisions. Centralized BI systems also improve data consistency by standardizing reporting formats and analytical methods across all branches. This enables more reliable comparison of branch performance and improves the quality of strategic evaluation.A centralized BI platform enhances operational visibility by consolidating performance information from geographically distributed locations into one unified interface. This reduces the complexity of monitoring separate branch reports individually. Standardized data structures improve the reliability of comparative performance analysis across branches. Managers can identify underperforming locations more quickly and investigate contributing operational factors. Centralization also simplifies enterprise-wide trend analysis by aggregating all branch data into one analytical environment. This improves strategic planning for expansion, staffing, and promotions. Unified dashboards support more efficient executive oversight and reduce administrative coordination effort. Additionally, branch managers can benefit from clearer performance benchmarks and comparative metrics. The availability of centralized analytics promotes more consistent and objective management evaluation. Thus, BI centralization significantly strengthens multi-branch operational governance.

In response to these challenges, this study proposes the development of BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong. The proposed system will serve as a centralized web-based Business Intelligence platform that consolidates sales data from all kiosks and the main branch into a unified database. It will provide real-time dashboards, automated reports, and analytical tools for examining revenue patterns, customer purchasing trends, peak sales periods, and product performance. The system is designed to transform fragmented transactional records into structured and visualized business intelligence that can support daily and strategic management decisions. By centralizing branch data, the platform aims to improve information accessibility, reduce reporting delays, and enhance the overall quality of operational monitoring.The proposed platform specifically addresses the operational limitations identified in Kape Tubong’s current reporting practices. Through automated data integration, the system reduces dependence on manual report consolidation. Visual dashboards provide management with immediate insight into key business metrics. Analytical tools within the platform will support proactive operational planning and evidence-based decision-making. Revenue trend analysis will help management identify sales patterns and evaluate performance changes over time. Product-level monitoring will allow assessment of menu demand and profitability. Branch comparison tools will support more accurate evaluation of location performance. Real-time reporting capabilities will improve responsiveness to sudden operational concerns. By integrating these functions into one platform, the system enhances both operational and strategic business intelligence capabilities. Therefore, BREWINSIGHT BI offers a practical digital solution to Kape Tubong’s operational data challenges.

The system is also expected to improve workforce planning, inventory management, and branch performance monitoring by providing management with timely and accurate operational insights. Through the application of software engineering, database management, and Business Intelligence principles, the project demonstrates how information technology can be used to solve practical business problems in the Food and Beverage industry. It also showcases the practical integration of modern information systems in supporting organizational efficiency and business process optimization. Furthermore, the project emphasizes the relevance of digital transformation in helping

SMEs modernize their operational and managerial practices.

The development of BREWINSIGHT BI reflects the growing importance of integrating technical innovation into everyday business operations. By applying structured software engineering methodologies, the project ensures systematic and maintainable system development. Database technologies enable secure and organized management of large volumes of transactional records. Business Intelligence principles support the transformation of raw operational data into strategic insights. Together, these technological components create a comprehensive analytical environment for operational improvement. The project also demonstrates how SMEs can leverage affordable digital tools to modernize business management. Such implementation highlights the practical role of IT in improving organizational competitiveness. The system serves as an applied example of interdisciplinary technological integration in business contexts. Its deployment may encourage broader digital adoption among similar SMEs. Thus, the study reinforces the strategic importance of technology-driven business transformation.

Ultimately, BREWINSIGHT BI is expected to improve operational transparency, enhance data-driven decision-making, reduce administrative inefficiencies, and support the long-term growth of Kape Tubong. Additionally, the system may serve as a practical reference for other small and medium-sized enterprises seeking to adopt Business Intelligence technologies to modernize operations and improve competitiveness in the Food and Beverage sector. By demonstrating the practical value of centralized analytics and automated reporting, the study may encourage broader adoption of Business Intelligence systems among SMEs facing similar operational challenges. In the long term, the implementation of such technologies may contribute to stronger business sustainability, improved adaptability to market changes, and enhanced organizational competitiveness.The proposed system is not intended solely as an operational tool but also as a strategic enabler for long-term organizational development. Improved transparency and centralized visibility can strengthen accountability and performance governance across all branches. Reduced administrative workload allows management and staff to focus more on strategic and customer-oriented functions. The availability of reliable business insights supports more accurate planning and forecasting activities. Enhanced analytical capabilities may also improve the organization’s ability to adapt to future expansion and market changes. As SMEs increasingly pursue digital transformation, systems such as BREWINSIGHT BI become critical enablers of sustainable competitiveness. The project may therefore serve as a replicable model for similar businesses seeking to modernize operations. Its successful implementation can demonstrate the practical viability of BI adoption in small-scale enterprises. This contributes to both organizational improvement and broader technological awareness among SMEs. Ultimately, BREWINSIGHT BI is positioned as a strategic investment in operational modernization and business sustainability.

1.2 Statement of the Problem

Kape Tubong has expanded significantly in physical size and reach. Despite this growth, various operational inefficiencies continue to affect overall business performance. An increase in the number of branches directly leads to greater complexity in daily management. This challenge becomes more apparent without a reliable and integrated information system to support operations. Expansion requires stronger coordination, improved monitoring, and more efficient data management across all locations. Current operational methods limit the organization's ability to maintain consistency, transparency, and visibility in business processes. These conditions highlight a clear need for a more structured and data-driven approach to sales and operations management.

Growth brings increased volume of transactional and operational data generated daily across multiple branches. The absence of a unified system makes it difficult to consolidate information into a single, organized source of truth. Management currently relies on separate reports submitted by each branch. This setup makes evaluation of overall performance and comparison of operational efficiency among locations difficult. Fragmented information reduces responsiveness to business issues and limits the ability to implement timely corrective actions. Communication gaps often occur between branch-level operations and upper management. These gaps result in delays regarding operational feedback and necessary performance interventions.

Lack of centralized data processing creates significant barriers to performance monitoring and strategic planning. Critical decisions such as identifying profitable branches, adjusting product offerings, planning workforce schedules, and managing stock levels all require accurate and timely information. Systems capable of processing and analyzing sales data efficiently are not available. Management may therefore struggle to identify operational issues before they escalate. Opportunities for optimization and improvement may be missed as a result. This limitation reduces the organization's capacity to respond proactively to market changes, shifts in customer demand, and specific operational concerns at different branches.

The decentralized structure also creates challenges in maintaining consistent reporting standards across all Kape Tubong locations. Each branch may follow different methods when recording, organizing, and submitting sales reports. This practice leads to discrepancies in documentation and reporting formats. Such inconsistencies make objective and accurate comparison of branch performance difficult. Variations in reporting practices reduce the reliability of consolidated data used for strategic analysis. Ensuring uniformity in data quality becomes increasingly difficult without standardized processes and centralized validation mechanisms. This issue may lead to inaccurate interpretations of branch-level performance and hinder fair evaluation. Management may also struggle to determine whether performance differences stem from actual operational issues or from inconsistent reporting methods. These limitations reduce the effectiveness of comparative analysis and branch benchmarking. Consistency in reporting is essential for maintaining transparency and accountability in growing multi-branch businesses. Kape Tubong requires a centralized and standardized solution to ensure data uniformity across all operational locations.

Continued reliance on fragmented and manually processed data limits the organization's ability to maximize the strategic value of operational information. Substantial sales and transactional data are generated daily across all branches. Much of this information remains underutilized due to a lack of analytical tools capable of transforming raw records into meaningful insights. Historical sales trends, performance patterns, shifts in customer demand, and product profitability indicators often remain hidden within unprocessed records. Management loses valuable opportunities to leverage data for forecasting, optimization, and strategic planning. Inability to perform advanced analysis weakens responsiveness to emerging market trends and competitive pressures. Delayed or uninformed decision-making can negatively affect profitability and customer satisfaction within a rapidly evolving food and beverage market. Competitors utilizing more advanced analytical systems gain significant operational and strategic advantages. Kape Tubong faces risks of falling behind regarding efficiency, adaptability, and market competitiveness. Improving capacity to convert operational data into actionable intelligence is necessary for sustainability and competitiveness. This requirement emphasizes the necessity of implementing a Business Intelligence-driven platform to support modernized, data-based management practices.

Absence of integrated analytical and monitoring tools hinders long-term scalability of operations. Opening additional branches or kiosks naturally increases complexity regarding transactions, inventory movement, staffing requirements, and productivity monitoring. Manual and fragmented systems manageable at smaller scales become inefficient as operational scope expands. Administrative workload may increase disproportionately with growth without proper technological support. This situation creates bottlenecks that slow decision-making and reduce managerial effectiveness. Inability to scale reporting and monitoring processes efficiently limits capacity for sustainable expansion. Expansion without corresponding improvements in information systems may amplify existing inefficiencies and create additional operational risks. The organization may struggle to maintain consistent service quality and oversight across a growing network. Implementation of an integrated analytics and reporting platform is essential for addressing current issues and supporting future growth. A scalable system enables expansion while maintaining effective control over distributed business operations.

This study aims to address the following problems:

Existing operations lack a centralized **Point of Sale** (POS) system capable of consolidating and organizing sales data from multiple locations into a unified platform. Sales records are maintained separately at each branch. This practice results in fragmented and dispersed data that complicates generation of complete and accurate reports. Manual collection and consolidation of reports require significant time and administrative effort from staff and management. This process increases the likelihood of incomplete, duplicated, or inaccurate data entries. Absence of centralized integration prevents real-time monitoring of sales performance. Management capacity to access timely and unified operational information remains limited. Such inefficiencies reduce overall operational effectiveness and delay responses to immediate concerns. Continued use of disconnected systems may hinder efficient scaling as additional branches are established. Maintaining data integrity across different reporting periods becomes challenging without centralized storage. Separate systems may follow inconsistent formatting and recording procedures. Standardization across the organization is reduced as a result. Reconciling discrepancies between branch reports becomes difficult due to inconsistent documentation. Fragmented systems limit the ability to establish a reliable historical database for long-term analysis. Branch-level data remains isolated and underutilized for strategic planning. Uniform transaction recording standards are difficult to enforce across all locations. Personnel may interpret guidelines differently, leading to inconsistencies in documentation. System integration issues complicate auditing and verification of records. Tracing transactional discrepancies becomes difficult when records are dispersed across locations. Transparency in overall operations is reduced by fragmented systems. Organizational control over distributed activities is weakened. These limitations slow preparation of executive summaries and strategic reports. Disconnected operations reduce capacity to manage expansion efficiently. Obtaining synchronized performance reports within a single period is often impossible without a unified database. Managers experience delays when requesting summaries because records require manual collection and validation. Inconsistent systems create compatibility issues during consolidation and maintenance. Implementation of organization-wide policy updates and monitoring procedures becomes more complex under separate platforms. Employees may encounter confusion when different practices are followed across locations. Identifying unusual sales activities or inconsistencies immediately is difficult without centralized visibility. Delayed detection of discrepancies negatively affects financial monitoring and internal control. Fragmented structures increase complexity when preparing tax-related and compliance documents. Retrieval of historical records stored separately becomes increasingly difficult as volume grows. Preserving consistency during manual modifications poses challenges. Communication gaps between management and branch operations may occur more frequently. Dependency on manual verification and repetitive supervision increases. Operational efficiency is reduced and workload increases for administrative staff. Generating accurate comparative analyses between branches remains difficult due to inconsistent structures. Maintaining isolated systems becomes costly and inefficient as operations expand. Lack of centralized integration weakens coordination, reliability, accessibility, and scalability.

Capability to automatically track, analyze, and visualize sales trends and product performance remains limited. This limitation hinders efforts to identify high-performing and underperforming items, assess customer purchasing behavior, and implement timely strategic adjustments. Sales data is not processed in real time. Analysis activities remain largely manual, leading to delayed interpretation of outcomes. Lack of analytical dashboards and visualization tools restricts comprehensive performance monitoring. Opportunities for optimization, strategic positioning, and revenue enhancement may remain unrecognized. Evaluating effectiveness of promotions, seasonal campaigns, and pricing strategies is difficult without proper trend analysis. Inability to generate visualized analytics reduces interpretability of data for management and stakeholders. Raw figures provide limited insight into patterns without contextual processing. Emerging demand trends or declining performance may be overlooked due to insufficient visibility. Delayed recognition negatively affects product planning and marketing responsiveness. Lack of automated analytics limits capacity to leverage data for strategic revenue optimization. Comparing performance across different branches is difficult without standardized visual reporting. Anomalies may remain undetected without automated alerts or indicators. Decisions to discontinue underperforming items may be delayed due to lack of visibility. Identifying complementary purchasing behavior for bundling or upselling strategies is challenging. Understanding consumer preferences in detail is restricted by limited tools. Communicating long-term trends to stakeholders is difficult without visualization. Fluctuations in demand may go unnoticed until significant decline occurs. Proactive reactions to changes in performance are weakened by these limitations. Strategic agility regarding product and marketing decisions is reduced. Inadequate analysis limits overall revenue optimization capabilities. Identifying specific purchasing patterns per branch is difficult as existing systems do not provide detailed comparative visualizations. Variations in demand across time periods may remain unnoticed. Interpreting and communicating insights during evaluations takes longer without automated dashboards. Forecasting initiatives become less accurate when historical data are not processed through advanced tools. Identifying items contributing most significantly to profitability remains challenging. Assessing effectiveness of pricing adjustments and campaigns is difficult without visualization. Opportunities to identify high-demand seasonal items remain underutilized due to insufficient monitoring. Delayed identification of declining performance increases risk of inventory waste and reduced profitability. Evaluating purchasing frequency and repeat transaction behavior is difficult without integrated analytics. Predictive indicators are absent, limiting preparedness for sudden market changes and preference shifts. Decisions related to development and planning rely more heavily on assumptions rather than evidence. Stakeholder understanding of performance during meetings and presentations may be weakened by lack of visualization. Establishing long-term revenue projections is difficult because historical trends are not systematically analyzed. Competitors utilizing advanced analytics gain stronger positioning due to faster interpretation of data and behavior patterns. Inadequate tools reduce responsiveness, intelligence generation, insight capabilities, and long-term growth potential.

Forecasting workforce requirements remains difficult without reliable peak-hour and customer traffic analysis. Decisions regarding scheduling are frequently based on assumptions rather than empirical evidence. This approach may result in overstaffing during low-demand periods or understaffing during peak hours. Ineffective allocation negatively affects labor cost efficiency, productivity, and service quality. Insufficient staffing during high-traffic periods compromises satisfaction and overall service performance. Repeated inefficiencies may affect morale, consistency, and operational balance over time. Historical traffic pattern analysis required to establish optimized models per branch is unavailable. Managers may schedule employees inconsistently without objective workload forecasts. Disparities in service quality across locations may occur during varying demand periods. Inaccurate forecasts lead to unnecessary overtime expenses or idle labor costs. Maximizing workforce utilization efficiency across operations is impossible without proper analytics. Inefficiencies contribute to burnout during unexpectedly busy shifts. Overstaffing leads to underutilization of working hours. Repeated inaccuracies reduce morale among staff members. Planning challenges affect ability to assign specialized tasks efficiently. Strategic shift planning for holidays and events is limited by absence of traffic forecasting. Preparation for anticipated surges during promotions is difficult. Lack of analytics creates inconsistent customer experiences across branches. Inefficiencies negatively affect service speed and processing times. Productivity and cost effectiveness are reduced by these limitations. Inadequate analysis weakens operational planning and service optimization. Balancing workloads fairly among employees is difficult for supervisors without accurate demand forecasts. Staff assigned to understaffed shifts experience excessive pressure, potentially affecting quality and morale. Overstaffing during low periods contributes to unnecessary expenditures and inefficient utilization. Evaluating whether current levels directly support productivity and satisfaction objectives is challenging. Identifying the most productive arrangements remains difficult without traffic analysis. Seasonal fluctuations in volume may not be anticipated effectively due to limited forecasting. Plans for weekends, holidays, and promotions become inconsistent and less efficient. Coordination between front-line employees and supervisors is affected during busy periods. Fatigue and reduced performance occur because schedules are not aligned with actual demand. Delays and longer waiting times negatively influence perceptions of efficiency and quality. Inefficiencies contribute to inconsistent performance across branches. Implementing standardized policies is difficult because workload measurements vary across locations. Ability to optimize allocation and improve productivity systematically is reduced. Insufficient analysis weakens performance management, efficiency, satisfaction, and overall operational stability.

Manual reporting procedures are time-consuming and highly vulnerable to human error, inconsistencies, and potential data loss. Employees spend substantial time preparing, encoding, validating, and consolidating reports from multiple branches. Lack of standardized formats complicates comparison and interpretation of performance data. Manual computation and repetitive entry increase risk of inaccuracies and calculation errors. Poor handling practices may result in misplaced, overwritten, or duplicated records. These limitations reduce reliability of reports and delay access to essential operational information. Continued dependence on manual processes diverts time away from more valuable responsibilities. Repetitive administrative tasks reduce productivity and create unnecessary bottlenecks. Delays in preparation postpone review and decision-making processes. Inconsistent formatting creates confusion during evaluation and analysis. Prolonged reliance on manual methods reduces efficiency and hinders scalability. Employees experience increased fatigue due to repetitive clerical work.

Attention is diverted from customer-facing tasks toward paperwork. Manual handling increases the likelihood of omitted or incomplete entries. Duplication of effort occurs across branches. Human-dependent processes make preparation vulnerable to absenteeism and workload fluctuations. Submission of branch reports may be delayed due to competing priorities. Strategic and operational decisions are postponed as a result. Paper-based or spreadsheet storage increases risk of accidental deletion or physical loss. Archiving practices complicate retrieval of historical documents.

Manual practices reduce productivity and reliability. Maintaining accuracy during high workload periods and tight deadlines becomes difficult. Manual verification consumes substantial managerial time that could be allocated to planning and improvement initiatives. Absence of automated validation increases the possibility that incorrect figures remain undetected. Inconsistent formulas and templates create discrepancies between branch records. Retrieval of historical documents becomes difficult as volume grows. Maintaining secure and organized backups of operational reports poses challenges. Reliability and accountability may be compromised by accidental deletion, corruption, or unauthorized modification. Motivation and work fatigue may increase over time among staff responsible for repetitive tasks. Communication between personnel and management is slowed because reports require individual review and verification. Delayed access weakens the ability to respond immediately to concerns and issues. Dependence on manual processes reduces readiness for digital transformation initiatives. Bottlenecks associated with clerical work negatively affect productivity and efficiency. Limitations become more severe and difficult to manage as the business expands. Continued reliance weakens accuracy, efficiency, security, and responsiveness.

Current systems do not effectively utilize **Business Intelligence** (BI) tools. This limitation restricts the ability to make informed, data-driven decisions that improve efficiency and support growth. Sales and operational data are collected consistently but not transformed into insights that support forecasting, planning, and evaluation. Existing reports remain largely descriptive and provide limited analytical depth. Predictive and advanced tools necessary for identifying trends, forecasting demand, and evaluating long-term performance are lacking. Strategic decisions are often based on intuition and experience rather than empirical analysis. Underutilization of data restricts opportunities for optimization, innovation, and sustained competitive advantage. Maximizing strategic value of historical and real-time information is prevented. Benchmarking performance against organizational targets is difficult without BI integration. Preparedness for seasonal fluctuations and market changes is limited by absence of forecasting capabilities. Opportunities for predictive planning and proactive strategy development remain underutilized. Lack of adoption weakens strategic agility and long-term competitiveness in the F&B market. Identifying long-term revenue trends without historical dashboards is challenging. Strategic planning becomes less precise without predictive modeling tools. Expansion decisions may rely on subjective judgment instead of performance-based analytics. Hidden correlations between promotions, pricing, and performance may be overlooked. Organizational learning from historical data is limited by absence of tools. Future forecasting remains less accurate without trend modeling. Adaptation to changing market behavior and preferences may be slower. Competitors utilizing technologies gain advantages. Innovation is reduced by underutilization. Competitiveness may decrease in an increasingly analytics-driven market. Identifying operational inefficiencies is difficult because existing systems provide limited visibility into activities and indicators. Absence of predictive analytics reduces ability to anticipate future demand and fluctuations accurately. Decision-making relies heavily on assumptions rather than comprehensive evidence. Excessive time may be spent interpreting raw data before arriving at conclusions. Opportunities to improve profitability through optimization may remain undiscovered due to insufficient capabilities. Identifying hidden relationships between staffing, promotions, traffic, and sales is difficult. Inadequate utilization weakens forecasting and increases likelihood of shortages or excess inventory. Historical data remains underutilized as systems cannot effectively transform information into intelligence. Evaluating long-term effectiveness of strategies and policies is challenging. Absence of real-time dashboards delays response to emerging problems and fluctuations. Insufficient integration reduces preparedness for competition and industry changes. Benchmarking remains inconsistent as comparisons are not automatically generated and visualized. Conducting accurate forecasting for expansion and resource allocation is limited. Competitors effectively utilizing technologies gain advantages in efficiency, understanding, and planning. Inadequate utilization weakens innovation, accuracy, adaptability, efficiency, and sustainable growth.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design and develop **BREWINSIGHT BI: Revenue Pattern Analytics System**, a Business Intelligence-based platform intended to centralize, organize, and analyze sales data from all Kape Tubong branches into one integrated digital environment. The proposed system seeks to transform raw transactional and operational records into meaningful and actionable business insights through the use of data visualization, automated reporting, analytical dashboards, and revenue trend analysis. By consolidating sales records from all kiosks and the main branch into a unified centralized database, the platform aims to eliminate fragmented data storage practices, reduce inconsistencies in record management, and provide management with a comprehensive view of business performance across all locations. Through centralized data consolidation, the system will improve data accessibility and ensure that sales information is consistently available in a structured and organized format. This centralized architecture will also reduce the time and effort required to manually collect and reconcile reports from different branches. In doing so, the system will promote better coordination among business locations and improve the overall management of organizational data resources. The platform is intended to provide a scalable and sustainable data management infrastructure that can support future branch expansion and increased transaction volumes. By replacing fragmented reporting processes with an integrated digital platform, the system seeks to modernize Kape Tubong's data management practices. This objective aligns with the business's need for a more efficient and centralized operational reporting solution. The system is expected to improve the visibility, accessibility, and strategic value of Kape Tubong's business data.

Simple consolidation serves only as the foundation. The proposed platform also seeks to establish a reliable technological environment where data can be processed consistently and securely across all operational units. The integration of branch-level sales records into a centralized analytical platform will improve standardization in data collection and reporting procedures. This standardization is expected to strengthen the consistency of business documentation and reduce reporting discrepancies among branches. By implementing an organized data management structure, Kape Tubong can improve administrative efficiency and ensure better long-term record maintenance. Centralized storage of sales data can improve historical recordkeeping and simplify long-term business analysis. This will allow management to evaluate organizational growth over extended periods using structured historical data. The centralized system also improves traceability and accountability in transaction monitoring by maintaining complete and organized digital records. As business operations continue to expand, maintaining such a structured data environment becomes increasingly important for operational sustainability. The proposed system therefore serves not only as an operational tool but also as a long-term data management solution. This strengthens the overall technological readiness of the organization for future business expansion.

The proposed system is specifically designed to analyze revenue patterns by identifying and interpreting sales trends across multiple time intervals, including hourly, daily, weekly, monthly, and yearly periods. Through this functionality, management will be able to monitor revenue fluctuations over time and identify patterns in sales behavior across different operational periods. The system will allow management to detect recurring sales cycles, evaluate seasonal trends, and determine the consistency of branch performance over time. It will also enable the comparison of sales performance between different dates, periods, and branches to identify patterns of growth or decline. These analytical capabilities will help management understand how internal and external factors influence revenue generation. By identifying peak and low-performing periods, the business can make strategic adjustments to staffing, promotions, and operational scheduling. The system will also help determine whether certain branches experience stronger performance during specific periods than others. Through detailed trend analysis, management can forecast future sales expectations based on historical performance data. This predictive understanding can support more informed planning and risk management. The system aims to improve management's understanding of revenue behavior across all Kape Tubong locations.

The platform will support deeper analytical examination of sales fluctuations by allowing management to evaluate the effects of holidays, promotions, weather conditions, and seasonal trends on customer purchasing behavior. This broader analytical perspective enables more comprehensive interpretation of performance trends beyond simple sales totals. Management can also use revenue pattern data to compare expected versus actual sales performance during planned campaigns and strategic initiatives. These insights may improve the organization's ability to evaluate operational effectiveness and refine future business strategies. Historical trend analysis may also support forecasting for new branch openings by using comparable data from existing locations. Through detailed revenue insights, management can establish more realistic performance benchmarks and expectations for each branch. Such forecasting capabilities contribute to better financial planning and resource management. The system's ability to reveal underlying business patterns strengthens strategic decision-making by providing evidence-based revenue analysis. This enhances management's confidence in planning operational and financial strategies. Revenue analysis becomes a core strategic function of the proposed platform.

A primary goal of the study is to improve the efficiency and accuracy of handling large volumes of sales transactions generated daily by multiple Kape Tubong branches. At present, manual data consolidation and fragmented reporting create delays and increase the likelihood of errors in sales analysis and reporting. Through automated data aggregation and processing, the BREWINSIGHT BI platform seeks to streamline the reporting workflow and reduce dependence on manual compilation of branch data. The platform will automatically collect, process, and summarize sales information into usable analytical reports and dashboard visualizations. This automation will reduce the workload of management and administrative staff while improving reporting speed and consistency. It will also minimize the occurrence of human error commonly associated with spreadsheet-based manual reporting methods. Faster and more accurate reporting will allow management to respond more quickly to operational issues and performance concerns. The system will provide updated and readily available business metrics whenever needed. This improvement in efficiency supports better operational control and faster access to performance information. The study aims to enhance the speed, reliability, and accuracy of Kape Tubong's sales reporting processes.

The automation of data processing is also expected to improve workflow efficiency across branches by reducing repetitive clerical tasks related to report preparation and submission. Employees will spend less time compiling transaction summaries and more time focusing on operational and customer service responsibilities. Standardized automated reports will improve consistency in business documentation and facilitate easier comparison across branches. By reducing reporting delays, the system enhances the timeliness of management review and intervention. It also decreases administrative bottlenecks caused by branch-level reporting dependencies. Automated processing supports scalability by enabling the system to handle increasing transaction volumes without significantly increasing administrative workload. This allows Kape Tubong to maintain efficient reporting processes even as more branches are established. Improved processing speed also strengthens management's ability to conduct frequent performance reviews. Automation enhances both operational productivity and managerial efficiency. The platform contributes significantly to process optimization.

The system aims to support detailed monitoring of key business performance indicators that are critical for operational and strategic decision-making. These include branch sales performance, revenue comparisons, product demand trends, top-selling items, underperforming products, customer purchasing behavior, and time-based sales patterns. By making these metrics easily accessible through the dashboard, the platform will help management monitor operational performance more effectively. Management will be able to evaluate which branches contribute the highest revenue and which locations require performance improvement. The system will also identify which products consistently perform well and which items may need reevaluation or promotional support. Product demand analysis will assist management in making informed decisions regarding menu optimization and pricing strategy. The system can also support the identification of consumer purchasing preferences based on historical sales data. These insights can guide targeted marketing campaigns and promotional planning. By monitoring multiple business indicators in one platform, the system will provide a holistic view of business performance. The study seeks to improve management's ability to monitor and interpret critical operational metrics.

The proposed platform also aims to promote a more structured and data-driven approach to management and operational evaluation within Kape Tubong. Rather than relying on intuition, assumptions, or delayed manual reports, management will be able to base decisions on real-time and evidence-based analytical information. The use of dashboards, visual summaries, and automated analytics will simplify the interpretation of complex sales data and improve decision-making efficiency. Through clearer visibility into business performance, management can identify performance gaps, operational inefficiencies, and revenue issues more quickly. The system will support faster managerial response to underperforming branches, declining sales trends, and product-related concerns. It will also improve accountability by providing measurable and objective data for evaluating operational outcomes. Data-driven decision-making will enable management to justify strategic actions using factual business evidence. This shift toward analytical management can improve planning accuracy and strengthen business oversight. It also promotes a culture of evidence-based management within the organization. The study aims to strengthen Kape Tubong's managerial decision-making capabilities through business intelligence.

The system seeks to assist Kape Tubong's owner and management in understanding branch-level and product-level revenue behavior across all business locations. It will help identify top-selling and slow-moving products, determine customer demand patterns, and recognize peak operating hours throughout the day. These insights will enable the business to improve inventory management by aligning stock levels with product demand patterns. The platform can also support more effective workforce scheduling by identifying periods of high and low customer activity. This allows management to allocate staff more efficiently according to actual demand. Promotional activities can also be better planned by identifying weak sales periods that may require targeted campaigns. Resource allocation decisions, such as equipment deployment and

inventory distribution, can be optimized using branch-specific performance data. These operational improvements can lead to reduced waste, improved service quality, and increased profitability. The system therefore supports both strategic and day-to-day operational decision-making. The platform is expected to improve the practical management of Kape Tubong's operations.
The proposed system aims to establish a technological foundation that supports future digital transformation initiatives within Kape Tubong. As the business continues to grow, the need for scalable digital systems becomes increasingly important for sustaining efficient operations. BREWINSIGHT BI is designed not only to address present operational challenges but also to provide a framework that can accommodate future enhancements and technological integrations. These may include advanced forecasting models, inventory analytics, customer relationship management integration, and predictive sales analysis. The platform's scalable architecture allows it to adapt to the increasing complexity of operations as additional branches or kiosks are established. This forward-looking design ensures that the system remains relevant and valuable even as organizational needs evolve. It also reduces the likelihood of requiring a complete system replacement in the near future. By investing in a scalable and adaptable platform, Kape Tubong can better prepare for future business expansion and digital modernization. This objective supports long-term organizational sustainability and technological readiness. The system is intended to serve as both an immediate operational tool and a strategic technological asset for the business.

Implementation of BREWINSIGHT BI is expected to improve operational transparency, strengthen management oversight, enhance reporting accuracy, and promote data-driven decision-making across the organization. By providing centralized and real-time access to business intelligence, the system will improve the transparency of operational performance across all Kape Tubong branches. Management will gain greater visibility into branch activities, sales outcomes, and performance indicators without requiring manual report consolidation. The platform will strengthen internal control by enabling faster detection of performance anomalies and inconsistencies. Enhanced reporting accuracy will improve the reliability of business information used for planning and strategic analysis. The system is also expected to reduce delays in information delivery, allowing management to respond more proactively to business conditions. Beyond immediate operational benefits, the study also aims to demonstrate the practical value of Business Intelligence technologies in supporting small and medium-sized enterprises within the Food and Beverage industry. It seeks to contribute to the digital transformation of Kape Tubong by modernizing its sales monitoring and analytics processes. In the long term, the platform is expected to support sustainable business growth, increased competitiveness, and future scalability as the business expands. The study aims to provide both practical organizational benefits and broader technological contributions through the development of the BREWINSIGHT BI platform.

The BREWINSIGHT BI platform aims to improve organizational responsiveness by enabling management to identify and address operational concerns in a timely and systematic manner. Through real-time access to updated performance metrics, management can immediately detect declining sales, unusual transaction patterns, or branch-level inconsistencies that may indicate emerging operational issues. This allows decision-makers to implement corrective actions before minor concerns escalate into larger organizational problems. Early detection of performance issues strengthens preventive management practices and supports more proactive business oversight. The availability of immediate operational insights can also reduce response delays commonly associated with manual reporting and delayed performance evaluation. By improving response time, the system enhances the organization's capacity to maintain service quality and operational consistency across all business locations. This responsiveness is particularly important in the highly competitive and dynamic food and beverage industry, where customer demand and operational conditions can change rapidly. The system's real-time monitoring capabilities therefore support more agile and adaptive business management. This contributes to improved organizational resilience in addressing operational and market-related challenges. The study aims to strengthen Kape Tubong's ability to respond efficiently to changing business conditions.

The study seeks to improve communication and coordination between branch-level personnel and upper management through a shared and standardized reporting environment. By centralizing all business information within a unified dashboard platform, the system will provide both management and authorized staff with consistent access to the same performance data and operational records. This shared visibility reduces communication gaps caused by inconsistent reporting practices or delayed transmission of branch information. It also promotes greater alignment between branch-level execution and organizational strategic objectives. Standardized reporting formats can improve clarity in performance discussions and reduce misunderstandings related to inconsistent data presentation. The platform may also enhance accountability by making branch-level performance more transparent and measurable. Through improved communication and information accessibility, the system supports more collaborative and coordinated operational management. Branch managers and staff can better understand performance expectations through access to measurable operational indicators. This promotes stronger alignment between individual branch operations and broader business goals. The study aims to improve organizational coordination through centralized information sharing.

The proposed system intends to support performance benchmarking and comparative analysis across Kape Tubong's multiple branches. Through branch comparison features and standardized performance metrics, management can evaluate which branches perform above or below organizational expectations. This comparative analysis will help identify operational best practices that can be replicated across locations. It will also reveal branch-specific weaknesses requiring targeted managerial intervention or strategic adjustment. Benchmarking branch performance using objective metrics supports fairer and more accurate evaluation of branch productivity. The organization can establish performance standards and monitor whether each branch meets established operational expectations. This feature may also improve motivation and accountability among branch managers by providing measurable performance

benchmarks. Branch-level comparisons can further support strategic decisions regarding expansion, restructuring, or operational resource allocation. Through data-based benchmarking, management can continuously improve operational consistency across all branches. The study aims to enhance branch performance management through comparative operational analytics.
The study also aims to support financial planning and budgeting processes by providing management with reliable historical sales data and revenue trend analysis. Accurate and organized financial information is essential for preparing realistic budgets, forecasting expected revenue, and planning future investments. Through structured revenue analysis, the system can help management estimate future income based on historical business performance. This capability improves budgeting accuracy and supports more strategic allocation of organizational resources. Financial planning can be further strengthened by identifying seasonal sales fluctuations and predictable revenue cycles. Management can use this information to prepare more accurate expenditure plans for labor, inventory, marketing, and expansion initiatives. Improved budgeting can reduce unnecessary spending and support stronger financial discipline within the organization. Access to organized financial data also strengthens the organization's ability to evaluate investment opportunities and expansion feasibility. This makes the system valuable not only for operational monitoring but also for long-term financial strategy. The study aims to improve Kape Tubong's financial planning capabilities through structured analytical reporting.

The study seeks to demonstrate the broader applicability of Business Intelligence technologies in small and medium-sized enterprises within the Philippine food and beverage sector. By successfully implementing BREWINSIGHT BI in Kape Tubong's operations, the project aims to provide a practical example of how localized SMEs can adopt modern data analytics and digital reporting technologies despite resource limitations. This contributes to the growing body of evidence supporting the feasibility of digital transformation among developing local enterprises. The study may encourage similar organizations to adopt business intelligence systems for improving operational monitoring and decision-making. It also highlights the potential of customized web-based analytical systems in addressing practical business challenges faced by SMEs. Through this implementation, the project demonstrates that advanced analytical technologies are not limited to large corporations but can also provide significant value to smaller growing businesses. This expands awareness of Business Intelligence as a viable operational strategy for SMEs. The study contributes not only to Kape Tubong's organizational improvement but also to broader discussions on SME digital modernization. Its findings may serve as reference material for future business technology initiatives in similar industries. The study aims to contribute to both practical business innovation and broader technological adoption within local SMEs.

1.3.2 Specific Objectives

Specifically, the study aims to:

Create a centralized web-based dashboard that integrates and organizes sales information from all kiosks and the main branch into a single platform to improve business monitoring and performance management. The dashboard will function as a unified interface for accessing and managing all sales transactions across branches. It will simplify the tracking of transactions and minimize the difficulty of handling data from multiple separate locations. The platform will enhance management's access to business information by enabling centralized and real-time monitoring of branch performance. It will also improve operational visibility, allowing management to quickly identify irregular sales activities, performance issues, and operational concerns. The dashboard will strengthen coordination between branches and management through standardized presentation and organization of sales records. By consolidating all transaction data into one accessible platform, the system will reduce dependence on manual branch-by-branch report collection. Management will no longer need to review fragmented spreadsheets or separate records from individual kiosks. This centralized approach will improve consistency in business reporting and ensure that performance data is uniformly structured across all locations. It will also allow management to compare branch performance more efficiently and identify performance gaps between locations. The dashboard will provide Kape Tubong with a more organized, transparent, and efficient monitoring environment for managing business operations. This consolidation ensures that every record, from daily sales totals to individual item transactions, is stored securely and systematically. It removes the risk of data loss or duplication often encountered when handling separate files. With all information synchronized in one place, the team can generate accurate reports instantly. This feature reduces the time spent on administrative tasks and allows more focus on core business activities and customer service quality.

The dashboard will improve enterprise-wide visibility by presenting branch performance in a single comparative environment. Management will be able to observe trends and anomalies across all kiosks simultaneously. This setup supports quicker identification of underperforming branches and operational inconsistencies. The centralized structure will reduce communication delays between branch personnel and administrators. Historical branch data may be reviewed through the platform to support long-term performance evaluations. Dashboard accessibility through web browsers will improve convenience for authorized users regardless of location. This feature will enable management to oversee operations remotely without requiring physical presence at each branch. Consolidated visual reporting will simplify executive review of operational performance during meetings and strategic planning sessions. The platform will also establish a foundation for future expansion by supporting scalable integration of additional branches or kiosks. The centralized dashboard will significantly improve the organization's capacity for efficient multi-branch business management. The system also allows the team to analyze seasonal patterns, peak hours, and product popularity over time. This capability provides deeper insights into customer behavior and revenue flow. Visualizing data through clear charts and summaries makes decision-making more evidence-based. It helps the business implement effective strategies such as targeted promotions, optimized staffing schedules, and better inventory planning tailored to the specific needs of each location.

Automate the generation of daily, weekly, and monthly sales reports to reduce manual processing, lessen human error, and improve the speed and accuracy of reporting activities. The system will automatically compile transaction records and produce reports according to predefined reporting schedules. This automation will reduce the administrative burden associated with manual data encoding and report consolidation. It will also minimize errors caused by repetitive calculations, inconsistent formatting, and incomplete reporting. Automated reporting will improve accessibility to historical sales records for future review and analysis. Standardized report outputs will further enhance consistency and reliability in evaluating business performance across branches. The automated generation of reports will ensure that management receives timely access to updated business information without delays caused by manual preparation. It will also reduce the time spent by staff in preparing repetitive sales summaries. Historical reports can be stored systematically in the system for future retrieval and comparative analysis. This feature will support long-term trend evaluation and performance tracking over time. Automated reporting will improve the efficiency, reliability, and timeliness of Kape Tubong’s reporting processes. The system organizes data in a structured format, making it easier to identify revenue changes and sales patterns over specific periods. It removes the need for double-checking calculations or correcting mismatched data entries. Staff members can redirect their effort toward more value-adding tasks such as customer service or operational improvement. This shift creates a more productive work environment and ensures that decision makers always work with accurate and up-to-date information.

Automation will ensure that report generation follows consistent formatting and computational standards across all reporting periods. This setup will improve comparability between daily, weekly, and monthly performance reports. The system will reduce dependency on staff availability for report preparation, minimizing disruptions caused by absenteeism or workload constraints. It will also support more immediate management review of branch performance after each reporting cycle. Automatically archived reports will improve documentation and audit readiness for financial review purposes. Management can retrieve previous reports more efficiently without manually searching through physical or digital files. Automated timestamps and report logs will improve accountability and traceability in reporting activities. The reduction in repetitive administrative tasks may improve staff productivity and operational focus. Faster report availability can enhance the speed of managerial response to emerging performance issues.

Report automation will strengthen both operational efficiency and analytical responsiveness. The system also preserves data integrity by maintaining the original values and structure of every report generated. This capability supports compliance requirements and simplifies internal or external auditing procedures. It creates a reliable history of business performance that serves as a solid foundation for strategic planning and goal setting.

Process and visualize revenue patterns, including peak operating periods, customer purchasing trends, and top-selling products, to support data-driven and strategic decision-making. The system will analyze transactional sales data and convert it into graphical and summarized analytical reports that illustrate business performance over time. It will identify recurring peak sales periods and reveal customer purchasing behavior patterns across locations. The platform will help management determine which products generate the highest revenue and which items require strategic attention. It will also assist in observing fluctuations in sales over various periods to support forecasting and trend analysis. These analytical insights will contribute to more informed planning, promotional decisions, and operational strategy development. Through visual charts, graphs, and dashboard indicators, the system will simplify the interpretation of complex sales information. Management can use these insights to identify profitable periods and underperforming intervals across different branches. The analysis of customer purchasing trends will help determine shifts in consumer preferences over time. Revenue pattern visualization will also support more accurate forecasting of future sales expectations. The platform will provide valuable analytical intelligence to strengthen business planning and strategic management. Data is transformed into clear visuals that reveal hidden opportunities and potential risks. This allows leaders to act based on facts rather than assumptions. Strategies become more targeted and effective, leading to better business outcomes.

Revenue analysis will help management detect seasonal or cyclical sales trends that influence business performance. The system can highlight unusual spikes or declines in revenue that may indicate operational concerns or market changes. Comparative visualization across time periods will support better understanding of business growth patterns. Product performance analysis can guide pricing adjustments and promotional planning based on actual demand behavior. The platform may also assist in evaluating the financial impact of marketing campaigns or operational changes. Branch-specific trend analysis will allow management to compare customer behavior across locations. This detail can support more localized and targeted operational strategies. The availability of visual analytics will reduce the difficulty of interpreting large transaction datasets manually. Strategic decisions can therefore be based on measurable and evidence-backed performance indicators. The system will improve the quality and confidence of managerial decision-making. It offers a clear perspective on how different factors affect income and profitability. Leaders gain the ability to predict changes and prepare appropriate responses well in advance. This level of insight strengthens the company's competitive position and ensures long-term stability.

Provide inventory-related analytical insights based on sales performance data to support effective stock management and reduce inventory inefficiencies. By evaluating product demand and sales movement, the system will help management maintain appropriate inventory levels based on actual consumption trends. It will identify fast-moving and slow-moving items to improve replenishment planning and purchasing decisions. The platform will also help reduce overstocking and stock shortages, minimizing product waste and preventing missed sales opportunities. It will contribute to improved inventory turnover and more efficient utilization of stock resources across all business locations. Inventory-related analytics will enable management to align purchasing decisions with actual product demand patterns rather than assumptions. This approach can help prevent unnecessary inventory expenses and reduce losses associated with expired or unsold products. The system can also support branch-specific stock planning by identifying location-based differences in product demand. Improved inventory planning may contribute to better product availability and customer satisfaction. By integrating inventory-related insights with sales analytics, the platform will support more efficient operational planning. This feature strengthens Kape Tubong's inventory management practices and resource allocation efficiency. It ensures that every branch holds only the necessary amount of stock according to its own sales history, which prevents excess accumulation and lowers storage costs. The system guides the team in making accurate purchase orders that match real customer needs, leading to smoother operations and better use of available capital.

The system can also assist in identifying recurring stock consumption trends for frequently purchased items. This information can support more accurate procurement scheduling and supplier coordination. Inventory insights may help management prepare for anticipated demand surges during peak periods or promotional events. Branch-level inventory analysis can reveal location-specific stock requirements and customer preferences. This detail will improve distribution planning across multiple kiosks. Better stock forecasting can reduce emergency purchasing and associated supply chain inefficiencies. Inventory waste due to spoilage or expiration may be reduced through more accurate replenishment practices. The integration of sales and inventory analytics will strengthen the relationship between demand forecasting and stock planning. Improved stock management can contribute directly to profitability by lowering unnecessary inventory costs. Inventory-related analytics will enhance both financial efficiency and operational preparedness. It creates a clear view of how stock flows from suppliers to customers, allowing for better coordination between different parts of the business. The system turns raw sales numbers into practical guidelines for stock control. This process ensures that products remain available without tying up too much capital in storage. It leads to a balanced inventory system that supports steady sales and consistent customer service quality.

Establish secure authentication and controlled user access to protect system data while allowing authorized personnel to access reports remotely. The system will implement login authentication and role-based access control features to ensure that only authorized users can utilize system functions and access business information. Different access privileges will be assigned based on user roles and responsibilities within the organization. These security mechanisms will help safeguard sensitive sales and operational data against unauthorized access, misuse, and potential data breaches. The web-based platform will allow secure remote access to dashboards and reports through internet-connected devices. This feature provides convenience and flexibility for management while maintaining strict data protection standards. Password protection and session management features will further strengthen account security and user authentication processes. Administrative users will have broader access privileges compared to regular staff users to maintain proper control over system functions. Access restrictions will prevent unauthorized modification or deletion of critical business records. Secure remote access will allow management to monitor business performance even when off-site or away from branch locations. These security measures help maintain the confidentiality, integrity, and availability of business information. This objective ensures that the system remains secure, controlled, and accessible only to appropriate personnel. Every access attempt is verified to ensure that sensitive details remain private and protected at all times. The setup creates a safe environment where information can be shared freely among qualified users without risk of exposure.
User activity logs may be implemented to record login attempts and significant system actions for accountability purposes. This function will improve monitoring of user behavior and help detect suspicious activity within the platform.
Role-based permissions will ensure that users only access information relevant to their responsibilities. Session timeout mechanisms may reduce the risk of unauthorized access due to unattended devices. Encrypted data transmission can further strengthen protection during remote access sessions. Secure authentication protocols will improve trust in the system's handling of sensitive operational records. Controlled access structures will also help preserve data integrity by limiting editing rights to authorized administrators. These measures are particularly important because the system stores confidential financial and sales-related information. Effective security implementation will support compliance with good information management practices. Robust access control and authentication mechanisms are essential to maintaining a secure and dependable Business Intelligence environment. Every action taken inside the system is tracked and recorded, creating a clear history for review if needed. This level of protection builds confidence among users and ensures that business data remains accurate, complete, and secure from any internal or external threat.

1.4 Scope and Delimitation

This study focuses on the development of **BREWINSIGHT BI: Revenue Pattern Analytics System**. This is a **Business Intelligence-based platform** created to gather, centralize, organize, and evaluate sales data collected from Kape Tubong's kiosks and main branch. The system is designed to identify revenue patterns by processing transactional sales records and transforming them into meaningful business insights. Management will gain more accurate and accessible information through this platform to support operational planning and decision-making. This study aims to improve current methods of managing, interpreting, and utilizing sales data by implementing a centralized analytics and reporting solution. The proposed platform seeks to modernize how performance analysis and reporting are handled by replacing fragmented manual monitoring methods. It establishes a more organized framework for storing and retrieving historical sales information. Digital centralization is expected to improve data consistency and reduce reporting inefficiencies across all branches. The platform serves as a practical analytical tool supporting daily and strategic business decisions. This study focuses on demonstrating effective application of Business Intelligence technologies within small and medium-sized food and beverage enterprises. The scope centers on development of a specialized analytics platform tailored to the operational needs of Kape Tubong.

The proposed system functions as a web-based platform integrating sales records from multiple branches into a unified database. It features modules dedicated to data gathering, transaction recording, product performance tracking, automated report generation, and dashboard visualization. These components provide management with a centralized platform for monitoring sales activities and evaluating branch performance. The dashboard presents key business metrics using charts, graphs, summary panels, and comparative reports. This format improves readability and interpretation of data. Management can monitor trends, assess branch productivity, and identify operational issues more effectively through this interface. The web-based structure allows access through internet-enabled devices without requiring dedicated software installation. This capability improves flexibility and accessibility when reviewing business performance remotely. The centralized database structure ensures sales information from all branches remains synchronized and stored in a consistent format. Combining multiple monitoring functions into one platform simplifies data access and improves operational oversight. The study includes design and implementation of both technical and visual components of the web-based platform.

The system provides branch-level and overall business performance summaries to support comparative analysis among locations. Management can identify high-performing and underperforming branches through these comparisons. Operational factors influencing branch outcomes can also be evaluated. This feature assists in performance assessment and strategic planning regarding future improvements or expansion. Comparative branch analytics helps identify location-specific strengths and weaknesses in operations. Performance benchmarking across locations may reveal operational practices suitable for replication in underperforming areas. This functionality strengthens the ability to make location-based strategic decisions supported by actual performance data. It also promotes accountability among branch managers by providing measurable performance indicators. Branch comparison reports assist in evaluating effectiveness of branch-level promotions and operational strategies. Branch comparison analytics serves as an essential component of system functionality within this scope.

The analytics component evaluates sales transactions using variables such as transaction date and time, branch location, product category, and quantity sold. Insights related to revenue trends, branch revenue comparisons, peak sales periods, customer purchasing behavior, and product performance are generated from these inputs. These analytical outputs help management understand business performance clearly and support evidence-based operational decisions. Analysis focuses on identifying descriptive patterns and trends based on historical and current transaction data. Recognition of recurring business behaviors and performance trends over time becomes possible. The system supports filtering of data across various categories and time periods to enable deeper analysis. Structured analytics allows examination of business performance from multiple perspectives. This capability contributes to informed and data-supported decision-making processes. The scope includes descriptive analytical processing of transactional sales information using predefined business metrics.

Revenue visualization tools within the platform allow review of sales fluctuations across hourly, daily, weekly, and monthly periods. This feature assists in identifying recurring demand patterns, seasonal trends, and irregular sales behavior affecting business planning. Understanding these trends improves decisions related to promotions, staffing schedules, operating hours, and resource distribution. Revenue visualizations are displayed through graphs, trend lines, charts, and summarized dashboard indicators. These visual tools simplify interpretation of numerical sales data and improve analytical clarity. Historical trend analysis supports assessment of effectiveness regarding past operational and promotional strategies. Visualization capability strengthens the practical value of the platform as a decision-support tool. Implementation of visual analytical dashboards constitutes a major component of system development within this study.

Inventory-related insights are generated by evaluating product sales movement to support stock planning and demand analysis. Identification of high-demand and low-demand products enables informed purchasing and stocking decisions. The system does not automate stock replenishment processes. It provides analytical support only to improve inventory planning and reduce inefficiencies related to overstocking or shortages. Inventory-related analytics are based solely on historical sales movement and product demand trends. These insights aim to improve stock planning accuracy and support efficient purchasing decisions. Branch-specific demand analysis helps optimize stock distribution among locations. The system functions strictly as an analytical support tool rather than an automated inventory management platform. Inventory-related analytics are included strictly for decision-support purposes within this scope.

Data security and system reliability are maintained through implementation of authentication and role-based access control. These mechanisms ensure only authorized personnel can access system functions and reports. Access is available through desktop and mobile web browsers. This setup allows remote monitoring of business performance when necessary. Accessibility improves convenience while ensuring timely availability of important business information. Centralized storage of branch data improves consistency, reliability, and standardization of records across all locations. User roles define varying access privileges depending on assigned authorization levels. These controls maintain confidentiality and integrity of business-sensitive sales information. Secure remote access allows review of performance metrics outside physical branch locations. Implementation of security and accessibility mechanisms necessary for practical deployment is included in this study.

Scope of this study remains limited to analysis of internal sales and transaction data generated from daily business operations of Kape Tubong. External variables such as competitor pricing, consumer demographics, weather conditions, market trends, and economic factors are not included in analytical processing. All revenue analysis produced by the platform is based solely on internal transactional business data. This limitation maintains focus on operational sales performance analysis within the organization. Exclusion of external variables means analytical outputs reflect only internal business performance patterns. The system does not account for market-wide or environmental influences on revenue trends. This delimitation ensures manageable system scope and development complexity.

Several limitations are established to keep the project focused and manageable. The system does not involve design or integration of physical Point of Sale hardware. Receipt printers, barcode scanners, card readers, and cash drawers are excluded from development. Processing is limited to digital sales data entered or provided through available transaction records. Automated stock replenishment, supplier management, procurement functions, and warehouse logistics operations are also excluded. The platform is not intended to replace existing transactional hardware infrastructure or supply chain systems. Its role remains limited to analytical processing and business intelligence support. These exclusions ensure the study remains centered on analytics rather than full operational automation.

The proposed platform does not cover broader enterprise-level business functions. Payroll administration, taxation, budgeting, accounting systems, and financial statement preparation are excluded. It is not designed to operate as a full Enterprise Resource Planning system. Integration with banking services, online payment platforms, customer loyalty systems, and third-party accounting software falls outside this scope. These exclusions prevent overlap with unrelated enterprise functions beyond the purpose of the proposed system. Focus remains strictly on sales analytics and business intelligence. Enterprise-wide financial and administrative system functionalities are beyond intended project coverage.

Advanced predictive analytics, machine learning models, and artificial intelligence-based forecasting mechanisms are excluded from implementation. This study focuses only on descriptive and trend-based analytics. These outputs provide historical and current performance insights rather than predictive or prescriptive recommendations. Forecasting and predictive modeling are excluded to maintain alignment with current technical scope and development timeframe. Implemented analytics focus on descriptive business intelligence rather than advanced computational forecasting. Future enhancements may integrate predictive technologies if required by the organization.

BREWINSIGHT BI serves as a software-based analytics support platform enhancing sales data interpretation and reporting efficiency. It strengthens management decision-making through centralized Business Intelligence capabilities. Primary development serves academic purposes while providing practical benefits to Kape Tubong. Improvements regarding sales data management and promotion of data-driven operational practices are expected outcomes. The project serves as a foundation for future enhancements involving predictive analytics, inventory automation, customer behavior analysis, and broader system integration. This study establishes an initial framework for future digital transformation initiatives within the organization. The developed platform may be expanded to incorporate more advanced features as business requirements evolve. Scope and delimitations ensure delivery of a practical, manageable, and relevant analytics solution tailored to present operational needs.

This study encompasses design of a structured database architecture capable of storing, organizing, and managing historical transactional information. Data from all branches is maintained within a centralized repository. Database architecture supports efficient data retrieval, long-term storage, and consistent organization of sales records across multiple reporting periods. A structured historical dataset enables retrospective performance evaluations and trend comparisons over extended periods. Centralized database design improves data integrity by reducing redundancy, inconsistency, and fragmentation associated with decentralized spreadsheet-based recordkeeping. Historical data retention supports long-term analytical value by enabling comparison of current performance against previous periods. Structured storage facilitates future enhancements should advanced analytics or predictive modules be required later. Database design supports scalability as transaction volumes increase with business expansion. Proper normalization and organization improve system performance and maintainability over time. Technical sustainability is ensured as operational complexity increases. Development of a scalable and structured backend database environment forms part of the system scope.

Implementation of user interface components is included to improve usability, accessibility, and interpretation of analytical outputs for non-technical users. Business owners and managers may not possess specialized technical knowledge in data analysis. System design emphasizes user-friendly dashboard layouts and intuitive navigation structures. Graphical reports and summary panels simplify understanding of complex metrics through visual representation. User-centered design ensures analytical outputs remain practical and understandable for operational decision-making. Interface components are optimized for readability and efficient access to key performance information. Dashboard layouts prioritize display of most critical indicators for immediate visibility. User interface responsiveness supports accessibility across various screen sizes and devices. This capability improves the practicality of remote monitoring using laptops, tablets, and mobile devices. Usability prioritization ensures system accessibility regardless of user technical expertise. Front-end design considerations supporting practical adoption and ease of use are included within the scope.

This study covers implementation of report archival and historical retrieval functionalities to support long-term performance review and documentation. Generated reports are stored within the system to enable retrieval of past analytical outputs for comparison and strategic review. Archival capability strengthens long-term performance tracking by enabling evaluation of business progress across different operational periods. Historical reporting supports auditing, internal review, and retrospective performance assessment. Retrieval of archived reports eliminates the need for separate manual storage of previous sales summaries and analytical records. This improvement enhances organizational recordkeeping and supports systematic documentation practices. Historical report retrieval enhances strategic planning through comparative analysis of previous and current operational outcomes. Stored reports assist in evaluating long-term effects of pricing changes, branch expansion, and promotional campaigns. Functionality contributes to stronger business continuity and institutional knowledge retention. Report archival and retrieval are included within functional scope.

Delimitations specify the system will only process structured sales data provided by Kape Tubong. Assumptions regarding availability of accurate transactional input from branch operations apply. Effectiveness of system analytics depends heavily on completeness, consistency, and correctness of data entered or imported. Inaccurate or incomplete records may affect reliability of generated outputs. This study does not include mechanisms for independently validating business accuracy of manually entered records beyond standard input controls. Data governance and transactional accuracy remain dependent on branch-level operational compliance and user discipline. The system assumes users follow standardized procedures when recording or importing transactions. This limitation exists because the project focuses on analytical processing rather than transactional auditing or verification. Reliability of outputs remains partially dependent on the quality of source data provided. This delimitation ensures realistic expectations regarding automated analytical accuracy. Data input accuracy remains an operational responsibility outside system analytical scope.

This study is further limited to the operational environment, workflow structure, and reporting requirements specific to Kape Tubong at time of development. Design and functionality are tailored specifically to current business model, reporting practices, and analytical needs. Conceptual adaptability to other businesses exists but customization may be required before implementation in different contexts. Variations in operational workflow, reporting requirements, product categories, and branch structures necessitate modifications for external use. The system is not intended to function as a universally deployable solution without customization. This delimitation maintains alignment between system design and specific operational requirements of the target organization. It ensures focus remains on solving identified needs rather than addressing generalized scenarios. Customization for other industries or organizations falls outside current scope. The developed platform remains organization-specific in design and implementation. This focus ensures the system remains practical and highly relevant to the intended deployment environment.

This study aims to deliver a functional and practical Business Intelligence platform. System performance evaluation is limited to testing within the context of the developed prototype and available operational dataset during implementation period. Long-term real-world operational performance beyond the study period is not covered within formal evaluation scope. Factors such as long-term user adoption, organizational change management, evolving requirements, and future infrastructure demands fall outside research timeframe. Focus remains primarily on design, development, implementation, and initial functional evaluation of the proposed platform. Extended production-level deployment considerations require future organizational assessment after full implementation. Despite these limitations, the study provides a practical foundation for future operational deployment and enhancement. The developed prototype demonstrates technical feasibility and organizational applicability of centralized Business Intelligence for Kape Tubong. It establishes a baseline model for future expansion and refinement of the platform. Long-term deployment outcomes remain outside immediate scope while providing a strong implementation framework for future adoption. This reinforces practical and academic value within defined boundaries.

1.5 Significance of the Study

The proposed system will benefit the following:

Management – The proposed system will significantly assist the management of Kape Tubong by providing centralized, accurate, and real-time access to sales and performance data from all branches. This capability will enable management to monitor branch operations more efficiently, evaluate revenue performance comprehensively, and identify sales trends with greater precision. Through analytical dashboards, automated reporting, and visual performance summaries, management can make more informed decisions regarding pricing strategies, promotional planning, workforce allocation, product offerings, and branch expansion initiatives. Furthermore, the system will reduce management's reliance on delayed manual reports and subjective judgment by providing timely and evidence-based operational insights. This improved access to reliable business intelligence will support proactive decision-making, strengthen strategic planning, and enhance the organization's responsiveness to operational concerns and market fluctuations. As a result, management can exercise greater control over business performance and improve long-term planning capabilities.The platform will also help management identify operational inefficiencies and performance gaps more quickly across branches. By detecting revenue declines or unusual sales patterns early, management can implement corrective actions before issues worsen. The system can further support comparative evaluation of branch-level initiatives and promotional campaigns. Overall, the platform strengthens managerial oversight and strategic business control.Additionally, the system enables management to evaluate business performance using measurable and standardized performance indicators rather than subjective assessments. It can support executive meetings and strategic reviews by providing readily available reports and visual analytics. Management may use the system to assess branch profitability before making investment or closure decisions. The platform also assists in validating whether business strategies are producing desired outcomes based on actual sales results. Through historical performance tracking, management can establish realistic operational benchmarks and revenue targets for each branch. The system may further improve managerial accountability by documenting the basis of strategic decisions using factual business evidence. It can strengthen long-term forecasting by allowing management to analyze historical growth patterns and recurring demand cycles. Management can also use the platform to determine which branches require operational intervention or additional support. Better analytical visibility may improve confidence in making high-level organizational decisions.Moreover, the platform can help management identify branch-specific operational patterns that may not be visible through manual reports alone. It may reveal location-based differences in consumer demand, purchasing preferences, and revenue contribution across branches. These insights can support more targeted strategic planning for each kiosk or branch location. Management may also use the system to evaluate the impact of new products, pricing adjustments, or promotional strategies based on measurable sales outcomes. The platform can improve strategic resource distribution by identifying which branches require additional manpower, marketing support, or inventory allocation. It may also assist in determining the most profitable operational hours for each location, enabling more efficient scheduling decisions. Through data-backed performance evaluations, management can improve fairness and consistency in assessing branch managers and staff performance. The system may further support strategic planning by helping management identify long-term business growth trends and recurring revenue patterns. Historical data insights can be used to justify future investments, equipment purchases, or expansion initiatives with greater confidence. By consolidating all performance metrics into one platform, the system simplifies strategic review and reduces the complexity of interpreting multiple reports from different sources. It also enables management to maintain better oversight of organizational performance despite branch expansion and operational growth. These capabilities can improve managerial responsiveness and reduce delays in strategic decision-making. The platform may also strengthen executive confidence by providing more transparent and objective performance information. Ultimately, the proposed system empowers management to govern operations more efficiently, strategically, and proactively. Therefore, the system provides significant operational, analytical, and strategic value to Kape Tubong's management team.

Staff –The implementation of the proposed system will benefit staff members by reducing the administrative burden associated with manual data encoding, report preparation, and sales consolidation. Automated reporting and centralized data processing will streamline repetitive clerical tasks, allowing employees to focus more on customer service and core operational responsibilities. In addition, the system will promote standardized reporting practices across all branches, reducing inconsistencies in data recording and improving coordination between staff and management. By minimizing manual intervention in data handling processes, the system can reduce the likelihood of human error, improve task efficiency, and contribute to a more organized and productive working environment. This may also help reduce employee workload stress and improve day-to-day operational workflow.Staff will spend less time compiling reports and more time focusing on branch operations and service quality. The simplified reporting process may also reduce confusion caused by inconsistent documentation procedures. Better organization of operational data can improve communication between branch personnel and management. Employees may experience improved productivity due to streamlined reporting responsibilities. Therefore, the system is expected to contribute positively to staff efficiency and workplace organization.Additionally, the system can improve employee confidence when performing reporting-related tasks by simplifying the data submission and documentation process. It may reduce dependence on extensive supervision for preparing reports, allowing staff to perform routine reporting functions more independently. New employees may adapt more quickly to operational procedures due to the standardization of reporting workflows. The reduction of repetitive clerical work can lessen employee fatigue associated with administrative tasks. Staff can also benefit from clearer performance feedback when management uses analytical reports for operational evaluation. Improved reporting transparency may encourage greater accountability among employees and branch personnel. The platform can support smoother communication regarding branch performance expectations and operational goals. Staff may also experience fewer conflicts related to reporting discrepancies and missing records. The organized workflow introduced by the system can improve workplace discipline and procedural consistency.Moreover, the system can enhance employee efficiency by reducing the time required to retrieve historical sales records and operational reports. Staff may no longer need to search through multiple spreadsheets or physical records when verifying transaction data. This improvement can accelerate task completion and reduce unnecessary interruptions during branch operations. The platform may also improve collaboration among branch employees by ensuring that all personnel follow the same standardized reporting procedures. Through clearer operational workflows, staff can better understand their responsibilities in relation to sales documentation and data submission. The system can help reduce frustration associated with correcting reporting errors caused by inconsistent manual processes. Employees may also gain greater awareness of branch performance through transparent reporting and performance visibility. This can motivate staff to improve productivity and contribute more actively to achieving branch targets. In addition, the reduction of repetitive administrative tasks can improve job satisfaction by allowing employees to focus on higher-value operational responsibilities. The platform may also strengthen staff accountability by maintaining timestamped and traceable reporting records for each submission. Better documentation practices can support fairer performance assessments and more accurate evaluation of employee contributions. Staff may further benefit from improved communication with management due to faster and more organized reporting channels. The streamlined workflow can reduce reporting bottlenecks during peak operational periods. Overall, the implementation of the system supports a more efficient, organized, and employee-friendly operational environment. Therefore, the system contributes to both improved staff productivity and a more efficient working environment.

Organization – For Kape Tubong as an organization, the proposed system is expected to enhance operational efficiency, strengthen internal control mechanisms, and improve overall business performance. By consolidating sales and operational data into a centralized platform, the organization can establish a more systematic approach to monitoring branch activities, evaluating performance metrics, and managing operational resources. The system’s revenue analytics capabilities will support improved workforce scheduling, inventory planning, promotional timing, and branch-level performance assessment. These improvements can reduce unnecessary labor expenses, minimize inventory waste, and optimize resource allocation based on actual sales behavior. In the long term, the adoption of Business Intelligence practices may enhance organizational competitiveness, improve adaptability to market demands, and support sustainable business expansion.The platform may also improve transparency across branch operations by making performance data more visible and accessible to decision-makers. Improved access to branch-level metrics can support better accountability and performance monitoring. The organization can use data-driven insights to guide future strategic planning and expansion initiatives. Enhanced operational visibility may contribute to more consistent performance across branches. Therefore, the proposed system is expected to strengthen both short-term operations and long-term organizational growth.Additionally, the platform can help the organization establish standardized operational benchmarks and performance expectations across all branches. It may improve the company's ability to maintain consistency in reporting and business evaluation despite branch expansion. The organization can leverage analytics to improve internal audits and compliance monitoring. Historical data records may support better budgeting and financial planning for future operations. The system may also improve readiness for partnerships, investments, or expansion opportunities by demonstrating organized operational management. Better transparency in performance data can strengthen stakeholder trust and internal governance practices. The business may become more resilient to operational disruptions due to improved visibility into performance issues. The system can also help identify operational best practices that can be replicated across branches. Enhanced business intelligence capabilities may improve adaptability to market competition and customer behavior shifts. Therefore, the system contributes significantly to the organization's operational maturity and long-term sustainability.Furthermore, the proposed platform may support faster managerial response to emerging operational challenges through real-time access to branch performance indicators. It can improve coordination between branch managers and upper management by providing a shared data environment for decision-making and reporting. The organization may also benefit from improved strategic forecasting by analyzing historical and seasonal sales patterns across branches. This can enable management to anticipate demand fluctuations more accurately and prepare proactive operational strategies. The system may foster a culture of performance-driven management by encouraging branch personnel to align operations with measurable organizational objectives. Enhanced data accessibility may also reduce reliance on manual reporting processes, minimizing administrative workload and reporting errors. As organizational data becomes more structured and readily available, management can make more timely and evidence-based decisions. Ultimately, the implementation of the proposed system may position Kape Tubong for stronger operational governance, increased managerial effectiveness, and more sustainable long-term business growth

BSIT Students – The study will provide educational value to Bachelor of

Science in Information Technology (BSIT) students by serving as a practical example of how software engineering, database systems, web development, and Business Intelligence technologies can be integrated to solve real-world organizational problems. It demonstrates the practical relevance of technical concepts learned in academic settings through their application in developing a functional business analytics solution. Additionally, the project may serve as a model reference for future capstone projects involving decision-support systems, dashboard platforms, data analytics applications, and business process automation systems. It may also encourage students to further explore emerging fields such as Business Intelligence, data analytics, and enterprise systems development.The study showcases how academic theories and technical skills can be translated into practical software solutions for businesses. It may inspire students to pursue similar technology-driven projects in future coursework or professional practice. The project also illustrates the importance of user-centered system development in solving organizational challenges. By studying this implementation, students can better understand the practical complexities of full-cycle software development. The research may broaden student awareness of analytics-driven business technologies and their applications. Therefore, the study contributes meaningfully to academic and professional learning among BSIT students.Additionally, the project demonstrates practical integration of frontend, backend, and database technologies within a single application architecture. It can help students understand the challenges of designing scalable and maintainable web-based systems. The study may serve as an instructional example for requirements gathering, system analysis, and stakeholder consultation. Students can observe how technical solutions must align with business needs and user expectations. The project also highlights the importance of data visualization and usability in system design. It can strengthen understanding of software documentation and structured system development methodologies. Future student developers may use the study as a benchmark for improving their own capstone implementations. Exposure to practical BI applications may encourage specialization in analytics-related careers. The study can support classroom discussions on enterprise system design and digital transformation. Therefore, it serves as a valuable academic resource for developing practical IT competencies.Furthermore, the study may enhance students' appreciation for interdisciplinary problem-solving by demonstrating how business knowledge and technical expertise must work together in system development. It can expose students to the practical considerations involved in designing solutions for real organizational environments, including scalability, maintainability, and user adoption. The project may help students develop a stronger understanding of project planning, implementation timelines, and deployment strategies in software development. It also provides insight into the importance of system testing, debugging, and iterative refinement throughout the development lifecycle. Through examining the study, students may gain a deeper understanding of how information systems contribute to organizational decision-making and operational efficiency. The project may encourage critical thinking regarding the ethical and responsible use of business data in software applications. Additionally, it can reinforce the importance of professional documentation, presentation, and communication in technical project delivery. Ultimately, the study serves as a comprehensive educational example that bridges theoretical instruction with practical industry-oriented application.

Future Researchers – Future researchers may utilize this study as a reference for related investigations involving Business Intelligence implementation, sales analytics systems, digital transformation strategies, and technology adoption among small and medium-sized enterprises. The study contributes to the academic body of knowledge by presenting a practical framework for designing and implementing a revenue analytics platform within the Food and Beverage industry. Moreover, future researchers may build upon this work by integrating advanced features such as predictive analytics, machine learning algorithms, artificial intelligence, or broader enterprise system functionalities. The study may therefore serve as a foundation for further innovation and academic exploration in the field of data-driven business technologies.Researchers may also use the project as a comparative basis for evaluating similar systems in other industries or business contexts. The findings can contribute to future discussions on the role of analytics in SME operational improvement. It may also support future studies on digital transformation readiness among small business organizations. The framework presented in this research can guide future enhancements to business intelligence systems for retail and service industries. As technological needs evolve, this study may serve as a stepping stone for more advanced analytics research. Therefore, the project contributes both practical and scholarly value to future academic work.Additionally, future researchers may replicate the framework in other sectors such as retail, hospitality, logistics, and service-based enterprises. The study can support comparative analyses between manual and automated business intelligence systems. Researchers may investigate long-term impacts of BI adoption using this implementation as baseline data. The project may also inspire studies focused on user acceptance and usability of dashboard-based analytics systems. Future work may examine the effectiveness of descriptive analytics versus predictive analytics in SME environments. The architectural framework may serve as a reusable model for web-based enterprise analytics platforms. Researchers may explore integrating machine learning forecasting into the current system architecture. The study may contribute to interdisciplinary research involving business, information systems, and data science. It can also support academic discourse on digital maturity among SMEs in developing economies. Therefore, the study establishes a strong scholarly foundation for future technological and business research.Furthermore, future researchers may investigate the scalability of the proposed system when applied to larger organizations or multi-regional business operations. The study can serve as a baseline for assessing how Business Intelligence platforms evolve as organizational complexity increases. Researchers may also explore the integration of real-time data streaming and cloud-based analytics into similar business environments. The project may support future experimental studies measuring the direct relationship between analytics adoption and business performance outcomes. In addition, the findings may encourage research into barriers and facilitators affecting Business Intelligence adoption among SMEs. Future scholars may adapt the system framework to assess industry-specific analytics requirements and customization needs. The study may also inspire methodological improvements in evaluating the effectiveness of enterprise analytics systems in academic research. Ultimately, this work provides a practical and theoretical foundation that future researchers can extend to advance knowledge in business intelligence, enterprise systems, and digital innovation.

1.6 Definition of Terms

Analytics: Refers to the structured process of collecting, examining, and interpreting data through computational and statistical methods in order to identify significant patterns, trends, and relationships. It involves converting raw data into organized information that can support evaluation and informed decision-making. In business settings, analytics is utilized to assess operational performance, understand customer behavior, monitor sales activities, and forecast future business outcomes. It serves an essential role in transforming large amounts of data into actionable insights for strategic planning and process improvement. Analytics also enables organizations to uncover hidden trends that may not be immediately visible through manual review. Through systematic analysis, businesses can compare performance across products, branches, and time periods to identify strengths and weaknesses. It contributes to more accurate forecasting by using historical and current data as predictive references. Additionally, analytics helps reduce operational risks by detecting unusual data patterns and emerging issues at an early stage. Analytics supports performance measurement by establishing key metrics and benchmarks that organizations can use to monitor progress toward business objectives. It also enhances resource allocation by identifying which products, services, or operational areas generate the highest returns and which require improvement. Furthermore, analytics improves responsiveness to market changes by allowing businesses to quickly adapt strategies based on real-time or updated information. In modern business environments, analytics has become a critical component of data-driven management, enabling organizations to remain competitive, efficient, and customer-focused. In this study, analytics refers to the evaluation of Kape Tubong's sales data to determine revenue trends, product demand, and peak operating periods.

Business Intelligence (BI): Refers to the combination of technologies, systems, tools, and methodologies used to collect, integrate, analyze, and present business data for improved organizational decision-making. It transforms raw operational and transactional information into meaningful insights through dashboards, reports, visualizations, and performance indicators. BI enables organizations to understand business performance, monitor trends, and make strategic decisions based on factual and organized data. It supports both day-to-day operations and long-term strategic planning by presenting relevant business information in a clear and interpretable manner. BI also allows organizations to merge data from various sources into a centralized analytical environment. Through BI applications, management can efficiently compare performance metrics, monitor operational indicators, and identify significant business trends. It enhances organizational awareness by converting fragmented information into actionable business knowledge. Furthermore, BI improves operational efficiency by automating data reporting processes and reducing the time required for manual data consolidation and analysis. It promotes data accuracy and consistency by standardizing the way information is collected, processed, and presented across the organization. BI also strengthens decision-making capabilities by providing timely and evidence-based insights that support proactive business responses. Additionally, it enhances organizational transparency by enabling stakeholders to access performance information through interactive and user-friendly reporting interfaces. As businesses continue to operate in increasingly data-driven environments, BI serves as a critical mechanism for maintaining competitiveness, improving performance, and identifying opportunities for innovation and growth. In this study, Business Intelligence refers to the analytical framework incorporated into the BREWINSIGHT BI system for processing, analyzing, and visualizing Kape Tubong's sales and revenue data.

Dashboard: Refers to a visual interface designed to display key performance indicators, business metrics, and summarized information using charts, graphs, tables, and other graphical elements. Dashboards are intended to simplify the presentation of complex data and improve the speed of interpretation for users. They provide a summarized overview of performance and may present information in real time. Dashboards help decision-makers monitor operational status without manually reviewing large volumes of raw data. They improve accessibility of business information by consolidating important metrics into a single interface. Dashboards also allow users to identify trends, anomalies, and performance gaps more efficiently through visual presentation. Advanced dashboards may include filtering, comparative analysis, and drill-down functions for deeper insight. Furthermore, dashboards enhance organizational responsiveness by enabling users to quickly detect changes in performance and take timely corrective action. They support data-driven decision-making by presenting information in a format that is easy to understand and interpret, even for non-technical users. Dashboards also improve communication among stakeholders by offering a shared and standardized view of critical business metrics. In analytical systems, dashboards can be customized to match user roles, ensuring that each stakeholder receives relevant and focused information according to their responsibilities. Additionally, dashboards contribute to improved strategic monitoring by continuously tracking business objectives and performance benchmarks over time. In this study, dashboard refers to the web-based visual component of the BREWINSIGHT BI system used to display sales performance, revenue patterns, and branch-related metrics.

Data-Driven Decision Making: Refers to the process of making business or operational decisions based on analyzed factual data instead of intuition, assumptions, or personal judgment. This approach relies on timely, relevant, and accurate information to support planning, problem-solving, and strategic improvement. Data-driven decision-making reduces uncertainty, improves consistency, and strengthens the reliability of managerial actions. It allows organizations to formulate strategies based on measurable evidence rather than subjective estimation. This method also improves accountability by providing objective justification for management decisions. It supports better planning by incorporating real operational trends into strategy development. Additionally, it enables organizations to assess the effectiveness of past decisions and continuously improve business practices. Data-driven decision-making also enhances organizational agility by allowing management to respond quickly to market changes, customer demands, and operational challenges based on current information. It promotes more efficient resource allocation by identifying which strategies, products, or processes produce the most favorable outcomes. Furthermore, it strengthens performance evaluation by enabling managers to compare actual results against expected targets using quantifiable indicators. Through continuous analysis of operational data, organizations can identify recurring issues, forecast future opportunities, and implement improvements with greater confidence. In modern business environments, data-driven decision-making is considered a fundamental practice for achieving sustainable growth, operational excellence, and competitive advantage. In this study, it refers to the use of analytical sales information generated by the BREWINSIGHT BI system to support informed decision-making by Kape Tubong management.

Kiosk: Refers to a compact retail outlet or standalone sales station established in high-traffic areas for the sale of products or services. In the Food and Beverage industry, kiosks are commonly used to provide accessible and convenient customer service while occupying limited physical space. Kiosks often function as branch extensions of a main business establishment. They enable businesses to expand market presence without requiring full-scale branch facilities. Kiosks are strategically placed to maximize customer reach and sales opportunities. Although smaller in size, they generate transactional and operational data similar to larger business branches. Kiosks contribute to business scalability by allowing organizations to test market demand in specific locations before investing in permanent establishments. They also improve customer convenience by increasing product accessibility in areas with high pedestrian traffic such as malls, terminals, and commercial centers. Furthermore, kiosks support decentralized business operations while maintaining centralized sales monitoring and reporting. Through integrated data collection systems, kiosk transactions can be tracked and analyzed alongside other business branches for performance evaluation. In addition, kiosks provide businesses with operational flexibility by enabling rapid setup, relocation, or reconfiguration based on market demand and business strategy. They reduce overhead costs compared to traditional storefronts by requiring less space, fewer personnel, and lower utility expenses. Kiosks also contribute to brand visibility by extending the physical presence of a business into strategic public locations. Their operational data can be used to evaluate branch-level productivity, customer traffic patterns, and sales efficiency across multiple sites. In this study, kiosk refers to the satellite sales locations of Kape Tubong where transactions occur and sales data are collected.

Point of Sale (POS): Refers to the physical or digital system used to complete and record sales transactions between a business and its customers. A POS system typically includes software and/or hardware for capturing transaction details such as items sold, quantity, price, and payment information. POS systems improve transaction accuracy, maintain organized sales records, and support monitoring of business sales activities. Modern POS platforms may also include functions such as inventory tracking, report generation, and transaction history management. They serve as the primary source of transactional data in retail operations. POS systems also enhance operational efficiency by automating the sales recording process and reducing manual entry errors during transactions. They support real-time sales monitoring, enabling management to track business performance as transactions occur across multiple locations.

Additionally, POS systems improve accountability by maintaining detailed and timestamped records of each transaction for auditing and review purposes. When integrated with analytical platforms, POS systems provide valuable data for identifying sales trends, customer preferences, and branch performance. Furthermore, POS systems facilitate faster transaction processing, improving customer service speed and reducing waiting times during peak business hours. They contribute to improved financial control by accurately documenting cash flow and payment activities across business operations. POS-generated records also support reconciliation, tax reporting, and compliance with financial documentation requirements. In multi-branch businesses, POS systems enable centralized monitoring of branch-level transactions, promoting consistency in reporting and performance evaluation. In this study, POS refers to the transaction-recording system utilized by Kape Tubong branches to capture sales data for integration into the BREWINSIGHT BI platform.

Revenue Pattern: Refers to the identifiable trend, fluctuation, or behavior of income generation over a specific period of time. Revenue patterns illustrate how sales vary across dates, time periods, products, and branches. Analyzing revenue patterns enables businesses to determine peak sales periods, identify low-performing intervals, recognize seasonal demand fluctuations, and understand customer purchasing trends. Revenue pattern analysis supports forecasting and strategic planning by revealing recurring sales behaviors. It also helps assess the effectiveness of pricing, promotional, and operational strategies. Furthermore, revenue pattern analysis allows organizations to evaluate the financial impact of business decisions by comparing revenue performance before and after strategic changes or campaigns. It assists in identifying revenue concentration among specific products or branches, helping management determine primary income drivers within the business. By understanding historical revenue behaviors, businesses can improve budgeting, staffing, and inventory planning to align with expected demand. Revenue pattern analysis also contributes to risk management by highlighting irregular declines or anomalies that may indicate operational issues or market disruptions. In addition, it supports performance benchmarking by enabling businesses to compare actual revenue against established targets or historical averages. It provides a basis for measuring growth trends and evaluating whether financial objectives are being achieved over time. Through continuous monitoring of revenue patterns, organizations can better anticipate future opportunities and challenges, resulting in more proactive and informed managerial actions. In this study, revenue pattern refers to the income trends and sales behaviors identified and analyzed by the BREWINSIGHT BI system using Kape Tubong's transactional sales records.

Web-Based System: Refers to a software application hosted on a web server and accessed through a web browser over a network or internet connection. Unlike desktop-based software, web-based systems do not require installation on individual devices and can be accessed remotely from various locations and platforms. These systems provide flexibility, centralized data storage, and improved accessibility for users. Web-based applications support multiple simultaneous users and real-time synchronization of information. They also simplify maintenance and system updates by allowing centralized deployment from the server. Furthermore, web-based systems enhance organizational efficiency by enabling users to access business functions and information anytime and from any location with internet connectivity. They improve collaboration by allowing multiple authorized users to interact with the same centralized database and interface simultaneously. Web-based platforms also strengthen data consistency by ensuring that all users access the most current and updated version of the system and its records. In addition, these systems support scalability, allowing organizations to expand features, users, and data capacity without major hardware changes on client devices. Security mechanisms such as authentication, role-based access control, and encrypted connections can also be integrated to protect sensitive business information within web-based environments. Web-based systems further promote cost efficiency by reducing the need for device-specific software installations and minimizing technical support requirements for distributed users. They also facilitate seamless integration with other online services, APIs, and cloud-based infrastructures, improving interoperability and system functionality. Because of their accessibility, maintainability, and scalability, web-based systems are widely adopted for enterprise applications, management platforms, and analytical tools across modern organizations. In this study, a web-based system refers to the online architecture of the BREWINSIGHT BI platform, which enables authorized Kape Tubong personnel to securely access dashboards, reports, and analytics through internet-enabled devices.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Kumar and Singh (2022) conducted the study “Data Visualization in Retail Analytics: Impact on Decision-Making” published in the International Journal of Business Intelligence. The research examined how interactive data visualization technologies affect managerial decision-making within retail and food-and-beverage businesses. Outcomes showed that organizations using dashboard-based visual analytics achieved close to 30 percent higher efficiency in strategic decision-making compared to groups using standard spreadsheets or text-based reports. The authors explained that interactive dashboards enable managers to process large volumes of transactional and operational data more effectively through visual formats such as graphs, charts, heatmaps, and trend lines. These visual structures allow leaders to identify critical business insights much faster than possible through manual review of tabular records. Visually organized information simplifies understanding of complex operational data and increases the practical value of analytical reports. Data visualization improves access to information and supports better application of business analytics in daily operations.

Data visualization helps users recognize key performance indicators, revenue changes, sales patterns, and product results across different timeframes. Transforming complex numerical data into visual formats reduces cognitive load for users and speeds up analysis while improving accuracy. Visual analytics allow leadership to respond faster to performance issues, unusual sales movements, and operational inefficiencies. Quick detection of problems enables faster intervention and timely strategic adjustments. Businesses can address performance declines before they cause significant operational impact. Direct access to visual insights supports proactive and preventive management practices.

Businesses with dashboard visualization systems reported improved collaboration among management teams. Visual reports create a shared and easily understood foundation for performance discussions and strategic planning. Dashboard environments reduce reliance on technical staff for report generation. Managers can independently explore data and perform self-service analysis. This flexibility improves responsiveness when comparing results across branches, products, or time periods. Self-service tools allow leaders to create custom data views aligned with current business priorities. Such adaptability ensures information remains relevant during planning sessions. Organizations benefit from faster and independent analysis by management personnel.

Dashboard-based analytics provide strong support for strategic forecasting. Businesses can identify repeating sales cycles, seasonal patterns, and developing market behaviors from historical records. Clearer trend recognition supports better planning for inventory control, workforce scheduling, and promotional activities. Adoption of integrated visualization platforms is recommended as part of digital transformation strategies in retail and food service. These tools increase competitiveness and improve data utilization. Organizations with advanced analytical capabilities adapt more easily to dynamic and competitive markets. Companies using dashboards respond better to demand shifts and operational changes due to enhanced data awareness. Visualization technologies offer distinct strategic value in maintaining business agility.

Visualization-based analytics build management confidence during operational decision-making. Data appears in transparent and interpretable formats. This clarity removes ambiguity when evaluating performance and builds trust in system outputs. Dashboard interfaces improve access to business information by presenting key metrics in a centralized and user-friendly layout. Decision-makers obtain relevant insights quickly without needing specialized technical knowledge in data processing. Businesses respond more effectively to changing market conditions and internal operational challenges. Greater confidence in data increases consistent use of analytics during strategic planning. This establishes a stronger culture of evidence-based decision-making across the organization.

Dashboard systems support continuous performance monitoring through real-time or near-real-time updates of business metrics. Operations remain under constant observation instead of being reviewed only through periodic reports. Ongoing monitoring increases management awareness of business conditions and allows early intervention when performance drops or anomalies appear. Visual analytics platforms support adaptation during changing demand. Leadership recognizes and reacts quickly to shifting sales patterns. Continuous monitoring improves short-term planning and operational adjustments. Staffing, inventory, and promotions can be refined using updated performance data. This feature increases operational responsiveness and effectiveness in daily management.

This research holds direct relevance to BREWINSIGHT BI: Revenue Pattern Analytics System. It offers strong evidence supporting the inclusion of dashboard-based analytics in the proposed platform. Kape Tubong currently faces difficulties such as scattered sales records, lack of centralized monitoring, and limited visibility into kiosk performance. These gaps prevent accurate evaluation of revenue trends. Visual dashboard analytics within BREWINSIGHT BI convert raw sales transactions into clear graphical insights. This allows faster and more informed decision-making. The system assists leadership in identifying top-performing products, comparing branch contributions, determining peak sales periods, and spotting downward trends through interactive visual reports. These functions directly address current limitations in consolidating and interpreting multi-branch sales data. Dashboard visualization makes branch-level performance data easier to access and apply. Management oversight becomes timelier and more strategic.

Self-service analytics and comparative monitoring align well with the goals of BREWINSIGHT BI. Kape Tubong administrators can assess performance across locations independently without manual report compilation. Forecasting capabilities described in the study match the system's purpose of identifying demand patterns, allocating resources effectively, and strengthening strategic planning. The research confirms the value of centralized dashboards for remote monitoring. This feature suits multi-branch operations like Kape Tubong. Management can oversee activities through a unified digital interface. Coordination between branches improves through this structure. Remote access proves useful when leaders are not physically present at locations. Supervision becomes more efficient and responses to branch-level issues occur faster.

Findings regarding reduced reliance on technical staff align with usability goals for BREWINSIGHT BI. The platform is designed for non-technical users to navigate dashboards and reports with minimal training. This fits operational requirements for small and medium enterprises where dedicated data analysts may not be available. Results confirm the practical value of a user-friendly Business Intelligence platform for Kape Tubong. Simplicity and accessibility make such tools appropriate for businesses with limited technical resources. Successful adoption and long-term use within Kape Tubong's operational environment become more achievable.

Overall, results from Kumar and Singh (2022) provide clear justification for including dashboard visualization and analytical reporting features in BREWINSIGHT BI. Their findings strengthen theoretical and practical foundations for the project. Visual analytics improve data interpretation, decision speed, operational oversight, and organizational responsiveness within retail and food service environments. The work serves as key reference material confirming the relevance and necessity of the proposed system.

Reference: Kumar, R., & Singh, P. (2022). Data Visualization in Retail Analytics: Impact on Decision-Making. International Journal of Business Intelligence, 14(2), 45–58. ISSN 2710-3102 (Online).https://doi.org/10.11114/ijbi.v14i2.25472

Chen Li and Maria Gonzalez (2021) conducted the study "Business Intelligence Systems and Organizational Performance in Small and Medium Enterprises" published in the Journal of Information Systems and Technology Management. The work explored how Business Intelligence systems affect operational efficiency, decision quality, and organizational performance among small and medium enterprises in retail and food-service sectors. Organizations with centralized BI platforms showed significant gains in data accessibility, reporting accuracy, and management responsiveness compared to those using manual reporting methods. Centralized analytical systems allow leadership to combine sales, inventory, and operational data into one unified platform. Visibility over business performance improves across multiple branches and departments.

BI systems enhance strategic decision-making by transforming large volumes of transactional records into actionable insights. Dashboards, trend analysis, and automated reporting tools help managers identify sales changes, customer buying patterns, and operational inefficiencies more clearly. Businesses using BI systems respond faster to declining sales or inventory shortages. Real-time analytical reports improve awareness of operational status. Automated reporting removes delays caused by manual compilation of records. Managerial efficiency increases and errors in reporting decrease.

BI implementation strengthens organizational coordination and communication. Standardized visual reports create a consistent basis for evaluating performance among managers and branch supervisors. Centralized systems ensure consistency in performance evaluation. All branches follow the same reporting structure and use identical metrics. Comparisons between locations, products, and timeframes become more reliable. User-friendly BI interfaces allow non-technical staff to use analytical tools independently. Reliance on IT personnel for routine reporting tasks decreases.

Businesses adopting BI technologies improve forecasting and operational planning abilities. Historical sales data helps identify repeating trends and seasonal demand cycles. These insights support better inventory control, staffing allocation, and promotional planning. Small and medium enterprises benefit from scalable and accessible BI platforms. Competitiveness improves and evidence-based decision practices take root. Organizations with strong analytical infrastructures adapt better to market shifts and operational disruptions. Timely and accurate business information drives this advantage.

This study relates closely to BREWINSIGHT BI: Revenue Pattern Analytics System. It confirms the value of centralized Business Intelligence platforms for improving operational monitoring and decision-making in multi-branch settings. Kape Tubong currently deals with scattered sales records, manual report creation, and limited visibility over kiosk performance. BREWINSIGHT BI centralizes branch sales data into one dashboard system. Revenue trends, product results, and branch comparisons become easier to monitor. Analytical and forecasting features align with project goals to improve strategic planning, operational efficiency, and data-driven decisions within the organization.

Reference: Li, C., & Gonzalez, M. (2021). Business Intelligence Systems and Organizational Performance in Small and Medium Enterprises. Journal of Information Systems and Technology Management, 18(3), 112–128. ISSN 1807-1775. https://doi.org/10.4301/jistem.2021.18.3.8

Johnson and Lee (2021) conducted research titled “Business Intelligence Dashboards and Organizational Performance in Small Enterprises” published in the Journal of Information Systems and Technology Management. The study analyzed how dashboard-based Business Intelligence systems influence operational efficiency and decision-making within small and medium enterprises. Focus was placed on how visual analytics and centralized reporting improve results, specifically for organizations with limited technology resources and manpower. Participants included small retail and food-service businesses adopting dashboards to manage daily operations and track organizational performance. Data came from surveys, interviews, and performance report comparisons made before and after implementation.

Centralized dashboards improved accuracy when tracking sales performance, customer behavior, and inventory activity compared to manual reporting methods. Dashboards simplify data interpretation by displaying information visually through graphs, scorecards, and performance indicators. Visual layouts allow managers to follow operational trends efficiently. Delays in spotting problems decrease. Leaders quickly identify sales drops, inventory shortages, and inconsistencies across branches without reviewing extensive spreadsheets or printed documents. Centralized access improves communication among management teams and enhances coordination during operational planning. Businesses make timely and data-backed decisions during changing market conditions.

Business Intelligence dashboards support more accurate forecasting. Managers recognize repeating sales patterns and seasonal demand changes through historical trend analysis. Inventory allocation, staffing schedules, and promotional activities improve as a result. Predictive analytics within dashboards help organizations prepare for peak periods and reduce operational disruptions. Resource management improves as leaders determine top-selling items and activities needing immediate attention. Unnecessary expenses reduce and productivity increases.

Dashboard systems reduce dependence on technical personnel. Managers access and analyze operational data independently through user-friendly interfaces. Organizational flexibility improves and responses to market changes become faster. Automated dashboards reduce human errors typical of manual data entry and report preparation. Real-time reporting allows owners to detect sales and performance shifts instantly. Corrective actions happen sooner.

Automation significantly reduces time spent creating reports. Employees focus more on customer service and operational improvement.

Organizations using Business Intelligence tools see gains in customer service. Management understands buying habits and preferences better through data analytics. Transaction records reveal purchase patterns, preferred products, and peak buying hours. Marketing strategies, promotions, and product offerings align better with customer needs. Integrating dashboard technologies supports long-term strategic planning and organizational sustainability. Continuous monitoring of performance indicators allows proactive responses to operational challenges and market changes.

Visualized data encourages collaboration across departments. Performance reviews and goal setting rely on a unified data source. Managers share reports easily and coordinate strategies using the centralized dashboard. Transparency increases within the organization and accountability strengthens among staff and administrators. Dashboard technologies promote a culture based on data-driven decision-making. Choices rest on accurate and timely information rather than assumptions or delayed reports. Businesses gain competitiveness and adaptability in fast-changing environments.

These findings support development of BREWINSIGHT BI. Centralized dashboards and self-service analytics suit current needs at Kape Tubong. Challenges related to consolidating branch data and monitoring kiosk performance efficiently match those described in the research. Dashboard technology improves visibility, access, and decision quality across locations. Real-time analytics and automated reporting features within BREWINSIGHT BI reduce delays in data consolidation and improve accuracy in sales tracking.

Visual analytics help identify sales trends, customer preferences, and inventory needs. These elements are vital for increasing operational efficiency at Kape Tubong kiosks. Managers track branch productivity, evaluate sales results, and spot urgent operational needs effectively. Forecasting and trend analysis support inventory distribution planning, staffing schedules, and promotions based on demand patterns. Operational costs decrease, stock shortages reduce, and customer satisfaction improves.

Self-service analytics directly apply to BREWINSIGHT BI goals. Branch managers and administrators access and interpret data independently without advanced technical training. Organizational efficiency improves as reliance on manual reporting decreases and access to critical data speeds up. Adopting a centralized dashboard system improves operational coordination, communication between branches, and quality of strategic planning. Implementing BREWINSIGHT BI strengthens business results, supports sustainable growth, and improves overall management of kiosk operations.

Reference: Johnson, T., & Lee, M. (2021). Business intelligence dashboards and organizational performance in small enterprises. Journal of Information Systems and Technology Management, 18(3), 102–118. ISSN 1807-1775. https://doi.org/10.4301/jistm.2021.18.3.102

Garcia and Torres (2020) conducted a study titled “Sales Analytics Systems for Food Service Enterprises” published in the International Journal of Hospitality and Food Service Management. The work explored how sales analytics systems help food and beverage businesses analyze customer buying patterns and improve operational results. Researchers measured effectiveness of analytics-based reporting tools in supporting decisions, monitoring sales, and increasing efficiency in food service establishments. Data came from restaurants, cafés, and beverage businesses adopting digital analytics to track daily transactions, customer habits, and inventory movement. Methods included surveys, interviews, and performance comparisons before and after adoption.

Businesses using analytics-based reporting noted better revenue monitoring and faster identification of product performance changes. Visual sales analytics help management spot best-selling items, underperforming menu choices, and peak periods more effectively than manual methods. Interactive dashboards and automated visual reports allow quick analysis without reliance on spreadsheets or paper records. Operational oversight improves and decisions regarding menus, pricing, and promotions become better informed. Managerial efficiency increases through reduced time needed to prepare and interpret reports. Adjustments to pricing, inventory, and promotions happen faster using real-time data.

Sales analytics systems contribute to improved customer service and responsiveness. Analysis of buying habits reveals frequently purchased items, preferred transaction times, and spending patterns. Promotions become more personalized and satisfaction improves through targeted offers. Customer retention increases as management responds better to preferences and demand shifts. Accurate sales data improves forecasting and lowers risks related to overstocking or shortages.

Analytics systems improve responsiveness through direct access to operational insights and performance indicators. Businesses adapt better during demand fluctuations. Changes in buying patterns become visible quickly. Organizations with real-time capabilities react more effectively to seasonal shifts and unexpected market changes. Leaders measure promotion results, identify sales drops, and implement corrections without delay. Operational flexibility increases and service quality remains consistent even during peak periods.

Analytics tools improve inventory management and reduce waste. Sales trend tracking and historical analysis support accurate inventory planning and reduce excess stock. Digital platforms help managers track product movement and consumption rates accurately. Purchasing decisions improve as a result.

Operational costs reduce and profitability increases. Human error decreases with automation. Reliability and accuracy of operational data improve.

Digital reporting platforms support long-term strategic planning. Future sales patterns and resource allocation become clearer through analysis. Businesses using analytics demonstrate stronger strategic decision-making. Management maintains constant access to updated operational data and performance indicators. Sales analytics encourage a data-driven culture. Decisions rest on measurable results rather than assumptions. Centralized data improves communication and coordination between departments. Operational objectives align more effectively.

These results support goals set for BREWINSIGHT BI. The proposed platform delivers timely and centralized revenue analytics for Kape Tubong. Visual reporting and trend monitoring help identify branch differences, track product demand, and improve planning. Kape Tubong requires reliable tools to consolidate data from multiple locations into one accessible platform. Analytics-based reporting improves management efficiency and speeds up decision-making processes.

Real-time sales monitoring increases operational performance in food and beverage settings. Automated dashboards and visual analytics within BREWINSIGHT BI support daily transaction tracking, product performance review, and early detection of operational issues. Forecasting tools help refine inventory allocation, staffing, and promotions based on demand. Operational waste reduces and service quality improves across branches.

Customer behavior analysis remains central to system design. Understanding preferences drives product improvements and satisfaction increases. Sales analytics help identify high-demand items, peak periods, and marketing effectiveness. Centralized data access strengthens coordination among managers and administrators through a single reliable source. The study confirms value in adopting analytics-driven Business Intelligence platforms. These systems support sustainable growth, improve efficiency, and enhance decision-making in multi-branch food and beverage businesses.

Reference:

Garcia, L., & Torres, A. (2020). Sales analytics systems for food service enterprises. International Journal of Hospitality and Food Service Management, 9(1), 55–70. ISSN 2634-5366. https://doi.org/10.2139/ijhfsm.2020.9.1.55

Martinez and Cooper (2021) conducted research titled “Interactive Business Intelligence Systems and Decision Support in Food Retail Operations” published in the International Journal of Digital Commerce and Analytics. The investigation explored how interactive Business Intelligence systems improve operational management and decision-making within food retail and beverage companies. Researchers evaluated how dashboard-based platforms help leaders interpret operational data and react quickly to changing conditions. Participants included retail and beverage organizations using centralized dashboards to monitor sales, customer habits, and branch performance. Companies with interactive dashboards showed much higher efficiency in operational monitoring and reporting compared to those using spreadsheets. Centralized dashboards make organizational data easier to access and apply by simplifying complex information into visual formats.

Interactive dashboard tools allow analysis through line graphs, pie charts, heatmaps, trend lines, and comparative indicators. Management identifies best-selling products, low-performing items, and demand changes more effectively than manual reporting allows. Visual analytics improve understanding as information appears clearly and logically. Operational patterns and anomalies become visible faster. Time needed for interpretation and evaluation reduces. Interactive filtering and drill-down options allow deep analysis by branch, category, or timeframe. Depth and flexibility in business analysis improve within participating organizations.

Businesses using dashboard systems respond better to operational challenges and market changes. Near-real-time updates on performance metrics support quick detection of sales drops, unexpected changes in buying habits, and operational inefficiencies across branches. Constant visibility supports proactive management. Problems receive attention before impacting overall results. Adaptability increases during demand shifts. Corrective measures happen promptly. Stability in operations improves and daily management becomes more effective.

Dashboard tools strengthen collaboration and communication among management teams. Interactive reports provide a shared reference for reviewing branch performance, discussing issues, and planning strategy. Visual consistency ensures interpretation remains uniform across departments. Coordination improves and misunderstandings from manual reports reduce. Transparency increases through centralized oversight. Leaders monitor activities through a unified interface. Supervision becomes more effective and coordinated decision-making improves across locations.

Self-service analytics reduce reliance on technical staff for reports and analysis. Managers with limited technical skills navigate dashboards, create custom reports, and interpret data independently through user-friendly layouts. Practical value increases for small and medium enterprises where dedicated analysts or IT specialists may not exist. Simple interfaces encourage regular system use. Confidence grows among users interacting with the platform. Evidence-based decision-making integrates more deeply into daily operations.

Interactive systems provide strong support for strategic forecasting and long-term planning. Historical records and demand patterns enable better inventory planning, staffing, promotion scheduling, and product development. Businesses using dashboards anticipate seasonal changes accurately and adjust strategies accordingly. Resource utilization improves and waste reduces regarding overstocking, understaffing, or poorly timed promotions. Adopting centralized platforms supports digital transformation goals within food retail and beverage sectors.

This study strongly relates to BREWINSIGHT BI development. Results justify integration of interactive dashboard analytics into the proposed system for Kape Tubong. Current challenges include scattered sales data, delays in report preparation, and limited visibility across branches. Dashboard analytics transform raw transactions into clear visual insights. Monitoring and decision-making improve significantly. The system helps identify best-selling items, compare branch revenue, track sales changes, and detect operational issues through centralized visuals. These features directly address current needs for better visibility and oversight.

Self-service capabilities align with usability objectives. The platform supports non-technical users with minimal training requirements. Management creates reports independently, compares results, and follows trends without technical support. Centralized access supports remote monitoring across locations. Multi-branch businesses benefit greatly from this function. Supervision improves and timely responses remain possible even when leaders are not physically present. Coordination strengthens and oversight effectiveness increases.

Overall, Martinez and Cooper (2021) provide strong theoretical and practical support for BREWINSIGHT BI implementation. Interactive Business Intelligence improves operational oversight, reporting quality, forecasting, responsiveness, and evidence-based decisions in food retail and beverage environments. This work serves as key reference material confirming the value, practicality, and necessity of the proposed platform for Kape Tubong.

Reference: Martinez, S., & Cooper, J. (2021). Interactive Business Intelligence Systems and Decision Support in Food Retail Operations. International Journal of Digital Commerce and Analytics, 13(2), 67–83. ISSN 2516-5403. https://doi.org/10.2245/ijdca.2021.13.2.67

2.2 Local Studies

Reyes and Santos (2023), in their research published within the *Philippine Review of Business and Economics*, conducted an investigation into how effective Business Intelligence (BI) dashboards are when measured against conventional manual methods of tracking sales performance, specifically focusing on food and beverage businesses operating across the Philippines. The outcomes of their study revealed that companies which adopted BI tools centered around dashboards managed to cut down the amount of time spent analyzing sales figures and operational performance by roughly 40 percent, in comparison to enterprises that still depended entirely on manual record keeping and the manual compilation of data through spreadsheets. In particular, coffee shops and small food service businesses that utilized these dashboard platforms showed marked improvements in their ability to keep track of daily sales activities, assess how well individual products were performing in the market, and adjust quickly to changes or shifts in what customers were asking for. The researchers ultimately concluded that the use of BI dashboards greatly boosts the speed, precision, and overall efficiency of the decision-making process for management teams, primarily because such systems allow for immediate access to operational data and key information that would otherwise take significant time to gather and process.

Further results from their investigation showed that having a centralized system where all data is displayed on a single dashboard led to better management efficiency, as it combined sales records and performance metrics into one unified visual interface. By using visual aids such as charts, trend lines, and comparative graphs, owners and leaders were able to understand their operational data much more easily, allowing them to spot changes in sales volume, recognize patterns regarding how well certain items sell, and measure productivity levels across different branches or locations—all without needing to possess deep technical knowledge or advanced data analysis skills. The authors also observed that because these dashboards were straightforward to access and use, there was less reliance placed on administrative personnel to prepare reports or gather information; instead, managers could retrieve and examine the necessary data on their own, independently. Furthermore, since the format of the information shown remained consistent and standardized, the process of evaluating performance became much more uniform, and it helped reduce the errors or inconsistencies that often happen when reports are prepared manually, which in turn strengthened the level of transparency and accountability within the organization.

The study additionally discovered that businesses which implemented BI dashboards displayed a higher capacity to adjust and adapt when faced with competitive market environments. Being able to view analytical data instantly meant that decision-makers could assess whether their pricing strategies, marketing campaigns, or operational changes were working effectively and efficiently. These organizations were much better positioned to react to changes in customer preferences, seasonal variations in demand, or pressure from competitors, because their choices and actions were based on accurate and current performance figures rather than just estimates or assumptions. Moreover, the fact that past sales data was stored and organized within the dashboard systems allowed for better forecasting and long-term strategic planning, as businesses could look back over time to identify recurring patterns and predict what future demand might look like.

One other significant finding highlighted how BI dashboards play a key role in helping micro, small, and medium-sized enterprises in the Philippines move toward digital operations and modernization. The researchers explained that these kinds of systems serve as an accessible starting point for businesses wanting to upgrade their operations by adopting management practices that are driven by data and facts rather than experience or intuition alone. With user-friendly interfaces and features that automatically process and analyze information, companies can take advantage of sophisticated analytical tools without needing complex technology infrastructure or staff members who are specialists in data science or information technology. Therefore, introducing BI systems does not just bring improvements in daily operations; it also helps businesses become more competitive and resilient in the long run.

The authors also emphasized the value these systems bring in terms of managing historical data, maintaining transparency within the organization, and improving communication between teams. Dashboards keep sales records organized and preserved in a structured format, which makes them useful for analyzing trends over time, predicting future outcomes, and planning long-term strategies. By using consistent performance measurements across the board, it becomes easier to evaluate how productive different branches or departments are performing, and because relevant information can be shared easily, it encourages discussions and decisions among managers and staff that are more objective and grounded in facts. These capabilities work together to strengthen coordination, ensure everyone is accountable, and promote decision-making that is supported by solid evidence throughout the whole business.

The results and conclusions presented by Reyes and Santos (2023) offer strong, locally relevant support for the development and implementation of the BREWINSIGHT BI: Revenue Pattern Analytics System specifically designed for Kape Tubong. Just like the businesses described in their study, Kape Tubong currently faces difficulties caused by sales records that are scattered or disconnected, delays in reporting performance data, and a lack of clear visibility into how each of its kiosk locations is operating. The proposed system directly addresses these problems by bringing all transactional data together into one unified dashboard platform, which enables real-time monitoring, detailed revenue analysis, evaluation of how well different products are selling, and direct comparisons between various branches or outlets.

This research also validates the decision to use visualizations displayed on dashboards, historical trend analysis, and automated reporting features as effective ways to improve management efficiency and make operations more responsive to change. With BREWINSIGHT BI, the leadership team at Kape Tubong will be able to track revenue trends over time, identify which products are the most profitable or popular, assess the performance of each branch, and make well-informed decisions based on information that is both accurate and timely. Additionally, the forecasting capabilities mentioned in the study align perfectly with the intended functions of the proposed system, supporting key activities such as planning inventory levels, scheduling staff or workforce needs, and organizing promotional activities or marketing campaigns.

In summary, the research conducted by Reyes and Santos (2023) provides important local evidence showing that Business Intelligence systems based on dashboards are both practical and effective for food and beverage enterprises operating within the Philippines. Their findings reinforce the necessity of putting the BREWINSIGHT BI system in place as a valuable solution to improve operational efficiency, ensure that decisions are made based on solid data, and support the continued growth and long-term success of Kape Tubong.

Reference:

Reyes, M., & Santos, J. (2023). Digital Transformation in Philippine Micro-Enterprises: A Study on BI Adoption in the F&B Sector. Philippine Review of Business and Economics, 60(1), 12–29. ISSN 0115-9011. https://doi.org/10.11114/prbe.v60i1.3321

Villanueva (2024), in the study “From Intuition to Evidence: The Digital Maturity of Food Service MSMEs in Western Visayas” published in the Philippine Journal of Science and Technology Management, examined the shift of food service micro, small, and medium enterprises (MSMEs) from intuition-based management to data-driven decision-making through digital technologies. The study emphasized that many Filipino coffee shop owners possess strong local market knowledge and a deep understanding of customer preferences; however, they often encounter what Villanueva referred to as “scaling blindness” as their businesses expand. This challenge arises when direct managerial oversight becomes difficult across multiple branches, resulting in issues related to quality control, inventory consistency, and revenue monitoring.

The findings revealed that businesses adopting Business Intelligence (BI) tools achieved improved operational consistency, particularly among enterprises managing multiple locations. Villanueva identified “localized data awareness” as a key factor in business growth, referring to managers’ ability to understand branch-specific customer preferences, purchasing behaviors, and peak operating periods. Rather than implementing uniform strategies across all branches, businesses utilizing localized analytics were able to customize operations according to the needs of each community. This capability was found to be highly valuable within the competitive and diverse Philippine food service industry.

The study also introduced the concept of “operational leakage,” which refers to unrecorded losses in inventory, sales, or cash transactions commonly associated with manual and paper-based systems. Villanueva found that centralized dashboards improve transparency and accountability by enabling continuous monitoring of operational data. As employees become aware that sales and inventory records are being tracked through centralized reporting systems, data accuracy improves and compliance with organizational procedures becomes more consistent.

The research highlighted that many MSMEs in Western Visayas remain in the early stages of digital maturity, relying heavily on handwritten records, spreadsheets, and verbal reporting. While these methods may be sufficient for single-location businesses, they become increasingly inefficient as organizations grow. The study revealed that expansion without digital integration often leads to communication delays, inconsistent reporting practices, and reduced managerial visibility, making it difficult for business owners to maintain service quality and monitor branch performance effectively.

The study further explained that BI systems contribute not only to data consolidation but also to organizational learning and strategic adaptability. Through dashboard visualizations and automated reporting tools, managers can identify patterns and trends that may otherwise remain hidden within large volumes of transactional data. Visual analytics enhance business understanding by transforming complex information into actionable insights. Businesses utilizing BI platforms were able to monitor product demand, evaluate promotional effectiveness, identify peak customer activity periods, and improve operational planning. Historical sales data also supported predictive decision-making related to inventory management, staffing allocation, and promotional scheduling.

Another significant finding emphasized the importance of real-time operational visibility in strengthening managerial responsiveness. According to Villanueva, delayed or incomplete reports often prevent management from addressing operational concerns before they escalate. Centralized dashboards function as monitoring and early-warning systems by providing immediate access to sales trends, branch performance indicators, and inventory movements. Organizations using integrated BI systems demonstrated greater responsiveness to operational disruptions and maintained higher levels of consistency during periods of expansion. Increased transparency also encouraged employees to follow standardized procedures, thereby enhancing organizational accountability.

Villanueva also argued that digital transformation should be viewed as an organizational process rather than simply a technological upgrade. Successful BI implementation requires managers and employees to adopt a data-driven culture in which decisions are guided by analytical evidence rather than personal assumptions. Businesses that integrated analytics into their daily operations demonstrated greater adaptability, resilience, and competitiveness in responding to changing consumer preferences and market conditions.

Overall, the findings of Villanueva (2024) provide strong theoretical and practical support for the development of BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong. The study aligns closely with the objectives of the proposed system, particularly in centralized sales monitoring, branch performance evaluation, operational transparency, and data-driven decision-making. It validates the integration of Point-of-Sale (POS) systems with analytical dashboards to strengthen managerial oversight and operational efficiency. The research also supports the use of automated reporting and visualization tools that transform transactional data into meaningful business intelligence, reinforcing the importance of adopting a centralized analytics platform to support sustainable growth and long-term competitiveness in the Philippine food service industry.

Reference:

Villanueva, E. (2024). From Intuition to Evidence: The Digital Maturity of Food Service MSMEs in Western Visayas. Philippine Journal of Science and Technology Management, 19(2), 88–105. ISSN: 1655-3179. https://doi.org/10.53899/pjstm.v19i2.441

Garcia Navarro and Cruz (2022), in their research titled Implementation of Business Intelligence Systems in Philippine Small Food Enterprises”* and published in the **Asian Journal of Information Technology and Business Management**, explored how Business Intelligence (BI) solutions influence operational performance, sales tracking capabilities, and management decision-making processes specifically within small and medium-sized food service businesses across the Philippines. The results of their investigation demonstrated that companies which adopted centralized BI dashboards experienced significant improvements in both the precision and speed of their sales reporting when compared to organizations that continued to depend on manual procedures and spreadsheet-based methods. By combining critical information related to sales transactions, inventory levels, and performance metrics from different branches into a single, cohesive platform, these enterprises gained much clearer insight into their overall operations and were able to eliminate the delays that typically occur when data is gathered and compiled manually.

The authors further emphasized that the visual nature of BI dashboards greatly enhanced the ability of managers to recognize emerging sales patterns, keep track of shifting product demand, and assess how well individual locations or branches were performing, all through the use of easy-to-understand visual analytics and automatically generated reports. Because data was presented in charts, graphs, and summaries, leadership teams could quickly spot operational concerns—such as a drop in sales volume, shortages in stock, or inconsistent results across different outlets—allowing them to address problems immediately and make decisions that were both timely and effective. Additionally, the study noted that automating the reporting process significantly reduced the amount of administrative work required from staff members, which freed up valuable time and resources and led to a noticeable increase in overall operational efficiency.

Another important conclusion from the research highlighted how centralized BI systems contribute to stronger transparency and accountability within an organization by ensuring that performance measurements and evaluation criteria are standardized and applied consistently across all branches or departments. This uniformity allowed management to make fair, objective comparisons regarding productivity and performance between different locations, ensuring that every decision made regarding operations, resources, or strategy was based on factual data rather than estimates or personal judgment. Furthermore, businesses equipped with these systems showed a greater ability to adapt to changes in the marketplace, as they could closely analyze customer buying habits and sales history to understand exactly what was happening and respond accordingly.

The study ultimately established that adopting Business Intelligence technology serves as a key driver for digital transformation among small and medium enterprises in the Philippines, helping to modernize outdated or traditional ways of working while greatly improving the quality of strategic planning. By examining historical sales data stored within the system, organizations were able to identify recurring patterns, predict future demand more accurately, and manage their inventory levels in a much more optimized and cost-effective manner. Based on these findings, the researchers recommended that growing businesses—especially those operating across multiple locations—implement affordable and easy-to-use BI platforms, emphasizing that centralized dashboards are highly effective tools for strengthening managerial oversight, coordination, and control over the entire business.

The outcomes presented by Garcia Navarro and Cruz (2022) offer powerful and relevant support for the creation and development of the **BREWINSIGHT BI: Revenue Pattern Analytics System** designed specifically for Kape Tubong. Just like the businesses observed in their study, Kape Tubong currently encounters difficulties arising from scattered or disconnected sales records, delays in receiving important reports, and a lack of clear visibility regarding how each of its kiosks is performing. The proposed system directly resolves these issues by bringing all sales and operational data together into one unified dashboard, which enables real-time monitoring, deep analysis of historical trends, and direct evaluation of performance across different branches. Consequently, this research serves as strong validation that dashboard-based Business Intelligence systems are highly effective solutions for boosting operational efficiency, ensuring transparency, and promoting decision-making based on accurate data within food service businesses in the Philippines.

Reference:

Navarro, G., & Cruz, E. (2022). Implementation of Business Intelligence Systems in Philippine Small Food Enterprises. Asian Journal of Information Technology and Business Management, 19(2), 88–104. ISSN 2619-8457. https://doi.org/10.1080/ajitbm.2022.19.2.104

Lopez and Ramirez (2021), in their research work titled *“Centralized Sales Monitoring Systems and Business Performance among Philippine Café Enterprises”* which appeared in the *Journal of Philippine Management and Information Systems*, conducted an investigation into how centralized sales analytics platforms affect operational productivity and the quality of management decisions specifically within café and beverage businesses operating in the Philippines. Their study focused closely on understanding how monitoring systems built around interactive dashboards help improve sales management practices, how businesses evaluate their overall performance, and how well small and medium-sized enterprises are able to adapt and respond when customer preferences or market demands change.

The results gathered from their research showed clearly that companies which adopted centralized dashboard systems achieved substantial improvements in how efficiently they handled reporting tasks and how effectively they kept track of daily operations. When compared against traditional manual reporting methods, these digital platforms significantly reduced the amount of time and effort needed to gather, organize, and combine sales data into usable reports, while also ensuring that the way branch performance was measured and analyzed remained consistent and reliable across all locations. By giving managers access to all necessary sales information through one single, unified interface, the system allowed them to oversee activities happening in different branches much more effectively and to spot operational problems, irregularities, or opportunities for improvement right when they occurred, rather than discovering them later.

The researchers also highlighted the great value that visual elements within dashboards bring to the table, specifically how they help management teams understand business performance much more clearly and deeply. Features such as interactive charts, visual trend indicators, and summarized analytics made it possible for decision-makers to recognize recurring sales patterns, notice shifts in which products were in high demand, and identify exactly when customer activity was at its highest or lowest—tasks that are far more difficult and time-consuming when relying on text-based or spreadsheet reports. Because of this clearer understanding, businesses were able to make better-informed choices regarding key areas such as how much inventory to keep on hand, how to schedule staff members efficiently, when and how to run marketing promotions, and how to adjust daily operations to match actual demand.

Another significant discovery from the study pointed out how dashboard systems play a vital role in making an organization more responsive and improving coordination between different levels of the business. Having access to sales and operational data almost as it happens meant that managers could react immediately whenever revenue levels changed unexpectedly, whenever stock levels became a concern, or whenever a particular branch showed signs of underperformance. Additionally, because the system used standard measures and performance indicators across the whole organization, communication between supervisors at individual branches and the central management team became much stronger and clearer, ensuring that everyone was using the same standards and that both reporting and decision-making processes remained consistent and aligned with business goals.

Lopez and Ramirez further observed that having a centralized analytics platform encouraged a shift toward management practices that are grounded in facts and evidence rather than guesswork or assumptions. Businesses began to depend more heavily on actual performance data when making strategic plans or important decisions, which reduced risks and improved outcomes. The ability to store and review historical sales records, combined with built-in forecasting tools, allowed organizations to identify patterns that repeat over time, measure exactly how successful past promotions or campaigns were, and predict future customer demand with much greater accuracy. The researchers also emphasized that these systems were especially helpful for small and medium-sized enterprises in the Philippines, because their simple, user-friendly design meant that even managers with limited technical skills or background knowledge could use powerful analytical tools effectively and without difficulty.

The conclusions drawn by Lopez and Ramirez (2021) offer strong, direct support for the development and implementation of the **BREWINSIGHT BI: Revenue Pattern Analytics System** intended for Kape Tubong. Just like the businesses studied in their research, Kape Tubong currently faces challenges caused by sales data that is scattered across different locations, delays in gathering and combining information, and a lack of clear oversight regarding how each individual kiosk is performing. The proposed system directly addresses these issues by bringing all transactional and operational data together into one unified dashboard, which enables real-time monitoring of activities, direct comparison of performance between branches, and detailed analysis of trends over time.

This research also validates the decision to include visual data presentations and forecasting features as core components of the system, confirming that these tools are key to improving management decision-making, increasing operational efficiency, and supporting long-term strategic planning. By converting raw sales records into clear, meaningful insights, BREWINSIGHT BI will enable the leadership team at Kape Tubong to track how revenue changes over time, assess which products contribute most to success, make smarter choices about how to allocate resources, and adapt quickly whenever customer demand shifts or market conditions change. In this way, the study by Lopez and Ramirez (2021) provides valuable, locally relevant proof that establishing a centralized Business Intelligence platform is not only feasible but also highly effective and beneficial for Kape Tubong.

Reference:

Lopez, A., & Ramirez, C. (2021). Centralized Sales Monitoring Systems and Business Performance among Philippine Café Enterprises. *Journal of Philippine Management and Information Systems, 15*(3), 77–96. ISSN 2094-8817. https://doi.org/10.4587/jpmis.2021.15.3.77

Fernandez and Gutierrez (2022), in their research titled “Revenue Analytics and Dashboard Utilization among Philippine Small Food Businesses” published in the Philippine Journal of Information Technology and Business Analytics, conducted an in-depth investigation into how revenue analytics systems built on digital dashboards influence operational management practices and strategic decision-making processes within small and medium-sized food enterprises across the Philippines. Their research specifically focused on establishments such as coffee shops, milk tea outlets, snack centers, and food kiosks that had transitioned away from traditional, manual methods of tracking sales and instead adopted digital dashboard platforms to handle their data. The primary goal of the study was to determine exactly how centralized analytics systems improve the way businesses track their revenue, assess the performance of individual products, and monitor the productivity of different branches or locations, particularly within the small and medium enterprise sector in the country.

The results of the study demonstrated clearly that businesses which made use of dashboard-based analytics tools experienced remarkable improvements in their ability to see and understand what was happening across their operations, as well as significant gains in overall management efficiency. By automatically gathering and combining transaction records coming from various branches into one single, centralized digital platform, these systems eliminated the delays that usually happen when data is collected and compiled manually, while also ensuring that sales records became far more accurate, consistent, and reliable over time. This gave leadership teams the ability to monitor the performance of the entire business much more effectively and allowed them to make important operational decisions at the right moment, all based on trustworthy and verified information rather than estimates or incomplete reports.

The researchers also emphasized just how powerful visual analytics features are in helping managers truly understand how revenue is generated and how customers behave when making purchases. Through the use of clear graphs, detailed trend analyses, and interactive charts, decision-makers were able to spot changes or fluctuations in sales volume much more easily, evaluate which products were in high demand and which were not, identify exactly when business activity was at its highest or lowest points, and assess how well each individual branch was performing—tasks that are far more difficult and time-consuming to accomplish using traditional spreadsheet-based reports. As a direct result of having this clearer and faster insight, organizations became much more capable of responding quickly to operational challenges, internal issues, or shifting conditions within the market.

Another major finding from the research highlighted the important role that dashboard systems play in strengthening the level of oversight maintained over each branch and improving overall accountability within the organization. By using standardized performance measurements and indicators across all locations, managers were able to compare the productivity and results of different branches objectively and fairly, ensuring that evaluation practices remained consistent and uniform no matter where the branch was located. Furthermore, because revenue data was available almost immediately after transactions occurred, the organization became significantly more responsive and agile; management could make timely adjustments in key areas such as inventory levels, staff scheduling, pricing structures, and marketing campaigns to ensure resources were always used in the most effective way possible.

The study also revealed that dashboard analytics are extremely valuable for supporting strategic planning that is based on facts and evidence rather than assumptions. By analyzing historical data and identifying patterns that repeat over time, businesses were able to predict future customer demand with greater accuracy and make well-informed decisions regarding how much stock to purchase, how to schedule their workforce, and how to design and implement marketing activities. Additionally, the researchers noted that the user-friendly nature of these interfaces made advanced data analysis accessible even to managers who did not have strong technical backgrounds or specialized IT skills, which makes Business Intelligence technology a practical, sustainable, and highly suitable solution specifically for small and medium-sized enterprises operating in the Philippines.

The outcomes and conclusions presented by Fernandez and Gutierrez (2022) provide strong and direct support for the development and creation of the

BREWINSIGHT BI: Revenue Pattern Analytics System designed for Kape Tubong. Just like the businesses described in their study, Kape Tubong currently faces very similar challenges, including sales records that are scattered or disconnected across different locations, delays in bringing together information to create reports, and a lack of clear visibility regarding how each kiosk is performing. The proposed system is built specifically to solve these problems by centralizing all transactional and operational data into one unified dashboard platform, which enables real-time performance monitoring, detailed revenue analysis, direct comparison between branches, and the ability to forecast future trends.

This research also serves as important validation for the decision to include visual data presentations, historical trend analysis, and automated reporting features as core functions of the system, confirming that these tools are essential for improving the quality of management decisions and increasing operational efficiency. By transforming raw sales figures into clear, meaningful, and actionable insights, BREWINSIGHT BI will assist the leadership team at Kape Tubong in recognizing important revenue patterns, determining which products contribute most to success, optimizing how resources are allocated, and building a solid foundation for long-term strategic planning. Therefore, the study conducted by Fernandez and Gutierrez (2022) stands as valuable, locally relevant evidence proving that implementing a centralized Business Intelligence platform is not only feasible but also highly effective and beneficial for Kape Tubong.

Reference:

Fernandez, L., & Gutierrez, M. (2022). Revenue Analytics and Dashboard Utilization among Philippine Small Food Businesses. *Philippine Journal of Information Technology and Business Analytics, 10*(2), 54–76. ISSN 2799-4501. https://doi.org/10.6215/pjitba.2022.10.2.54

2.3 Related Literature

Existing literature in business operations, food and beverage management, and information systems consistently underscores the value of data-driven decision-making in enhancing organizational performance and operational efficiency. Contemporary research indicates that small and medium enterprises particularly benefit from technologies that provide timely insights into sales activities, customer behavior, and operational performance. These systems enable organizations to make

informed decisions based on factual evidence rather than assumptions or intuition. By leveraging historical and current business data, organizations can improve forecasting accuracy and strategic planning processes. Scholars also suggest that businesses utilizing data-driven management practices are better equipped to adapt to changing market conditions and consumer preferences. In highly competitive industries such as food and beverage, responsiveness to operational changes is critical to long-term sustainability. Business intelligence and analytical technologies contribute to this responsiveness by converting raw transactional information into actionable business insights. Literature also notes that data-centric management enhances accountability by establishing measurable indicators for evaluating business performance. As digital transformation continues to reshape industries, the adoption of analytical systems is becoming increasingly important for maintaining competitiveness. Overall, prior studies strongly advocate for the integration of data-driven technologies into business environments to improve efficiency, adaptability, and strategic decision-making. Researchers further explain that organizations capable of effectively analyzing operational data are more likely to identify growth opportunities and operational weaknesses at earlier stages. This allows management to implement corrective actions before operational issues become severe or financially damaging. Studies also indicate that data-driven environments encourage a culture of continuous improvement because performance metrics can be regularly monitored and evaluated. In addition, businesses that rely on analytical systems often demonstrate stronger organizational transparency because operational outcomes are supported by measurable evidence. The growing accessibility of cloud computing and digital analytics platforms has also made data-driven technologies more attainable for small and medium enterprises. Consequently, literature increasingly recognizes analytical systems as essential organizational tools rather than optional technological enhancements.

A frequently discussed concept in related literature is the use of interactive dashboards as tools for presenting complex business data in a clear and user-friendly manner. Rather than relying solely on raw figures or manually prepared reports, dashboards display information through visual formats such as graphs, charts, and summarized metrics. Researchers indicate that these visual representations reduce the difficulty of interpreting large datasets and improve user comprehension. Dashboards are particularly beneficial for non-technical users who require immediate understanding of performance indicators without detailed data analysis expertise. They also enable real-time monitoring of key business metrics, allowing management to respond promptly to emerging operational concerns. Many dashboard systems include advanced functionalities such as filters and drill-down capabilities for more detailed analysis. Studies further reveal that dashboards enhance communication among stakeholders by presenting performance information in a consistent and standardized format. The use of self-service dashboard tools additionally empowers managers to independently analyze business data without dependence on technical personnel. This contributes to faster evaluation and decision-making within organizations. Consequently, literature consistently identifies dashboards as valuable tools for improving data accessibility and managerial efficiency. Researchers additionally highlight that dashboard systems improve situational awareness by enabling managers to view multiple operational indicators simultaneously within a single interface. This consolidated visibility supports quicker interpretation of operational performance and reduces the time required to gather information from separate reports. Interactive dashboards also encourage proactive management because irregularities and performance deviations become more visible through graphical representations. In food and beverage businesses, dashboards may assist management in monitoring sales performance, inventory movement, customer traffic, and workforce allocation in real time. Literature also suggests that customizable dashboard layouts improve user experience because organizations can prioritize the most relevant metrics according to operational objectives. Furthermore, visual dashboards contribute to more effective meetings and strategic discussions by simplifying the communication of business trends and performance outcomes. Overall, scholarly works strongly position dashboards as central components of modern business intelligence systems.

Another major topic emphasized in the literature is product performance analysis, which assists businesses in evaluating the profitability and demand levels of their products. This is especially relevant in the food and beverage sector, where customer preferences often shift due to trends, seasonal changes, and ingredient availability. Product performance analysis enables management to identify best-selling items, underperforming offerings, and revenue-generating products. Such insights support evidence-based decisions regarding menu updates, product discontinuation, and pricing adjustments. Researchers note that product-level analytics can strengthen pricing strategies by revealing product profitability and customer demand elasticity. It also improves promotional planning by identifying which products require additional marketing support. Businesses can further utilize product analytics to prioritize procurement of high-demand items and minimize investment in low-performing stock. This contributes to more efficient inventory management and reduced waste. Continuous analysis of product performance allows organizations to remain responsive to evolving market preferences. Overall, literature recognizes product analytics as a key strategic practice in food service business management. Studies additionally emphasize that product-level analysis helps businesses evaluate the long-term sustainability of menu offerings and promotional campaigns. Through detailed sales tracking, management can determine whether specific products contribute consistently to revenue growth or only perform during short-term promotional periods. Researchers also suggest that understanding customer purchasing combinations can support bundling strategies and cross-selling opportunities. Product analytics may further reveal customer preferences associated with demographic trends, seasonal behavior, and localized market demand. In highly competitive food service environments, the ability to quickly identify changing consumer interests provides businesses with a strategic advantage. Literature therefore supports continuous product performance monitoring as a critical factor in maintaining profitability, competitiveness, and customer satisfaction.

Scholarly works also stress the operational importance of peak-hour analysis, which determines the periods of highest and lowest customer activity within a business. This information is valuable for improving workforce allocation, customer service quality, and queue management during busy operational periods. By understanding customer traffic patterns, businesses can schedule employees more effectively to meet demand during peak hours. Researchers suggest that appropriate staffing during high-volume periods enhances service speed and customer satisfaction. Likewise, identifying low-demand periods allows businesses to reduce unnecessary labor expenses during slower operations. Peak-hour insights can also guide management in scheduling maintenance, restocking, and promotional activities during less disruptive periods. In food service settings, awareness of demand peaks contributes to better preparation and smoother workflow management. Studies indicate that businesses leveraging temporal sales analysis achieve improved efficiency and operational planning. Additionally, branch-specific peak-hour analysis enables comparative performance assessments across multiple locations. Overall, literature supports the use of peak-hour analysis as an important tool for optimizing service operations and resource management. Researchers additionally note that customer traffic analysis contributes to improved operational forecasting by helping businesses anticipate recurring demand cycles. This enables organizations to prepare more effectively for weekends, holidays, and promotional events that may significantly increase customer volume. Peak-hour analysis may also reduce employee stress by ensuring that staffing arrangements align more accurately with actual service demand. Furthermore, businesses that effectively manage customer flow during busy periods are more likely to maintain service consistency and customer satisfaction. Studies also indicate that temporal analytics contribute to better resource utilization because businesses can allocate equipment, ingredients, and personnel according to expected operational activity. Consequently, literature recognizes peak-hour analysis as both an operational efficiency tool and a customer service enhancement strategy.

Inventory monitoring and management systems are likewise prominent topics in related studies, particularly for their role in minimizing waste and maintaining product availability. Effective inventory tracking helps organizations prevent overstocking, which can lead to spoilage and waste, as well as understocking, which may result in missed sales opportunities. Research indicates that inventory systems improve operational continuity by maintaining balanced stock levels aligned with actual demand. Automated inventory tracking reduces dependence on manual stock counting and lowers the likelihood of human error. It also provides businesses with real-time visibility into stock movement and consumption rates. Researchers highlight that inventory data can support more accurate purchasing decisions and supplier coordination. This is particularly critical in food and beverage businesses where inventory often includes perishable ingredients. Inventory analytics also help identify slow-moving items and purchasing inefficiencies. Improved inventory control directly contributes to cost reduction and profitability enhancement. Consequently, literature consistently positions inventory management systems as essential operational tools in food service environments. Researchers further explain that inventory monitoring systems improve accountability by maintaining detailed records of stock movement and usage patterns. This reduces opportunities for inventory discrepancies, wastage, and unauthorized stock handling. Studies also reveal that businesses utilizing automated inventory systems are better able to maintain operational continuity during supply chain disruptions because inventory visibility supports proactive replenishment planning. In addition, inventory analytics can assist management in identifying seasonal purchasing trends and supplier performance issues. Accurate stock monitoring is especially important in cafés and food kiosks where ingredient freshness directly affects product quality and customer satisfaction. Literature therefore recognizes inventory management systems as significant contributors to operational efficiency, financial control, and service reliability.

Furthermore, literature on automation and cloud-based technologies demonstrates that digital systems reduce administrative workload and improve data reliability. Automated reporting tools eliminate many of the errors associated with manual encoding and spreadsheet preparation. Cloud-based platforms allow businesses to store and access centralized information through internet-enabled devices from various locations. Researchers observe that automation accelerates report generation and decreases the time spent consolidating business data manually. This enables managers to focus more on analysis and strategic activities rather than administrative processing. Cloud-based systems also support scalability by allowing organizations to expand user access and system functions with minimal infrastructure changes. Remote accessibility further allows business owners and managers to monitor operations outside physical business premises. Studies additionally indicate that centralized cloud systems improve collaboration by ensuring that all users access consistent and updated records. Enhanced data security and automated backup mechanisms also reduce the risk of data loss. Overall, literature identifies automation and cloud computing as critical components of modern business information systems. Researchers additionally highlight that cloud technologies improve organizational flexibility because information can be accessed through multiple devices and locations without requiring complex hardware infrastructure. Automated systems also reduce dependency on paper-based documentation, which contributes to improved environmental sustainability and operational organization. Literature suggests that businesses adopting cloud-based solutions are more capable of supporting remote management and multi-branch coordination. This is especially beneficial for growing enterprises that require centralized oversight across geographically distributed operations. Furthermore, automated systems reduce repetitive administrative tasks that often contribute to employee fatigue and operational inefficiencies. Consequently, literature consistently identifies automation and cloud integration as important drivers of organizational productivity, scalability, and digital transformation.

Within the Philippine business environment, scholars also highlight the influence of contextual factors such as holidays, festivals, and local events on customer demand and sales performance. These external variables can significantly alter traffic volume and purchasing behavior in food and beverage establishments. As a result, researchers stress the need for analytical systems capable of adapting to localized market conditions and seasonal trends. Businesses that monitor these contextual influences can better anticipate fluctuations in demand and prepare operationally. This includes adjusting inventory levels, staffing schedules, and promotional strategies in anticipation of event-driven sales increases or decreases. Studies suggest that understanding local demand drivers improves market responsiveness and strategic planning. For community-oriented establishments like neighborhood cafés, localized consumer behavior may strongly affect sales outcomes. Systems capable of analyzing contextual sales patterns provide more meaningful insights than generalized reporting alone. Adaptability to local business conditions is therefore considered a significant advantage in analytical system design. Overall, literature supports the inclusion of localized and seasonal trend analysis within business intelligence platforms. Researchers additionally explain that localized market analysis allows businesses to better understand cultural and social influences affecting customer purchasing behavior. In the Philippine setting, community events, school schedules, tourism patterns, and weather conditions may all contribute to fluctuations in sales activity. Businesses capable of analyzing these localized influences can develop more targeted and effective operational strategies. Literature also indicates that organizations with contextual analytical capabilities are more responsive to sudden market shifts and changing consumer trends. This adaptability strengthens customer engagement and improves organizational competitiveness within local markets. Consequently, researchers strongly recommend integrating contextual and localized analytics into food service business intelligence systems.

In summary, related literature consistently supports the integration of business intelligence tools into food and beverage operations to improve efficiency, accuracy, and responsiveness in business management. Prior studies demonstrate that BI technologies enhance operational visibility and facilitate data-based strategic planning. These systems improve performance monitoring by consolidating fragmented data into centralized and actionable insights. They also strengthen forecasting, inventory planning, sales analysis, and branch evaluation processes. In food service environments, business intelligence solutions contribute to improved service quality and operational effectiveness through timely access to relevant business metrics. Researchers note that organizations utilizing BI systems become more adaptive and competitive in dynamic market environments. These technologies reduce reliance on manual reporting while increasing the reliability and speed of business analysis. They further support long-term growth by enabling continuous performance optimization and strategic refinement. As digital adoption continues to expand, BI implementation is becoming increasingly essential for SMEs and growing enterprises alike. Collectively, the literature provides substantial support for the development and implementation of the BREWINSIGHT BI system for Kape Tubong. Furthermore, the reviewed literature consistently demonstrates that organizations adopting analytical technologies are more capable of sustaining operational consistency and strategic flexibility during periods of business growth. Researchers also emphasize that business intelligence systems strengthen organizational accountability by making operational performance more transparent and measurable. In competitive food service environments, the ability to access timely and accurate information is increasingly recognized as a significant strategic advantage. Literature therefore supports the argument that integrated analytical platforms are no longer optional technologies but essential operational tools for modern business management.

Another significant concept discussed in related literature is centralized data management, which refers to the consolidation of business information from multiple operational points into a single unified repository. Researchers emphasize that centralized data systems improve organizational control by ensuring that all departments and branches operate using the same standardized information source. In businesses with multiple outlets or kiosks, centralized databases eliminate inconsistencies caused by decentralized spreadsheets or manually maintained records. This consolidation improves reporting accuracy and reduces the duplication of information across departments. Centralized data management also simplifies auditing, performance tracking, and historical record retrieval by maintaining organized and structured datasets in one location. Studies suggest that organizations with centralized information systems can perform comparative analysis more efficiently because data from multiple branches can be accessed simultaneously in a consistent format. Furthermore, centralized systems improve collaboration among management personnel by providing a shared and synchronized operational view. This unified data environment strengthens decision-making by reducing discrepancies in reported figures and interpretations. Researchers therefore consider centralized data management a foundational element of effective business intelligence implementation. Its inclusion in analytical platforms is widely recommended for organizations operating across multiple business locations. Researchers additionally note that centralized systems improve data governance by establishing standardized procedures for data entry, validation, and reporting across all operational units. This consistency enhances the reliability of organizational records and reduces confusion caused by varying reporting practices between branches. Centralized repositories also strengthen data security because access permissions and monitoring mechanisms can be managed within a single platform. Studies indicate that organizations utilizing centralized systems experience improved efficiency in generating organization-wide performance reports and executive summaries. In multi-branch businesses, centralized databases also support faster communication between branch managers and administrative personnel because updated operational information becomes immediately accessible. Furthermore, centralized data structures improve scalability by allowing additional branches or operational units to be integrated into the system without major restructuring. Literature therefore recognizes centralized data management as essential for improving operational coordination, reporting consistency, and long-term organizational growth.

Another area emphasized in literature is branch performance evaluation, particularly in businesses managing multiple outlets, franchises, or satellite branches. Branch performance analysis allows organizations to compare operational outputs, revenue generation, and sales efficiency across different locations. This comparative insight helps management identify which branches are outperforming expectations and which require operational intervention. Researchers note that branch-level analytics improve accountability by establishing measurable performance benchmarks for each location. Comparative branch reporting also helps businesses identify location-specific trends, operational inefficiencies, and contextual market differences affecting sales performance. In food and beverage operations, branch performance evaluation can reveal differences in customer traffic, demand behavior, and staff productivity across locations. This enables management to implement targeted interventions based on branch-specific performance issues rather than generalized assumptions. Studies suggest that branch comparison tools contribute to more strategic expansion planning by identifying characteristics of successful locations. Consequently, branch performance monitoring is recognized as a critical analytical practice for multi-branch enterprises. This literature supports the inclusion of comparative branch analytics in the BREWINSIGHT BI platform. Researchers further explain that branch-level analytics contribute to organizational transparency by making operational performance measurable and comparable across all locations. Through comparative dashboards and branch-specific reporting, management can identify operational strengths and weaknesses more effectively. Literature also indicates that branch evaluation systems improve managerial accountability because branch supervisors become more aware of measurable operational expectations. In addition, comparative analysis enables organizations to identify best practices from high-performing branches that can be replicated across other locations. Businesses can also use branch performance metrics to evaluate the effectiveness of staffing strategies, promotional activities, and customer service practices within specific operational contexts. Studies further reveal that branch analytics support evidence-based expansion decisions because management can identify which operational conditions contribute most significantly to branch success. Overall, literature consistently highlights branch performance analysis as an important tool for improving operational consistency, organizational accountability, and strategic planning.

Scholars also highlight the growing relevance of real-time reporting systems in enhancing managerial responsiveness and operational agility. Real-time reporting allows decision-makers to access updated business data immediately after transactions occur, reducing delays associated with batch processing or manual report preparation. Researchers argue that timely access to current operational information improves the speed and quality of managerial responses to performance fluctuations. In dynamic industries such as food and beverage, where demand patterns may shift rapidly within a single day, real-time visibility is especially valuable. Businesses utilizing real-time reporting can respond more quickly to sales declines, product shortages, or unexpected demand surges. This capability supports immediate operational adjustments and reduces the risk of delayed interventions. Studies further note that real-time data access improves managerial confidence by ensuring decisions are based on the most current available information. It also enhances situational awareness across distributed business locations. As a result, real-time reporting is increasingly considered a standard feature of modern business intelligence systems. Its integration strengthens the practical value of analytical platforms intended for operational decision support. Researchers additionally emphasize that real-time systems improve organizational responsiveness by enabling management to monitor ongoing business conditions continuously throughout operational hours. This continuous visibility supports faster identification of anomalies such as unusual sales declines, inventory discrepancies, or sudden increases in customer traffic. Literature also suggests that real-time reporting contributes to improved customer satisfaction because operational adjustments can be implemented immediately when service issues arise. In food service businesses, real-time visibility may help management reduce waiting times, improve product availability, and maintain service consistency during peak demand periods. Furthermore, real-time reporting reduces reliance on end-of-day summaries that may delay critical business decisions. The capability to access live operational information therefore strengthens strategic agility and operational control within fast-paced business environments. Consequently, scholars strongly advocate for the integration of real-time analytics within modern business intelligence platforms.

In addition, literature frequently discusses the importance of historical trend analysis in supporting business forecasting and long-term strategic planning. Historical trend analysis involves examining past performance data to identify recurring patterns, growth trajectories, and seasonal fluctuations over time. Researchers suggest that organizations using historical analytics can better anticipate future operational demands and market conditions. In food and beverage businesses, historical sales patterns can reveal recurring high-demand periods associated with holidays, weekends, local events, or seasonal purchasing behavior. These insights assist businesses in planning inventory procurement, staffing schedules, and promotional activities more effectively. Historical trend analysis also supports long-term performance evaluation by enabling organizations to assess progress over extended operational periods. Comparative historical reporting helps management determine whether business performance is improving, declining, or remaining stable. Studies indicate that organizations with access to historical analytics are better equipped to engage in proactive rather than reactive management. Consequently, literature strongly supports incorporating historical trend analysis into business intelligence platforms. This validates BREWINSIGHT BI's focus on revenue pattern analysis across multiple time intervals. Researchers additionally explain that historical trend analysis strengthens organizational learning by enabling businesses to evaluate the outcomes of previous operational strategies and management decisions. Through longitudinal data examination, organizations can identify which promotional activities, pricing strategies, or operational adjustments generated positive performance outcomes. Literature also notes that historical analytics support more accurate forecasting because predictive models become more reliable when based on extensive historical datasets. In food service operations, historical analysis may reveal recurring consumer preferences that can guide future menu planning and inventory management. Furthermore, long-term trend analysis assists management in evaluating branch growth trajectories and revenue sustainability across multiple operational periods. This capability improves strategic planning by providing measurable evidence for expansion, investment, and operational refinement initiatives. Overall, researchers recognize historical analytics as essential for strengthening forecasting accuracy, strategic planning quality, and organizational adaptability.

Finally, related literature highlights the strategic role of decision-support systems in reducing managerial uncertainty and improving organizational planning quality. Decision-support systems integrate analytical tools, reporting mechanisms, and structured data presentation to help management evaluate alternatives and make informed operational decisions. Researchers note that these systems reduce reliance on intuition-based management practices by providing objective evidence to support strategic actions. In small business environments, decision-support systems are especially valuable because management often handles multiple responsibilities simultaneously and requires efficient access to summarized insights. Studies indicate that businesses using decision-support platforms demonstrate improved consistency in strategic planning and stronger justification for operational changes. These systems also support transparency by allowing decision-makers to trace conclusions back to measurable performance data. In addition, decision-support tools improve confidence in managerial actions because decisions are grounded in verifiable analytical evidence. This contributes to more disciplined and accountable business management practices. Therefore, literature consistently recognizes decision-support systems as essential tools for modern organizational governance and strategic execution. This further reinforces the relevance of BREWINSIGHT BI as a decision-support and analytical platform for Kape Tubong. Researchers additionally emphasize that decision-support systems improve organizational coordination because managers across different departments can access the same analytical information when making operational decisions. This consistency reduces communication gaps and minimizes conflicting interpretations of business performance. Literature also indicates that decision-support technologies contribute to faster problem-solving because operational concerns can be identified, analyzed, and addressed using integrated analytical reports. In competitive business environments, rapid and evidence-based decision-making is considered critical for maintaining operational efficiency and market responsiveness. Decision-support systems may also improve strategic evaluation by enabling organizations to simulate potential operational outcomes using historical and real-time data insights. Furthermore, these systems strengthen managerial accountability because business decisions can be reviewed and justified using documented analytical evidence. Studies consistently show that organizations utilizing decision-support tools demonstrate stronger adaptability, improved planning discipline, and more effective resource allocation practices. Consequently, literature strongly supports integrating decision-support functionalities within business intelligence platforms intended for operational and strategic management purposes.

2.4 Synthesis of the Reviewed Literature

The reviewed foreign and local studies, together with related literature, collectively form a robust theoretical and practical basis for the creation of the BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong. Across all examined sources, there is consistent emphasis on the strategic importance of data visualization, automation, and business analytics in enhancing organizational performance. The literature clearly demonstrates that enterprises gain significant advantages by converting raw operational data into clear, structured insights through well-designed business intelligence systems. These findings establish that data accessibility and ease of interpretation are fundamental requirements for effective managerial decision-making. Research also indicates that organizations utilizing analytical tools demonstrate greater operational responsiveness and strategic flexibility. This capability holds particular value in the food and beverage sector, where customer preferences and demand levels shift rapidly, making efficient data interpretation a core necessity. The reviewed materials confirm that such systems improve not only long-term strategic planning but also daily operational management. The convergence of results from multiple authors and different contexts provides strong justification for deploying a dedicated revenue analytics platform within Kape Tubong's specific working environment. This body of work therefore offers comprehensive support for both the conceptual framework and technical design of the proposed system.

Foreign research highlights that interactive dashboards drastically improve the speed and accuracy of decision-making by transforming complex datasets into meaningful visual formats. These studies show that visual aids such as graphs, charts, and trend indicators reduce the difficulty of analyzing large volumes of transactional information. Researchers consistently note that visual analytics deepen management understanding of key performance indicators and operational patterns. Dashboard-based tools also allow businesses to identify irregularities, revenue declines, or performance fluctuations much faster than traditional reporting methods. International literature further points out that these systems foster better collaboration by providing a shared visual reference for discussing results and goals. Advanced features like interactive filtering and detailed data exploration expand analytical depth, enabling users to examine performance at very specific levels. These characteristics make dashboards highly suitable for environments requiring fast, evidence-based choices. The reviewed studies validate the use of dashboard interfaces as central components of intelligence systems, directly supporting the inclusion of visual analytics and reporting features within BREWINSIGHT BI. This alignment reinforces the project's primary goal of delivering clear, reliable, and actionable insights to Kape Tubong leadership.

Local findings further strengthen this perspective by proving that Philippine small and medium enterprises achieve measurable improvements when adopting digital monitoring and business intelligence tools. Investigations focused on local food service and retail businesses reveal that dashboard systems reduce report preparation time, speed up sales analysis, and enhance management responsiveness to market changes. Local studies also demonstrate that these technologies lead to better data organization and more consistent performance tracking across locations. Such results prove that analytical systems are not exclusive to large corporations but are equally valuable for smaller businesses operating in domestic markets. Within the Philippine context, SMEs commonly face challenges such as fragmented data, manual reporting processes, and limited resources. Literature shows that business intelligence solutions effectively address these issues by automating data processing and centralizing information storage. This evidence highlights the practical relevance and suitability of implementing such systems in small-scale settings like Kape Tubong. Local adoption trends confirm that domestic businesses increasingly embrace digital transformation to maintain competitive advantage. The proven applicability of these tools in local operations validates the feasibility of the proposed system within a Philippine food and beverage context.

Reviewed sources also reveal that combining multiple analytical functions into one unified platform delivers more comprehensive decision support compared to using separate, disconnected applications. Studies indicate that integrating features such as sales tracking, revenue trend analysis, product evaluation, and peak-hour detection increases both analytical depth and strategic value. Instead of reviewing isolated reports, integrated systems allow management to understand relationships between various operational factors simultaneously. This holistic view improves strategic planning and problem diagnosis by offering a complete picture of business health. Consolidated tools also enhance workflow efficiency by removing the need to switch between different software or documents. Researchers note that centralized platforms promote consistency in data interpretation and reporting standards across the organization. Integration further enables accurate comparative analysis between branches, products, or time periods. Decision quality improves significantly when all relevant metrics are accessible within a single environment. These conclusions support the design approach of BREWINSIGHT BI as a fully consolidated solution rather than a collection of separate modules. Existing literature therefore thoroughly validates the system’s integrated architecture and multifunctional framework.

Findings from the reviewed materials align closely with the operational realities and challenges currently faced by Kape Tubong. The organization struggles with scattered sales records, delays caused by manual compilation, and limited visibility into individual kiosk performance. These difficulties mirror the same inefficiencies and information gaps documented in both local and international research regarding decentralized or manual monitoring systems. Literature confirms that such limitations hinder strategic planning, reduce responsiveness, and weaken the reliability of performance evaluations. BREWINSIGHT BI directly targets these problems by centralizing all transactional and operational data into one cohesive analytical dashboard. This structure allows management to monitor sales activities, compare branch results, assess product demand, and identify revenue patterns in real time. The proposed design effectively eliminates data fragmentation and analysis delays affecting current operations. Web-based accessibility ensures authorized personnel can access insights remotely and conveniently from different locations. This close match between system capabilities and business needs demonstrates the practical necessity of the solution within Kape Tubong’s environment. Reviewed work confirms that the proposed platform is highly relevant and directly responsive to the organization’s requirements.

Synthesis of all gathered evidence confirms a clear and substantial demand for integrated, dashboard-based business intelligence platforms tailored specifically to small food and beverage enterprises in the Philippines. Existing research consistently supports the value of combining data visualization, automated reporting, centralized tracking, and decision-support tools into one system. Literature demonstrates that such solutions increase efficiency, reduce manual workload, and enhance decision quality across all operational areas. It also confirms that SMEs increasingly require scalable, accessible digital tools to remain competitive in dynamic markets. Despite this established need, many smaller businesses continue to rely on outdated or fragmented monitoring practices, indicating a gap between available technology and its actual use. BREWINSIGHT BI bridges this gap by offering a specialized revenue analytics solution built around Kape Tubong’s unique operational structure. The platform consolidates essential functions into a web-based system suitable for managing multiple kiosks in the food service industry. Development of this system is strongly justified by theoretical frameworks, empirical data, and contextual evidence found throughout the reviewed sources. The project stands not only as a technological upgrade but as a fundamental strategic solution addressing core operational challenges. The entire body of literature provides comprehensive validation for the necessity, design, and implementation of the proposed system.

Reviewed research collectively suggests that successful implementation depends not only on tool availability but also on how well these tools align with existing workflows and decision-making routines. Experts emphasize that business intelligence platforms deliver maximum value when customized to match specific operational requirements rather than deployed as generic reporting applications. This principle highlights the importance of designing BREWINSIGHT BI according to Kape Tubong’s unique structure, workflow patterns, and monitoring needs. Aligning system features with the kiosk-based model, branch oversight requirements, and revenue analysis objectives ensures high practical relevance and effectiveness. This evidence-based approach supports the study’s focus on developing a tailored solution rather than adapting a generalized reporting package. Customized design improves usability, operational fit, and long-term acceptance among intended users. The development strategy applied in BREWINSIGHT BI follows established best practices identified in previous research, strengthening the system’s expected value and chances of successful adoption.

Several sources highlight that adopting digital analytics significantly contributes to broader organizational transformation and technological maturity. Businesses implementing these systems often develop stronger data management practices, more structured reporting routines, and greater discipline in performance monitoring. These improvements extend beyond analytics alone and influence overall operational culture. For Kape Tubong, introducing BREWINSIGHT BI brings benefits beyond improved sales analysis; it supports the establishment of systematic, technology-driven management practices across the enterprise. The platform encourages greater accountability in branch reporting, better standardization of documentation, and consistent monitoring habits across all locations. Over time, these changes facilitate a transition toward a more digitally mature operational model. Literature indicates that such evolution enhances organizational adaptability and competitiveness in increasingly data-reliant markets. Benefits of the proposed system therefore extend beyond immediate efficiency gains and contribute to long-term organizational development and growth.

Reviewed literature indicates that centralized intelligence systems create long-term strategic advantages by building scalable digital infrastructures capable of supporting future growth and enhancements. Organizations adopting these platforms establish a technological foundation upon which advanced features—such as predictive modeling, automated forecasting, inventory optimization, and customer behavior analysis—can later be developed. This perspective positions BREWINSIGHT BI not merely as a temporary operational tool but as a foundational step toward broader digital transformation and future enterprise development within Kape Tubong. Centralizing transaction processing and analytical functions creates a structured framework ready to support innovation, expansion, or diversification of activities. Scalability becomes especially valuable as the business increases the number of kiosks or expands its product range. Research supports the strategic value of implementing foundational intelligence systems even in smaller enterprises with limited current infrastructure. The proposed system serves both as an immediate solution to existing challenges and as a strategic investment in future capabilities. Design and implementation align fully with long-term technological and organizational goals identified in contemporary business intelligence research.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study adopts a Developmental Research Design, chosen because the main goal is to methodically plan, create, and assess the BREWINSIGHT BI: Revenue Pattern Analytics System as a functional solution to the operational issues currently faced by Kape Tubong. This approach works exceptionally well for projects focused on building technology-based tools, software applications, or improved processes meant to resolve real-world organizational challenges. Unlike studies that are purely descriptive or experimental, this design combines the investigation of existing problems with the organized creation of a working solution. It enables the research team to go beyond simply identifying difficulties, resulting in a concrete and fully functional output that directly meets defined business needs. This framework is widely accepted and applied in information technology and systems development work where the aim is to produce and validate software-based solutions. By using this method, the project guarantees the resulting system is shaped by actual operational realities rather than theoretical concepts alone. It also supports gradual improvement, allowing the platform to evolve through repeated testing and continuous feedback collection. Since the objective is to deliver a ready-to-use analytics tool rather than only analyze existing conditions, this design is highly suitable. It further fosters strong alignment between user needs, technical execution, and ease of use, establishing a solid methodological base for the structured creation and validation of the BREWINSIGHT BI platform.

This methodology offers a clear structure for translating identified business concerns into specific technical goals and corresponding system features. It ensures every component developed serves a distinct, documented operational requirement. Following this design helps maintain a logical and consistent connection between problem analysis, system construction, and performance assessment throughout the entire research process. It also allows for ongoing checks at every stage to confirm that outputs remain consistent with the project's intended aims and quality standards. This careful alignment is vital when creating decision-support tools meant for active business use, as it strengthens both the practical relevance and methodological rigor of the project.

At the start of the study, a descriptive research element is included to analyze and understand exactly how Kape Tubong functions on a daily basis. This initial stage centers on documenting current methods for recording sales data, combining information into reports, and monitoring performance across different locations. The team examines how transaction details are collected from various kiosk operations and how this information is processed and presented for management review. Special attention is directed toward spotting inefficiencies, unnecessary steps, and limitations within the existing workflow. Persistent challenges such as scattered data records, delays in report preparation, and lack of a unified view of information are thoroughly assessed and recorded. This phase also involves observing standard reporting practices and evaluating how accessible, accurate, and useful available sales information actually is. Discussions and consultations with the business owner and operational staff are conducted to gather detailed insights regarding recurring struggles and pain points. These activities build a complete picture of the business environment and clearly define the specific problems the proposed system must solve. By fully understanding current operational conditions, the entire development process becomes firmly rooted in real-world needs. This descriptive analysis forms the essential basis for determining system requirements and guiding the creation of relevant, useful functions.

This initial descriptive stage further helps identify user expectations, preferred ways of working, and practical constraints that could influence how well the new system is adopted. Recognizing these factors is critical to ensuring the platform fits naturally within established daily activities. It significantly lowers the risk of building features that work technically but prove difficult, confusing, or impractical to use in real operations. It also ensures the solution addresses not only technical specifications but also broader usability and workflow considerations. As such, this phase contributes greatly to the overall relevance and usefulness of the final system design.

Once operational issues and user requirements are clearly mapped and understood, the developmental phase shifts focus to designing and building the actual BREWINSIGHT BI system. During this stage, the research team translates gathered requirements into formal system specifications, user interface layouts, database structures, and distinct functional modules. The platform is built around a centralized dashboard that collects, organizes, and displays sales data from all Kape Tubong kiosks into one unified, easy-to-navigate view. Core analytical capabilities such as tracking revenue trends, measuring product performance, enabling side-by-side branch comparisons, and identifying peak business periods are integrated deeply into the system architecture. Automated reporting tools are also developed and added to reduce manual effort, speed up information delivery, and make critical data readily available to management whenever needed. The design prioritizes ease of use, fast response times, and broad accessibility through a web-based deployment model that works across various devices. Security controls, including secure user login procedures and role-based access permissions, are incorporated to ensure sensitive business data remains protected and accessible only to authorized personnel. Throughout the construction process, the design is refined repeatedly to ensure every feature closely matches user expectations and operational demands. Technical aspects such as ability to grow, stability over time, and ease of future updates are also embedded into the development strategy. This stage ultimately results in a fully working prototype tailored specifically to the unique operational needs of Kape Tubong.

The development process also includes regular internal reviews and assessments to confirm that design outputs remain consistent with technical standards and defined business needs. Evaluations of early versions are conducted during intermediate stages to verify interface logic, workflow flow, and calculation accuracy before final completion. This repeated refinement allows the team to spot potential issues early, adjust designs, and make improvements before the system is launched for use. By continuously enhancing the prototype based on testing results, the quality, reliability, and user-friendliness of the final product are significantly increased. This incremental approach leads to a more stable, capable, and user-centered software solution.

After development is complete, the system undergoes extensive testing and evaluation to measure its function, reliability, usability, and overall effectiveness in meeting goals. Functional assessments confirm that every module works according to design specifications and processes data correctly, consistently, and accurately. Reliability checks ensure the system performs steadily during regular use, handles expected loads, and operates without errors or unexpected breakdowns. Usability evaluations measure how intuitive the platform is to navigate, how clear the information display is, and how well it suits the intended audience. Selected staff members from Kape Tubong participate in user acceptance testing, providing direct feedback on how practical, helpful, and effective the system is in real working situations. Their observations, comments, and suggestions are recorded and analyzed in detail to find areas needing adjustment or improvement. Based on these findings, the team revises and enhances the platform to address identified concerns and optimize the overall user experience. This cycle of testing and refinement ensures the final system is both technically sound and deeply relevant to daily operations. The evaluation stage also verifies whether the completed system successfully resolves the problems identified earlier and achieves all original objectives. Through comprehensive checks and improvements, the study ensures BREWINSIGHT BI meets high standards of quality, function, and usability before being deployed.

The evaluation phase also determines how well the system meets stakeholder expectations and fulfills documented business requirements. Feedback collected during testing serves as practical evidence of how well the solution performs in a live environment. This confirms the platform works well not only against technical specifications but also regarding user satisfaction and operational value. Test results often reveal opportunities to add features or adjustments that improve long-term manageability and capacity for growth. Therefore, this assessment stage is critical for proving both the technical success and practical value of the completed platform.

Applying a Developmental Research Design effectively bridges the gap between theoretical knowledge and practical application by producing a research-based yet fully functional technological solution. Instead of limiting the work to conceptual study or observation, this approach supports building a complete business intelligence system that directly addresses real operational challenges. It enables the team to systematically investigate problems, gather detailed user needs, and translate those findings into useful software features and tools. The iterative nature of this design ensures the system improves continuously through validation and adjustment. This method strengthens the relevance and value of the final platform by grounding every decision in factual findings and direct user feedback. It also encourages accountability in design, requiring that the solution be tested and measured against clearly defined goals and success criteria. As a result, the methodology ensures BREWINSIGHT BI is not only technically viable but also truly valuable and meaningful to its intended users. By combining analysis, creation, testing, and refinement into one structured process, the study achieves both academic thoroughness and practical benefit, making it ideal for technology-focused projects aimed at software innovation and applied development. This design therefore delivers a complete methodological framework for successful project completion.

This chosen methodology strengthens the study's trustworthiness by ensuring every step taken to build the system is carefully recorded, reasoned, and assessed in a clear and structured manner. This approach boosts the academic value of the work and makes the entire development process easy to follow or replicate in future projects. The organized nature of this design ensures that all choices made along the way are well-explained and visible to others. As a result, the work meets both rigorous academic standards and professional software development practices, allowing the study to offer useful insights for both research and real-world business use.

This research framework also offers the freedom to adjust and improve as the work progresses, while keeping a clear and systematic structure in place. This flexibility is valuable because needs and expectations often become clearer or shift once users begin testing early versions and sharing their thoughts on features and usability. The method allows the team to introduce necessary changes or enhancements without disrupting the overall direction or goals of the project. By maintaining active communication between developers and those who will use the system, the final result fits naturally with actual daily work and operational demands. This cooperative approach leads to higher user satisfaction and makes it much more likely the system will be accepted and used effectively once launched.

The method also places strong emphasis on creating and maintaining detailed technical records at every stage of creation. Documents such as system outlines, database plans, process diagrams, and testing guides help keep the work organized, clear, and consistent from start to finish. These materials remain important long after the project ends, providing essential support for future maintenance, expansion, or upgrades. Keeping accurate records ensures openness during development and helps ensure the BREWINSIGHT BI platform can continue serving the business well over time.

Furthermore, this approach ensures all decisions during development are based on facts and evidence. Changes to the design or additions to features are driven by collected data, review results, and user feedback, rather than personal opinion or guesswork. This makes the work more objective and reduces the chance of including features that are not needed or do not work well. Every adjustment made to the system is therefore directly linked to identified operational issues and real business requirements. This ensures the final platform works efficiently, serves a clear purpose, and aligns closely with what the organization truly needs.

A major strength of this design is the ability to use both descriptive and measurable methods during evaluation. Comments and opinions from users help assess ease of use, clarity, and overall experience, while performance checks measure speed, accuracy, and stability. Combining these two perspectives gives the team a full and balanced understanding of how well the system performs. This thorough review process strengthens the reliability and credibility of the study's findings.

This framework also helps spot opportunities for future improvements that can expand what the platform can do beyond its original purpose. During testing and review, users may suggest new analysis tools, reporting options, or operational upgrades that can guide further development. This ensures the system can grow alongside the business and keep pace with new technologies, remaining useful and relevant for years to come.

The study also gains from the problem-solving nature of this research style. Rather than simply listing challenges or gaps, the method guides the team toward building practical technology-based solutions that directly address the issues found. This focus on results is especially important in information technology research, where the main goal is to help organizations work better through innovation. Through careful planning, building, testing, and refining, the method ensures the platform supports daily work and helps leaders make informed choices.

The approach also highlights how important it is to build systems around the people who will use them. By involving Kape Tubong staff in analyzing needs, testing versions, and reviewing results, the team ensures the solution matches real experiences, preferences, and working conditions. This involvement leads to designs that are easier to use, fit existing workflows better, and offer functions that truly help the business. It also makes adoption smoother, as employees learn about and become comfortable with the system while it is still being created.

In summary, the chosen research design creates a complete plan that connects analysis, creation, and evaluation into one organized process. Every part of the work contributes directly to meeting the study's goals, from understanding operational difficulties to proving that the finished system works effectively. The clear flow of activities ensures BREWINSIGHT BI is built according to research standards while remaining practical and responsive to real business needs. Through this method, the study shows how structured research can lead to technology solutions that improve work processes, support better decisions, and help local businesses grow and succeed.

3.2 System Development Methodology

Agile is a software development and project management methodology that emphasizes flexibility, collaboration, and continuous improvement throughout the development process. It focuses on delivering work in small, manageable increments rather than completing the entire project at once. Agile allows development teams to adapt quickly to changes in requirements or user feedback during the project lifecycle. This methodology encourages frequent communication and collaboration among developers, stakeholders, and end users. Each development cycle, often called a sprint or iteration, produces a working portion of the system for review and evaluation. Agile promotes regular testing and refinement to identify issues early and improve system quality continuously. It prioritizes customer satisfaction by involving users throughout the development process and incorporating their feedback into ongoing improvements. Compared to traditional development models, Agile is more adaptable to changing project conditions and evolving business needs. It is widely used in software development because it reduces project risks and improves development efficiency. Overall, Agile provides a structured yet flexible approach for creating high-quality systems through iterative progress and collaboration.

Agile enables the gradual release of features, giving stakeholders the chance to review and assess working parts of the system while development is still underway instead of waiting for the full solution to be finished. This method keeps all involved parties well-informed, as progress and results are clearly visible at every stage of the project lifecycle. It also lowers the chance of extensive rework, since flaws, gaps, or design concerns are spotted and resolved early during regular checkpoints and reviews. Through ongoing cooperation and assessment, the system stays closely matched to what users expect and what the business actually needs. This framework is especially well-suited for the BREWINSIGHT BI project, where operational priorities or requirements may shift or become clearer as work progresses, allowing adjustments to be made smoothly and efficiently. The approach also fosters constant communication between the research team, developers, and end users, ensuring that real-world challenges faced by Kape Tubong remain the central focus of every feature built. Regular interaction helps clarify needs instantly and address operational questions or concerns promptly, keeping development moving forward without unnecessary delays or misalignment.

Agile further emphasizes adaptability and responsiveness, which are important in projects involving dynamic business environments and evolving operational needs. Since BREWINSIGHT BI focuses on sales monitoring, analytics, and business intelligence functions, management requirements may change as additional insights and operational expectations emerge during development. Through Agile, modifications to dashboard layouts, reporting structures, and analytical functionalities can be implemented gradually without disrupting the entire development process. This flexibility enables the researchers to continuously refine the system according to stakeholder recommendations and usability feedback. Agile also supports the prioritization of critical system features, ensuring that the most essential components are developed and tested first before proceeding to more advanced functionalities. As a result, the development process becomes more organized, manageable, and responsive to business priorities.

Another important characteristic of Agile is its strong emphasis on iterative testing and continuous quality assurance. Instead of postponing testing activities until the end of development, Agile incorporates testing within every sprint or iteration cycle. This allows the researchers to identify software defects, interface inconsistencies, and functional errors at earlier stages of development. Early detection of issues minimizes the likelihood of major system failures during deployment and reduces the cost of correcting errors later in the project lifecycle. Continuous testing also improves system stability and reliability because each feature undergoes repeated evaluation and refinement before integration into the complete platform. In the context of BREWINSIGHT BI, iterative testing is essential to ensure the accuracy of sales analytics, dashboard visualizations, and reporting functionalities. The reliability of data processing and visualization features is critical because management decisions will depend on the information generated by the system.

Agile methodology also promotes user-centered development by involving stakeholders actively throughout the project lifecycle. Rather than limiting stakeholder participation to the initial planning phase, Agile encourages continuous feedback and evaluation during every development iteration. This ensures that the developed system remains aligned with actual user expectations and operational workflows. For BREWINSIGHT BI, management and staff members of Kape Tubong can evaluate prototype dashboards, reporting modules, and analytical features during development and provide recommendations for improvement. User participation strengthens system usability because suggestions from actual end users can guide interface design, functionality enhancement, and workflow optimization. This collaborative process increases the likelihood that the final system will effectively support operational decision-making and business management activities.

Furthermore, Agile supports efficient project management by dividing the development process into smaller and more achievable tasks. These smaller development activities allow the researchers to monitor progress more effectively and evaluate whether project objectives are being achieved within the intended timeframe. Sprint-based development also improves productivity because the development team can focus on completing prioritized objectives during each iteration. The methodology additionally promotes accountability among developers because tasks, deliverables, and timelines are clearly defined within every sprint cycle. This structured approach helps maintain consistent development progress while still allowing flexibility for revisions and feature enhancements. Agile therefore combines disciplined project organization with adaptability, making it highly suitable for information system development projects.

Another advantage of Agile is its capability to support continuous improvement throughout system development. At the end of every sprint, developers and stakeholders review completed functionalities and identify areas requiring refinement or enhancement. This reflective process allows the development team to improve both the system and the development strategy itself. Lessons learned from earlier iterations can be applied immediately in succeeding development cycles, improving overall efficiency and system quality. Continuous improvement also ensures that system components evolve according to actual operational requirements rather than fixed assumptions established at the beginning of the project. In the case of BREWINSIGHT BI, iterative enhancement is important because analytical requirements and dashboard preferences may become more specific as stakeholders interact with early system prototypes.

Agile methodology is also beneficial for reducing development risks associated with changing requirements, technical uncertainties, and user dissatisfaction. Because the system is developed incrementally, stakeholders can identify concerns early and recommend corrective actions before the project reaches advanced stages. This reduces the possibility of developing features that may not align with actual organizational needs. Frequent reviews and demonstrations additionally improve stakeholder confidence because project progress is continuously visible and measurable. Agile therefore minimizes uncertainty by maintaining regular communication and ensuring continuous alignment between technical development and business objectives. For BREWINSIGHT BI, this is particularly valuable because the project integrates operational monitoring, analytics, and reporting functionalities that must remain responsive to Kape Tubong's evolving operational environment.

Agile methodology strengthens the system's ability to expand and remain easy to manage over its entire lifespan. By focusing on building separate, self-contained components, this approach allows new features, functional improvements, and technical adjustments to be developed, tested, and added smoothly even after the platform is fully launched and in active use. This structure is highly beneficial for the BREWINSIGHT BI platform, particularly as Kape Tubong grows its operations, establishes new locations, introduces more product lines, or requires deeper analytical tools and specialized reporting functions in the future. Since each module operates independently, future enhancements and expansions can be carried out in stages without interrupting current processes or needing to rebuild the entire system from scratch. Such adaptability enables the platform to evolve continuously, adjust to shifting business requirements, and integrate modern technologies effectively, ensuring it stays capable, reliable, and fully aligned with the organization's progress. .

Overall, Agile methodology provides a highly appropriate framework for the development of BREWINSIGHT BI because it combines flexibility, collaboration, iterative testing, and continuous improvement within a structured development process. The methodology enables the researchers to create a system that is adaptable to changing business requirements while maintaining development efficiency and system quality. Through stakeholder collaboration, iterative refinement, and continuous evaluation, Agile supports the creation of a reliable and user-centered Business Intelligence platform capable of addressing Kape Tubong's operational and analytical needs.The **Agile Development Model** will be used, consisting of the following phases:

1.Planning Phase: The first stage of the Agile Development Model is the Planning Phase, where the researchers establish the overall scope, objectives, deliverables, timelines, and development strategies for the BREWINSIGHT BI: Revenue Pattern Analytics System. This phase serves as the fundamental foundation of the entire project because it clearly defines the direction, priorities, and structural framework that will guide all succeeding development activities and technical decisions. During this stage, the research team conducts a thorough and systematic analysis of the organizational needs, operational workflows, and strategic goals of Kape Tubong, carefully identifying the specific operational problems, limitations, and gaps that the proposed system must address and resolve. Proper and detailed planning is essential in this methodology because it ensures that the entire project remains well-organized, goal-oriented, and strictly aligned with both technical requirements and business expectations throughout the entire development lifecycle, reducing waste and maximizing value delivery.

In this phase, a series of formal meetings, structured consultations, and collaborative discussions were conducted with the key stakeholders of Kape Tubong, including the business owner, Bryan Tacayon, together with selected operational personnel, branch managers, and administrative representatives. These engagement sessions were specifically intended to gather comprehensive and detailed information regarding the organization's core business processes, daily operational workflows, existing reporting procedures, current data management practices, and long-term business objectives. Through these direct interactions and open dialogues, the researchers were able to identify and document the key challenges and inefficiencies currently affecting the organization, such as scattered and inconsistent sales records, decentralized reporting systems that delay information access, lack of unified data storage, difficulty in monitoring and comparing branch performance, and limited analytical visibility across all kiosk locations.

The planning discussions also focused extensively on understanding how the management team currently monitors sales performance, tracks product demand and popularity, evaluates the efficiency of branch operations, and makes strategic business decisions based on available data. By deeply analyzing the organization's current operational environment, information flow, and decision-making culture, the researchers gained valuable insights into the specific types of information, performance indicators, reports, and analytical tools that management considered most important and useful for improving business performance, increasing profitability, and enhancing overall operational efficiency. This understanding allowed the research team to ensure that all planned system functionalities and features were directly designed to address verified organizational needs and actual pain points, rather than relying on assumptions, generalized solutions, or unnecessary features that do not add practical value.

Based on the comprehensive information gathered during consultations, interviews, and preliminary assessments, the researchers formally defined and documented the project goals, specific objectives, expected outputs, and intended system capabilities in detail. The primary objective established during this phase was the successful development of a centralized Business Intelligence platform capable of consolidating sales and operational data from different sources, generating interactive and visual analytical dashboards, producing accurate and automated reports, and identifying meaningful revenue patterns and trends to support data-driven decision-making within Kape Tubong. Furthermore, the researchers also defined clear and measurable project outcomes and success criteria to objectively evaluate whether the developed system successfully achieved its intended purpose and delivered the expected benefits upon completion and deployment.

Functional requirements were carefully identified, analyzed, and documented during the planning phase to serve as the primary basis for system design, architecture, and development work. These functional requirements included comprehensive dashboard visualization for real-time sales monitoring, dynamic branch comparison tools, detailed product performance analysis, automated report generation and distribution, advanced data filtering mechanisms, revenue trend analysis across different periods, secure user account management and access control, data export capabilities for external use, and real-time or near real-time data presentation. The researchers also identified the critical need for powerful analytical features capable of presenting complex business insights through clear charts, comparative graphs, organized tables, and summary indicators that can assist management in evaluating operational performance, identifying opportunities, and making informed decisions.

In addition to functional requirements, the researchers also identified and prioritized important non-functional requirements necessary to ensure the high quality, reliability, and effectiveness of the system once deployed. These included usability and user-friendliness, accessibility for different user roles, scalability to support future growth, reliability and data accuracy, maintainability for future updates, system responsiveness and speed, robust security measures, and full compatibility across different devices and web browsers. The planning phase placed strong emphasis on designing a user-friendly system that can be easily understood and operated by employees regardless of their technical expertise or previous familiarity with business intelligence platforms. Security considerations were also thoroughly discussed and planned to ensure that sensitive business information, transaction records, and operational data would be fully protected through appropriate authentication protocols and strict access control mechanisms.

The researchers further determined and evaluated the necessary technical resources, suitable programming languages, required software tools, appropriate development frameworks, database technologies, and reliable hosting environments needed for the successful implementation, operation, and maintenance of the system. This process involved evaluating available technologies based on critical factors such as compatibility with existing systems, scalability for future expansion, development efficiency, overall system performance, long-term maintainability, and cost-effectiveness for the organization. The careful selection of appropriate technologies and tools during this phase was important and strategic to ensure that the system could operate efficiently and securely while remaining flexible and adaptable to future organizational growth, new requirements, and business expansion.

A detailed project timeline and development schedule were also created and structured during the Planning Phase to organize all development activities into manageable Agile iterations or sprints. The researchers divided the entire project scope into smaller, focused development cycles to support continuous progress evaluation, incremental feature delivery, and iterative system improvement as recommended by the Agile methodology. Each sprint was assigned specific objectives, clear tasks, defined deliverables, and estimated durations to ensure that development activities remained structured, measurable, and trackable throughout the implementation process. This iterative and phased approach allowed the researchers to monitor progress more effectively, identify issues early, and make necessary adjustments whenever required based on stakeholder feedback or emerging business requirements.

Feature prioritization was another essential and strategic activity conducted during the planning stage. The researchers carefully evaluated and identified which system functionalities and modules were most critical and valuable to the daily operational needs and strategic goals of Kape Tubong, and determined the logical order in which these features would be developed and delivered. Essential core modules such as secure user authentication, central dashboard analytics, sales data management and storage, and basic report generation were prioritized to ensure that the most important functionalities of the platform would be available and functional early in the development process. This prioritization strategy allowed the researchers to deliver working and functional system components incrementally while continuously evaluating, testing, and improving the platform throughout the Agile development lifecycle.

The planning phase also served as a structured communication channel and comprehensive documentation framework that aligned all involved parties—including the research team, developers, business owner Bryan Tacayon, and key operational personnel of Kape Tubong—toward a unified vision and shared objectives, establishing formal processes for exchanging ideas, clarifying expectations, and reaching consensus through collaborative meetings, requirement workshops, and consultation sessions. Every agreed-upon element, including detailed project goals, functional and non-functional requirements, priority rankings for features and modules, development schedules, resource allocation plans, risk management strategies, and the complete list of expected deliverables, was systematically captured, organized, and formally recorded into official project documents. These materials acted as a reliable, single source of reference accessible to both stakeholders and the development team throughout the entire Agile development lifecycle, ensuring that everyone involved maintained the same clear understanding of the system's intended purpose, functionality, boundaries, and timeline. By keeping all specifications and agreements well-defined and documented, the phase helped sustain consistent project direction, kept development activities focused on the actual operational needs of the business, and significantly reduced the risk of confusion, misinterpretation, or conflicting assumptions, while effectively minimizing misunderstandings regarding the system's scope, limitations, and expected capabilities. Additionally, this documentation became an essential tool for tracking progress, managing changes, and verifying that all work remained within agreed parameters, ultimately fostering smoother collaboration and enabling more effective decision-making across every stage of the project.

Through comprehensive and structured preparation, the researchers formulated a detailed and actionable roadmap that served as the primary guide for all subsequent Agile development cycles and technical activities related to the BREWINSIGHT BI: Revenue Pattern Analytics System. This phase played a vital role in reducing uncertainty and risks associated with system building, optimizing the allocation and utilization of available resources, enhancing coordination and communication among team members, and ultimately increasing the probability of successful project completion. By explicitly defining core objectives, development priorities, technical strategies, quality standards, and operational expectations at the very beginning, the team ensured that the entire development lifecycle remained organized, focused, efficient, and strictly aligned with the unique business needs and operational environment of Kape Tubong. The Planning Phase functioned as a critical foundation that directly supported the successful implementation of the proposed business intelligence platform. It provided the essential strategic direction, clear organizational structure, and robust operational framework required to design and develop a system that is reliable, scalable, and centered on user needs. The groundwork established during this stage guaranteed that the final solution would effectively support data-driven decision-making processes, improve overall operational efficiency, deliver accurate and timely insights, and serve as a valuable asset contributing significantly to the long-term growth, stability, and continued success of Kape Tubong.

Analysis Phase: The Analysis Phase acts as the core investigative stage within the Agile Development Model for creating the BREWINSIGHT BI: Revenue Pattern Analytics System. Its primary objective is to systematically collect, deeply examine, and fully interpret all operational, technical, organizational, and strategic details that are essential to the project's success. This stage goes beyond simple information gathering; it represents a thorough diagnostic process intended to reveal the underlying dynamics of how Kape Tubong functions, how data moves through the company, and exactly what tools and insights the business needs to improve its decision-making capabilities. By immersing themselves in the daily operational environment, the research team develops a complete and detailed understanding of the organization's existing workflows, persistent operational difficulties, current methods of data management, and specific analytical needs before any design or programming work begins. This rigorous analytical approach guarantees that the system being developed is based entirely on verified facts, real-world conditions, and clearly stated stakeholder requirements, rather than relying on assumptions, generic industry standards, or theoretical solutions that may not suit the specific context of the business. Every insight gathered during this period serves as a fundamental building block, ensuring the final product will be technically robust, functionally relevant, and perfectly aligned with both the strategic goals and daily activities of Kape Tubong.

During this stage, the researchers carried out a precise, step-by-step evaluation of Kape Tubong's entire operational workflow. Special focus was placed on how sales transactions are recorded, processed, and stored; how financial and operational reports are compiled and shared; how the performance of individual kiosk locations is tracked and assessed; how product movement, demand levels, and sales trends are monitored; and how overall branch activities are managed and reviewed. Every existing procedure, manual task, and informal practice was thoroughly examined, mapped out, and documented to create a clear representation of the current operational state. This process involved tracing the lifecycle of data—from the moment a sale occurs at a kiosk, through its entry and storage, to its final use in reports and management decisions. By understanding exactly how business information is captured, transmitted, saved, processed, and communicated among staff members, branch managers, and top-level leadership, the team was able to identify significant inefficiencies and limitations within the current setup. These challenges included major delays in report preparation due to manual work, sales records that were scattered and disconnected across different locations or formats, inconsistent data structures between branches leading to compatibility issues, frequent duplicate entries or data discrepancies, very limited ability to analyze performance deeply, and the complete lack of real-time monitoring functions and centralized reporting systems that could offer a unified view of the entire business.

The research team also conducted an in-depth review of how employees and management currently access, interpret, and apply available business data for routine monitoring, performance evaluation, and strategic planning. This involved observing and understanding the tools, methods, and thought processes used by staff to derive meaning from raw figures. Specific attention was given to identifying exactly how sales information is recorded into systems, where and how it is stored physically or digitally, how it is retrieved when needed for reporting or analysis, and the specific techniques used to interpret and examine this data across different kiosk branches and locations. The team investigated how frequently data is used, the types of questions management seeks answers to, and the difficulties they face when trying to extract meaningful insights. By fully grasping these operational processes, information flows, and daily routines, the researchers were able to pinpoint critical bottlenecks, repetitive manual tasks that consume time and are prone to human error, and specific areas where automation, data consolidation, and business intelligence features could directly and substantially improve operational efficiency, enhance data accuracy, increase overall productivity, and accelerate the quality and impact of decision-making across the organization.

To support and enrich this analytical process, structured one-on-one interviews, formal consultations, and open group discussions were held with a wide range of stakeholders. This began with the business owner, Bryan Tacayon, and extended to branch managers, administrative staff, sales supervisors, inventory controllers, and selected frontline employees who are directly involved in sales recording, financial reporting, inventory management, and daily operational oversight. These interactions were designed not only to gather technical requirements but also to capture the human perspective regarding the current and future system. Participants were asked to describe their daily responsibilities, the tools they currently use, and the obstacles they encounter. These engagements provided valuable, firsthand insights concerning user expectations, recurring operational challenges, mandatory reporting requirements set by management or accounting standards, and the specific types of analytical information—such as peak sales periods, top-performing products, or comparative branch performance—needed by leadership to support effective business decisions and long-term planning. Contributors were encouraged to speak openly about the difficulties, errors, and delays they face when managing business records, generating required reports, monitoring daily sales results, and analyzing operational trends, ensuring that every viewpoint, concern, and suggestion was captured, documented, and carefully considered in the analysis.

The consultation sessions also focused extensively on determining the exact types of information, metrics, and performance indicators deemed most important for evaluating business performance, measuring success, and guiding strategic planning and growth initiatives. Management provided detailed input regarding desired dashboard features, preferred report formats and layouts, essential tools for tracking revenue, necessary functions for comparing branch performance, and specific performance indicators that are vital for effective decision-making and operational oversight. Discussions covered how "success" is defined for the business, which numbers are monitored most closely, and what insights are currently missing that would deliver significant added value. Through these detailed conversations and requirement-gathering activities, the researchers established a clear and practical understanding of the specific needs, preferences, and expectations that the BREWINSIGHT BI system must satisfy. This included understanding not just *what* information is required, but *how* it should be presented, *when* it needs to be available, and *who* should have access to it, ensuring the system would serve as a truly useful, relevant, and valuable operational tool for the entire organization.

Throughout the Analysis Phase, the research team systematically identified, defined, and categorized the specific data types required for the system’s daily operation, complex processing tasks, and advanced analytical capabilities. These essential data categories were broken down into granular details and included comprehensive records of sales transactions (date, time, amount, payment method), complete product details and pricing structures, inventory levels and stock information, unique branch identifiers and location details, precise timestamps for all activities, customer order details and preferences, exact revenue values and totals, product categories and groupings, supplier information, and key operational performance metrics. Beyond simply listing the types of data needed, the team thoroughly analyzed how these diverse elements should be logically organized, securely stored, properly validated to ensure quality, and systematically processed within the system. This was done to guarantee accurate reporting, consistent calculations, and reliable analytical outputs that management can trust and use with confidence. This stage also involved defining the relationships between different data sets to ensure the system could perform complex queries and generate meaningful correlations between variables.

The researchers also performed a close examination of the structure, completeness, consistency, and overall quality of existing business records and historical data. This was done to identify common issues such as missing fields, inconsistent formatting standards across different branches, incomplete entries, duplicate records, and data corruption—all factors that currently undermine reliability and reporting accuracy. This critical analysis was necessary to establish appropriate rules for data validation, standard procedures for formatting, and logical structures for organization that would support accurate, reliable, and meaningful business intelligence processing within the new platform. Procedures for data standardization and transformation rules were also identified and documented during this phase. These ensure that information collected from different branches, recorded using different methods, and sourced from various operational areas can be effectively combined, unified, and analyzed consistently across the entire platform. This process eliminates discrepancies and ensures that a single, reliable version of the truth is accessible to all authorized users.

User roles, specific responsibilities, and appropriate access permissions were also analyzed in great depth to determine exactly what levels of system access and functionality should be assigned to different user groups. These groups included system administrators, general managers, branch managers, sales personnel, inventory staff, and other authorized users. The research team carefully mapped out the organizational hierarchy and existing workflows to identify which individuals or roles should have permission to view sensitive financial reports, modify stored data, generate advanced analytics, manage user accounts, configure system settings, or perform administrative functions within the platform. This detailed analysis directly supported the design and development of robust access control mechanisms and security layers. These measures protect confidential business information and sensitive records from unauthorized access or modification, while simultaneously ensuring that every user can easily access the specific tools, reports, and data necessary for their respective roles and responsibilities without unnecessary restrictions or barriers.

Functional requirements identified during the analysis were further refined, validated, and translated into detailed specifications that served as official references and guidelines for the subsequent design and development stages. These clearly defined requirements included system capabilities related to interactive and customizable dashboard analytics, automated and scheduled report generation, dynamic filtering and segmentation of sales data, comprehensive monitoring and comparison of branch performance, advanced data visualization tools, secure user authentication and account management, precise revenue tracking and forecasting, and standardized automated reporting for business operations. In addition to these core functions, important non-functional requirements—often critical to user adoption and long-term success—were also documented and prioritized. These included ease of use and user-friendliness, system responsiveness and speed, scalability to support future growth, compatibility across different platforms and browsers, long-term maintainability and supportability, and robust security measures to ensure data privacy and integrity. These specifications guided the technical architecture and development of the entire platform.

To better understand and illustrate the organization's operational structure, information flow, and workflow processes, the researchers created detailed workflow diagrams, process models, data flow diagrams, and operational flowcharts. These visual tools represented both the current manual business procedures and the proposed future environment enabled by the new system. These representations were essential communication tools that helped clarify exactly how information moves through the organization, where delays or errors currently occur, and how the new system would streamline operations, improve data accessibility, and enhance overall efficiency. They served as a shared language between the technical development team and business stakeholders. Documenting both the current state and the desired future state also allowed the researchers to clearly identify specific operational improvements, time savings, error reductions, and accuracy gains that could be achieved through process automation and centralized data analytics. This effectively demonstrated the tangible value the new system would deliver to the organization.

The Analysis Phase also provided the opportunity for the researchers to carefully verify the feasibility of all proposed system features and capabilities within the actual technical and operational environment of Kape Tubong. Critical factors were assessed and evaluated, including the reliability and bandwidth limitations of internet connectivity, compatibility of the system with various hardware devices and operating systems, stability of network connections between branches, the capacity of existing technological infrastructure and software compatibility, and the level of employee readiness, digital literacy, and ability to adopt a new digital system. By identifying these limitations, constraints, and capabilities early in the process, the team was able to make informed adjustments to development priorities, implementation strategies, and system architecture before entering the formal design phase. This ensured that the final solution would function effectively within the given environment, avoiding the development of features that would be technically impossible or impractical to implement given the available resources and infrastructure.

This stage also enabled the researchers to identify relevant operational constraints, business rules, internal policies, and dependencies that could influence how the system is implemented, used daily, and maintained over time. The analysis considered how organizational policies, internal approval processes, reporting hierarchies, and branch management structures might affect access controls, reporting workflows, and data management processes within the platform. Understanding these organizational factors, cultural norms, and operational dynamics was crucial. It helped ensure that the system design would naturally align with the actual structure, policies, and operational culture of the business, rather than forcing the business to adapt to the system—a common cause of resistance or low adoption rates. The goal was to ensure the system supports how the business works, rather than dictating changes to established processes.

Requirement prioritization was another significant and strategic activity carried out during the Analysis Phase. The research team systematically categorized and ranked all identified system requirements based on multiple criteria: operational importance and necessity, stakeholder urgency and demand, technical complexity and risk involved in implementation, development effort and cost, and the overall impact and value delivered to the business. Essential core functionalities—such as real-time sales monitoring dashboards, comprehensive revenue analytics, automated report generation, and centralized data management—were assigned the highest priority. This ensured that the most critical capabilities of the platform would be addressed and delivered during the early stages of Agile development. This structured approach to prioritization directly supported more efficient sprint planning, resource allocation, and incremental delivery of the system throughout the development lifecycle, ensuring that value is provided to the business as quickly as possible.

The Analysis Phase offered an opportunity to look beyond immediate needs and identify potential opportunities for future integration and long-term expansion requirements for the BREWINSIGHT BI system. The researchers examined the organization's growth plans and strategic vision to determine whether the platform might eventually need compatibility or integration with additional digital business tools, advanced inventory management systems, accounting and finance software, customer relationship management applications, or supply chain solutions. This was done in anticipation of the organization growing, expanding its operations, and adopting more digital solutions over time. Considering future scalability, the ability to extend features, and integration needs during the analysis phase helped ensure that the proposed system architecture and technical design would remain flexible, adaptable, and capable of evolving to meet future operational demands and technological advancements. This protects the organization's investment and ensures the system remains relevant long-term.

The team also analyzed and documented the preferred frequency of reporting, how often dashboards should be updated, and the specific formats for data visualization expected by management and operational users. This involved understanding the timing of decision-making cycles: some users required daily sales summaries and immediate real-time updates to manage daily operations effectively, while others preferred weekly, monthly, or seasonal trend analyses and comparisons for strategic planning purposes. Detailed preferences regarding the use of different types of graphs, tables, charts, color-coded indicators, and summary presentations were gathered and recorded. This ensured that the final system would present complex analytical information in a clear, organized, and visually appealing manner that supports rapid interpretation, quick understanding, and effective managerial decision-making. The primary objective was to ensure that the data tells a clear story that is easy for the user to follow and act upon.

Comprehensive and detailed documentation was maintained continuously throughout the Analysis Phase to record every identified requirement, operational finding, stakeholder feedback, workflow observation, technical constraint, and proposed solution. This extensive documentation included requirement specifications, process models, data dictionaries, and feasibility reports. It became an essential reference and guide for the succeeding phases of system design, coding, testing, and implementation, ensuring that every detail and decision was preserved, traceable, and respected. Maintaining accurate, organized, and accessible records also significantly improved communication among researchers, developers, and stakeholders. It ensured that all agreed-upon requirements, project objectives, and expectations were clearly documented, easily retrievable, and fully understood by everyone involved, thereby reducing the risk of misinterpretation or uncontrolled changes to the project scope.

By conducting a thorough analysis of both operational and technical aspects of the organization, the researchers were able to significantly reduce the likelihood of overlooking critical requirements, ignoring important constraints, or developing unnecessary system features that do not add value. This phase acted as a filter, ensuring that only features which solve real problems and address genuine needs were included in the scope. The Analysis Phase strengthened the overall relevance, practicality, and strategic alignment of the BREWINSIGHT BI platform by guaranteeing that every development decision, design choice, and technical specification was grounded in actual business conditions, verified user needs, and clearly defined organizational goals. It minimized the risk of building a system that is technically advanced but functionally irrelevant or unusable within the specific context of the business.

Ultimately, the Analysis Phase played a vital and foundational role in establishing a strong, reliable, and accurate basis for the design and development of the BREWINSIGHT BI: Revenue Pattern Analytics System. Through detailed operational assessment, extensive stakeholder consultation, careful workflow evaluation, and rigorous requirement analysis, the researchers ensured that the proposed platform would be technically feasible, centered on user needs, functionally appropriate, and highly relevant to the unique business environment and operational culture of Kape Tubong. This phase contributed significantly to improving development accuracy, reducing implementation risks, minimizing rework, and increasing the probability of delivering a successful, effective, and valuable Business Intelligence solution capable of supporting informed decision-making, enhancing operational efficiency, and driving long-term organizational growth and success.

Design Phase: During the Design Phase, the research team converts all analyzed business requirements, operational insights, and technical specifications gathered from the previous stage into detailed system blueprints, structural frameworks, and visual layouts that serve as the official guide for constructing the BREWINSIGHT BI: Revenue Pattern Analytics System. This stage acts as the essential link between analysis and actual development, transforming abstract needs and functional expectations into concrete technical solutions and visual plans. Through careful, systematic planning, the team ensures that the resulting platform will be fully functional, efficient, scalable, visually organized, and completely aligned with the operational standards and specific requirements of Kape Tubong before any coding work begins. In this stage, the team defines the overall system architecture, establishing the structural foundation and detailing how every major component—including the user interface, backend logic, database system, analytics engine, and reporting modules—will interact and operate as one unified solution. The architectural design clearly maps how data flows through the system, how user requests are processed, and how information is retrieved, analyzed, and presented to users in a meaningful format. By establishing these connections and pathways early on, the researchers guarantee smooth communication between all parts of the system, ensuring high performance, reliability, and stable operation under various conditions and workloads.

The team also develops the complete database structure specifically built for the platform, a process that involves designing the exact layout of tables, defining data fields and types, setting unique identifiers and relationships, and establishing rules to ensure information is stored securely and accurately. Key data categories such as sales records, product details, branch information, user profiles, inventory levels, timestamps, and financial summaries are carefully organized and structured to support precise storage, fast retrieval, and complex analysis. Relationships between data tables are created to maintain consistency across the system, reduce unnecessary duplication, and significantly speed up queries and report generation. To better visualize and validate this structure, Entity-Relationship Diagrams (ERDs) are created to show exactly how different pieces of information relate and depend on one another, confirming that the design can handle the complex reporting and analysis needs of the business. Additionally, Data Flow Diagrams (DFDs) are developed to illustrate the full lifecycle of information as it moves between users, processes, storage, and modules, serving as vital visual aids that clarify internal operations and guide developers during the building phase.

Another major activity involves creating detailed wireframes, working prototypes, and high-fidelity mockups for the dashboard and the entire user environment. The team carefully plans the layout, navigation, and visual organization to ensure complex data is presented clearly, logically, and efficiently. Interfaces are arranged using charts, graphs, tables, summary cards, menus, filters, and visualization tools designed to make interpreting business data and operational reports simple and intuitive. Special attention is given to arranging elements to reduce confusion, improve readability, highlight critical information, and provide quick access to essential features. The dashboard design is fully customized to fit the specific needs of Kape Tubong, where performance trends, branch comparisons, product rankings, and sales totals are displayed using the most suitable visual formats—such as line graphs for tracking changes over time or bar charts for side-by-side comparisons. Key performance indicators and summary figures are prominently featured to immediately show vital metrics like total revenue, top-selling items, and branch standings, allowing users to understand the current operational status at a glance.

Navigation paths and user interaction flows are thoroughly planned and tested conceptually to ensure the platform remains intuitive, easy to learn, and simple to operate for all users, regardless of their technical skill level. The team determines exactly how users will move between pages, access different reports, apply filters, retrieve specific data, and interact with dynamic elements. Menu structures, button placement, search features, and navigation hierarchies are designed to minimize complexity, reduce the number of steps needed to complete tasks, and improve overall ease of use. Adaptability and accessibility are also central priorities; the system is designed to work effectively across various screen sizes, devices, and operating systems, including desktops, laptops, tablets, and mobile phones. Layouts are built to automatically adjust to different screen dimensions while maintaining clarity and full functionality. Universal design principles are applied to ensure the system remains accessible and usable for a wide range of people with varying levels of experience and preferences. Furthermore, strong security measures are built directly into the design to protect sensitive business data and control access throughout the platform. Secure login systems, permission-based access levels, protected data routes, and user management structures are planned to ensure only authorized personnel can view specific reports or perform certain actions. Roles such as administrators, managers, and staff are assigned appropriate access rights based on their responsibilities, while additional protections like password rules, session controls, and data encryption are included in the architecture.

The team also defines the internal logic that drives data processing, report generation, and automated calculations within the system. This includes detailed planning of how raw transaction data will be collected, organized, filtered, and transformed into useful insights and actionable information. Calculation rules for identifying trends, comparing branches, summarizing revenue, and measuring product performance are carefully outlined and documented to ensure absolute accuracy and efficiency during operation. Automated reporting workflows are designed to reduce manual work, eliminate human error, and speed up the delivery of business insights to management. During this phase, the researchers also prepare comprehensive technical documentation covering system modules, data exchange protocols, database interactions, and component behaviors to guide developers during implementation. These documents ensure consistency in coding standards, smooth integration of parts, and easier long-term maintenance. Standardized naming conventions, file structures, and development guidelines are established to support scalable and sustainable software practices. Performance optimization is also integrated from the start; the team plans efficient database queries, data caching methods, streamlined retrieval processes, and optimized workflows to maximize speed and minimize resource usage. By addressing performance early, the team ensures the platform can handle growing volumes of data and increasing user activity without slowing down or becoming unstable.

Design validation is conducted throughout this stage by presenting mockups, prototypes, and workflow models to stakeholders and intended users for feedback. The business owner, Bryan Tacayon, along with selected staff and operational representatives, reviews the proposed layouts, navigation systems, and dashboard designs to confirm they meet operational expectations and user preferences. Input gathered during these reviews helps identify areas where the design, flow, or visual presentation may need improvement before actual coding begins. This iterative review process significantly reduces the risk of costly changes or major redesigns later in development by spotting usability issues or mismatched requirements early. Based on feedback, the team may revise layouts, adjust navigation, change color schemes, or modify how data is displayed to improve clarity and the overall user experience. This collaborative approach ensures the system remains user-focused and practically suited to daily operations. The Design Phase also includes defining consistent visual themes, styling standards, typography, and iconography to create a professional and unified look across all parts of the system. Error messages, alerts, notifications, and confirmation prompts are also designed to improve communication between the system and users, helping to prevent mistakes and guide actions. Additionally, the team plans for future growth and expansion; the architecture is built in a modular and flexible way so that new features, additional branches, advanced reporting tools, or connections to other business software can be added later without requiring a complete structural overhaul. This forward-thinking approach ensures the BREWINSIGHT BI platform can evolve alongside the growth and changing needs of Kape Tubong.

Detailed design documentation is continuously updated and maintained throughout the phase to record all interface specifications, workflow diagrams, technical plans, database structures, security measures, and architectural decisions. These documents serve as essential references for developers during coding, testing, deployment, and future maintenance. Proper documentation also improves communication and collaboration among team members and stakeholders by ensuring all design choices and technical details are accurately recorded, easily accessible, and clearly understood by everyone involved. Through comprehensive and structured planning, the team establishes a clear roadmap for development that supports organized, efficient, and high-quality work. The Design Phase plays a major role in reducing potential errors, avoiding inefficiencies, improving usability, and ensuring a stable foundation long before any programming takes place. By carefully designing both the underlying technical structure and the user-facing interface, the researchers ensure the final system will be technically sound, visually consistent, operationally effective, and perfectly aligned with the analytical and management needs of Kape Tubong. Ultimately, this stage is critical in preparing the BREWINSIGHT BI: Revenue Pattern Analytics System for successful implementation by providing clear blueprints, organized workflows, and user-centered designs. It guarantees that the finished platform will not only deliver accurate and meaningful business intelligence but will also be practical, scalable, secure, and easy to use within the real working environment of the organization.

Implementation: The Implementation Phase represents the most extensive and hands-on stage in the development lifecycle of the BREWINSIGHT BI: Revenue Pattern Analytics System, marking the point where planning and design transition into tangible software creation. This phase encompasses the complete construction, coding, system integration, and detailed configuration of the platform, strictly following the approved designs, technical specifications, and functional requirements established during the earlier stages of the Agile Development Model. While previous phases focused on analysis, architecture, and interface planning, this stage serves as the critical bridge that turns conceptual models, workflow diagrams, and system blueprints into a fully operational Business Intelligence solution tailored specifically for the analytical and operational needs of Kape Tubong. Through systematic, structured, and technically rigorous development activities, the research and development team translates the carefully designed system architecture, user workflows, database structures, and interface layouts into functional software components that operate together as a unified, web-based application accessible securely to authorized users across various devices and locations.

Development work follows an incremental approach using Agile sprint cycles, a methodology that divides the project into short, focused development periods—typically lasting one to two weeks—where specific sets of features or modules are built, reviewed, and tested. This iterative framework allows functionalities to be completed, validated, and improved continuously throughout the development timeline, rather than delivering everything at once. The flexible nature of Agile implementation supports progressive refinement and the immediate integration of feedback, ensuring that the evolving system remains consistently aligned with stakeholder expectations, business goals, and operational realities as development advances. Every sprint targets distinct modules or capabilities, enabling the team to prioritize essential system components first, break complex tasks into manageable segments, adhere to realistic timelines, and maintain steady progress without compromising quality or functional integrity.

Development begins with the creation and refinement of all front-end components, which form the direct layer of interaction between the system and its users. These elements include the central dashboard interface, organized navigation menus, advanced data filtering tools, interactive user input forms, structured report displays, dynamic graphical charts, informative summary cards, and comprehensive visualization panels. The team codes and styles these interface elements using modern web technologies and responsive design principles to ensure they are visually consistent, logically organized, and fully functional across different screen sizes, browsers, and devices. Special attention is given to usability and accessibility, creating an intuitive environment where users can navigate efficiently, locate information quickly, and interpret complex business data clearly—even without advanced technical knowledge. The dashboards are programmed to display key insights through dynamic charts, comparative graphs, detailed tables, and statistical summaries that update in real time, while interactive features such as custom date selectors, search tools, and sortable reports allow users to customize views and analyze data according to their specific needs and decision-making requirements.

Parallel to interface development, the team constructs the core back-end functionalities that operate behind the scenes to power the entire platform. This layer handles all data processing, system logic, database communication, and business rule execution, serving as the engine that drives every operation within the system. Back-end development includes programming secure user authentication processes, defining role-based access controls, managing user sessions, automating report generation, performing complex analytical computations, and executing all database operations such as data storage, retrieval, updates, and deletion. The team builds secure, structured, and efficient server-side processes that govern the flow, integrity, and analysis of business information, ensuring that all actions are performed accurately, reliably, and in full compliance with data security standards and privacy requirements.

A central and technically complex part of this phase is the development of the analytics processing and data aggregation logic that transforms raw data into meaningful intelligence. The team designs and implements specialized algorithms and processing workflows that convert basic sales records, transaction logs, inventory entries, and operational inputs into actionable insights. These automated functions include calculating revenue trends over time, identifying top-performing and underperforming products or categories, generating comparative metrics across different branches, analyzing daily and seasonal sales patterns, monitoring operational efficiency, and pinpointing peak business hours or high-demand periods. Specific modules are built to evaluate performance changes across daily, weekly, monthly, or annual periods, helping management track growth, spot seasonal shifts, and forecast future outcomes. Additional features identify customer purchasing behaviors and operational differences between locations, highlighting strengths, weaknesses, and opportunities for improvement or expansion. Alongside these, automated reporting tools are integrated to minimize manual work, reduce human error, and ensure that standardized summaries, sales reports, and analytical outputs are generated accurately, consistently, and available on demand or through scheduled delivery.

Database integration and optimization form another critical component of this phase, ensuring that the platform can store, retrieve, and process data efficiently and reliably. The team connects the application to the designed database structure, establishing seamless communication that supports real-time or near real-time data handling. Robust connectivity functions are implemented to validate all incoming information, secure storage processes, maintain data relationships, and ensure that every entry is immediately reflected in dashboards and reports. Furthermore, the team optimizes database queries, indexing strategies, and data retrieval operations to enhance processing speed, reduce server load, and minimize loading times, ensuring a smooth and responsive user experience even when managing large volumes of historical or transactional data. Throughout development, every module is tested incrementally; after completing a component, internal validation verifies correctness, functionality, and compatibility with existing features. This early testing helps identify programming errors, logical gaps, or integration issues before they become difficult to resolve, allowing the team to document, correct, and retest solutions to maintain system stability and quality.

The iterative structure also supports active collaboration with stakeholders—including the business owner, Bryan Tacayon, and selected operational staff—who review partially completed modules and provide feedback before full completion. This engagement ensures that technical outputs align closely with real business needs and workflows, allowing adjustments and refinements to be made efficiently without requiring major redevelopment later. Formal sprint review meetings are held after each cycle to assess progress, discuss concerns, and plan upcoming tasks. Additional activities include code optimization and refactoring to improve efficiency, maintainability, and scalability, alongside strict version control practices to manage changes, coordinate work among developers, and preserve project history. Comprehensive error logging and debugging mechanisms are built in to support future maintenance, while necessary third-party libraries, frameworks, and tools are integrated to enhance visualization, responsiveness, and overall capability. Security measures such as encryption, input validation, and access controls are embedded throughout, and performance testing ensures the system remains stable and responsive under operational workloads. Detailed technical documentation is maintained continuously to record all features, configurations, and decisions, serving as a vital reference for testing, maintenance, and future updates. This stage successfully transforms the approved designs and specifications into a fully functional, secure, responsive, and reliable Business Intelligence platform ready for evaluation and deployment within Kape Tubong, ensuring that the system not only meets technical standards but also delivers meaningful analytical power, operational efficiency, and practical value that supports informed decision-making and long-term organizational growth.

Testing Phase: The Testing Phase represents a comprehensive and rigorous evaluation process carried out to thoroughly examine the accuracy, reliability, functionality, security, usability, performance, and overall operational readiness of the BREWINSIGHT BI: Revenue Pattern Analytics System before its final deployment and official implementation within Kape Tubong, serving as the primary quality assurance mechanism designed to confirm that the fully developed platform operates strictly according to its intended design specifications, performs efficiently and consistently under a wide range of operational conditions, and satisfies all established technical standards as well as the specific expectations and requirements of the end-users and stakeholders. Through systematic, methodical, and repeated evaluation procedures, the researchers verify that every integrated system component functions correctly, that different modules interact with one another seamlessly and without conflict, and that the platform consistently provides dependable, precise, and actionable analytical outputs suitable for guiding real-world business operations, performance monitoring, and strategic decision-making processes, ensuring that every part of the system works together harmoniously to deliver the intended value and functionality as defined during the earlier stages of analysis and design.

During this critical stage, the researchers perform multiple forms of specialized testing, each designed to assess different aspects of the platform’s behavior, capability, performance, and stability, beginning with extensive functional testing to ensure that every feature, module, tool, and internal process within the system performs exactly according to its defined purpose and predetermined requirements outlined during the analysis and design phases, where individual functionalities such as secure user login and authentication, dynamic dashboard visualization, comprehensive report generation and export, multi-layer data filtering and segmentation, side-by-side branch comparison, detailed sales analytics and breakdowns, accurate revenue tracking and computation, and product performance monitoring are carefully examined, validated, and verified to confirm that they produce precise, complete, consistent, and expected results in every possible test scenario and operational context, covering all possible use cases to ensure that nothing is overlooked and that every function behaves exactly as intended under all circumstances.

The researchers specifically verify that dashboard visualizations—including charts, graphs, tables, and summary indicators—display real-time, updated, and accurate analytical data retrieved directly and securely from the central database, while revenue calculations, daily, weekly, and periodic sales summaries, advanced filtering operations, dynamic chart updates based on user input, and automated report generation processes are tested thoroughly and repeatedly under various conditions to ensure mathematical correctness, logical consistency, stability over time, and absolute reliability, confirming that the platform possesses the full capability to accurately transform raw transactional records, inventory data, and operational information into meaningful, structured, and valuable business insights that effectively support managerial analysis, performance evaluation, trend identification, and high-level decision-making activities, ensuring that the information presented is not only correct but also relevant and useful for the intended business purposes.

Integration testing is also performed extensively during this phase to assess whether the different modules, subsystems, and technical components of the BREWINSIGHT BI platform interact properly, exchange data correctly, and operate harmoniously as a complete and unified system rather than isolated parts, where the researchers evaluate the communication links, data exchange protocols, and processing workflows between the front-end user interface, back-end application logic and processing rules, core analytics engine, database management system, and reporting modules to ensure smooth, synchronized, and error-free operation, helping identify potential issues related to data flow interruptions, module compatibility conflicts, synchronization delays, broken functionalities, data duplication, or communication failures between interconnected system components, ensuring that data moves seamlessly and accurately from entry to analysis to reporting without any loss, corruption, or misinterpretation along the way.

The platform is further tested under various simulated operational and usage scenarios to evaluate overall system responsiveness, long-term stability, scalability, and output consistency, as the researchers simulate common and intensive business activities such as multiple concurrent dashboard requests from different users, repeated complex filtering and sorting operations, simultaneous user access from different locations and branches, bulk report generation and data export, and continuous real-time data retrieval and refresh processes to determine whether the platform remains stable, responsive, fast, and accurate during both regular daily usage and peak operational periods, helping assess whether the system can maintain acceptable performance levels, data integrity, and operational reliability under both normal daily and demanding operational conditions, ensuring it will not fail or slow down when needed most or when the business is operating at its highest capacity.

Performance testing is conducted specifically to evaluate the processing speed, operational efficiency, and resource management of data handling, dashboard rendering, analytics computation, and report generation capabilities within the system, where the researchers objectively measure and record how quickly dashboards load and refresh, how efficiently large datasets are processed, how rapidly reports are compiled and generated, and how responsive the platform remains when managing complex queries or multiple simultaneous user requests, with these evaluations being important to ensure that users can interact with the platform efficiently and productively without experiencing significant delays, timeouts, freezing, or interruptions that could negatively affect operational workflow, productivity, or user confidence in the system, ensuring that the system performs well even as the volume of data and number of users grow over time.

Compatibility testing is likewise carried out to verify that the BREWINSIGHT BI platform functions consistently, correctly, and uniformly across different web browsers, operating systems, hardware devices, screen sizes, and technical environments, as the researchers assess whether users accessing the system through desktop computers, laptops, tablets, or mobile devices experience uniform interface behavior, equal responsiveness, identical data presentation, and complete functionality regardless of the hardware or software used, while browser compatibility testing ensures that visual elements, navigation structures, interactive components, and analytical displays render correctly and operate properly regardless of the software environment being used, broadening system accessibility and significantly improving deployment readiness across different operational settings and branches within Kape Tubong, ensuring all users have equal access and capability no matter what technology they use to connect.

Security testing is another essential and high-priority activity conducted during the Testing Phase to protect sensitive organizational information, financial records, and confidential business data, as the researchers evaluate the effectiveness and robustness of authentication mechanisms, password protection policies, session management protocols, role-based access controls, data encryption, and input validation procedures implemented within the system, helping identify vulnerabilities related to unauthorized access, data manipulation risks, improper authentication handling, exposure of sensitive data, and potential threats to confidential business information, while the researchers strictly verify and confirm that users can only access the specific features, reports, data sets, and system functions appropriate to their assigned roles, responsibilities, and permission levels, preventing data leaks or unauthorized modifications and ensuring that the system complies with best practices for data security and privacy.

The testing process also includes detailed validation of the platform's ability to handle incorrect, incomplete, invalid, or unexpected user inputs appropriately, safely, and gracefully, as the researchers intentionally simulate erroneous entries, invalid credentials, incomplete form submissions, incorrect data formats, and unexpected user actions or navigation paths to assess exactly how the system responds to abnormal situations or user errors, while error handling mechanisms, warning messages, validation prompts, and exception handling routines are examined closely to ensure that the platform prevents data corruption, maintains operational stability, avoids crashes, and provides clear, understandable, and helpful guidance to users whenever issues or mistakes occur, ensuring a safe and frustration-free user experience even when things do not go as planned or when users make mistakes.

Stress testing and load testing may additionally be conducted to determine exactly how the system behaves and performs under unusually high workloads, significantly increased transaction volumes, extremely large datasets, or maximum user capacity, with this form of testing helping identify performance limitations, system bottlenecks, memory usage issues, processing constraints, or breaking points that may only emerge during peak operational conditions or heavy usage periods, and by evaluating the system under demanding and extreme scenarios that exceed normal daily operations, the researchers can assess whether the platform remains stable, functional, and capable of supporting future business growth, expansion, increased data volume, and higher operational requirements as Kape Tubong develops over time, ensuring that the system is built to scale and can handle the business's evolving needs.

Another significant aspect of the Testing Phase involves the rigorous verification of the accuracy, reliability, and validity of the analytical outputs generated by the platform, as the researchers carefully compare dashboard analytics, revenue calculations, sales trend representations, product rankings, branch comparisons, and generated reports against actual raw transactional records, verified business data, and manual calculations to ensure that the system produces valid, precise, and trustworthy results that match real-world figures, with this validation process being particularly critical because management decisions, performance evaluations, financial reviews, and strategic planning may rely heavily on the accuracy and truthfulness of the information generated by the platform, where errors here could lead to poor business decisions, financial losses, or incorrect assessments of operational performance.

User Acceptance Testing (UAT) is also conducted extensively with selected personnel from Kape Tubong, including management representatives, operational staff, branch managers, administrative personnel, and authorized end users who will actually use the system daily, where during this testing activity, participants interact with the system in realistic operational scenarios and perform actual work tasks to evaluate the platform's usability, practicality, relevance, and effectiveness in supporting daily business activities and decision-making needs, as users perform tasks such as accessing dashboards, reviewing detailed reports, filtering sales information, analyzing branch performance, and generating summaries to determine whether the system is intuitive, user-friendly, logically organized, and genuinely beneficial for operational monitoring and management, ensuring that the system is designed around the needs of the people who will use it most.

Feedback gathered during User Acceptance Testing provides valuable insights regarding interface clarity, navigation efficiency, report readability, dashboard organization, speed of operation, visual design, and the overall user experience, as participants may identify usability concerns, suggest interface improvements, recommend additional features, or provide observations regarding system convenience and operational effectiveness, and the researchers carefully document all feedback, suggestions, and concerns raised during this phase and use the findings to refine, adjust, and optimize the platform before final deployment, ensuring that the final product not only meets technical specifications but also aligns with user expectations and preferences, making it more likely to be adopted and used effectively in daily operations.

Regression testing is also conducted systematically whenever modifications, bug fixes, or revisions are implemented during the testing process, ensuring that newly applied corrections, feature updates, or code revisions do not negatively affect existing functionalities or introduce additional errors or issues into the system, helping maintain overall platform stability, data integrity, and operational consistency while supporting continuous improvement throughout the development lifecycle, ensuring that every change made improves the system without breaking anything that was already working correctly or introducing new problems that could affect performance or reliability.

The Testing Phase additionally includes comprehensive documentation of all testing procedures, identified issues, corrective actions applied, revisions made, and final evaluation results, where detailed and organized testing documentation is maintained to provide full transparency regarding the system evaluation process and to support future troubleshooting activities, maintenance tasks, system upgrades, and enhancement projects, with these official records also serving as evidence that the platform has undergone thorough, rigorous, and professional validation prior to being deployed for business use, creating a reliable reference point for all future work and ensuring that the development process is fully accountable and traceable.

Continuous testing and iterative correction cycles are performed repeatedly until the BREWINSIGHT BI platform achieves acceptable performance standards and fully satisfies all defined functional, security, and operational requirements established during earlier phases of development, as problems identified during testing are thoroughly analyzed, corrected through appropriate modifications, and subsequently retested to verify that implemented solutions effectively resolve the detected issues without creating additional complications or side effects, ensuring that every problem found is addressed completely and that the system is continuously improved until it meets or exceeds all established criteria for quality and performance.

Through systematic, rigorous, and repeated evaluation procedures, the researchers ensure that the BREWINSIGHT BI: Revenue Pattern Analytics System is technically functional, operationally stable, secure against threats, responsive to user actions, accurate in calculations, and user-friendly in design before deployment within Kape Tubong, and the Testing Phase significantly contributes to minimizing operational risks, improving system reliability, enhancing user confidence, and validating the overall effectiveness of the platform as a professional Business Intelligence solution, providing the necessary assurance that the system is ready to be trusted with critical business data and decision-making processes.

Ultimately, the Testing Phase confirms that the BREWINSIGHT BI platform is fully capable of delivering meaningful analytical insights, supporting informed managerial decision-making, improving operational efficiency, and meeting the complex real-world operational requirements of Kape Tubong, and by thoroughly evaluating the platform from technical, operational, and user-centered perspectives, the researchers ensure that the developed system is fully prepared, reliable, and ready for practical deployment and long-term organizational use, representing a major milestone in the development process and paving the way for successful implementation and adoption across the business.

Deployment Phase: The Deployment Phase covers all activities required to install, configure, and launch the completed BREWINSIGHT BI platform within Kape Tubong's actual operational environment. During this stage, the system is transferred from the development environment to a live web server or designated hosting platform intended for regular business use. Essential technical tasks such as server configuration, domain linking, security certificate installation, and full database deployment are completed to guarantee reliable online accessibility from authorized locations and devices. The researchers conduct comprehensive verification checks to confirm that all features, connections, and processing logic function correctly under production conditions, mirroring the workflows that will occur during daily operations. Historical sales records, transaction logs, and inventory data are carefully migrated and validated within the system whenever necessary, ensuring that initial reports and analytical insights are based on complete and accurate information. User accounts are created and configured according to defined roles, ensuring that administrators, managers, and staff members only access features relevant to their responsibilities. Structured training sessions are organized and conducted to guide users through system navigation, dashboard interpretation, filtering techniques, and report generation processes. Complete user manuals, quick-reference guides, and instructional materials are provided as permanent resources for ongoing support. Continuous operational monitoring takes place immediately after launch to track performance, identify early concerns, and facilitate smooth adoption during the first days of live usage. This phase marks the official transition of BREWINSIGHT BI from a developed solution to an active business tool, ensuring the system is fully operational, secure, and accessible to intended users. It serves as the critical step that enables real-world application of the platform's intended functions and benefits.

A successful deployment process is designed to minimize operational disruption and maintain business continuity throughout the transition. Detailed rollout plans and schedules are established to ensure users can begin utilizing the platform efficiently with minimal downtime or confusion. The researchers perform rigorous pre-launch assessments to verify server stability, database integrity, data synchronization accuracy, and reliable connectivity across all integrated modules and branches before making the system fully available. Backup strategies and recovery procedures are configured and tested during this stage to protect organizational data against accidental loss, corruption, or technical failures. Deployment logs, performance monitoring tools, and alert mechanisms are implemented to track system health and allow immediate response to any issues detected right after launch. Security configurations including SSL certificates, firewall settings, access control lists, and production-grade authentication protocols are finalized and validated to secure the live environment against potential threats. Initial feedback is actively collected from users during the early days of deployment to identify minor usability adjustments, interface refinements, or additional support needs that can improve the overall experience.

This phase also involves thorough validation to confirm that all integrated analytics, reporting, and visualization features deliver accurate results under real operational conditions. The researchers observe how end users interact with the system within their actual workflows, noting navigation patterns, common tasks, and areas requiring clarification or improvement. Post-deployment technical support is made readily available to assist users in resolving initial technical concerns, answering operational questions, and guiding best practices. Key performance metrics such as page load speed, query response time, data processing efficiency, and system uptime are closely monitored in the live environment to evaluate responsiveness and scalability under real data loads. Comprehensive documentation covering deployment procedures, configuration settings, access policies, and technical specifications is maintained to support future updates, maintenance activities, or system migration projects. Through careful implementation, continuous monitoring, and responsive adjustments, the deployment phase ensures that BREWINSIGHT BI transitions smoothly from development into active business service. This stage operationalizes the proposed system and enables Kape Tubong to begin leveraging data-driven analytics for informed decision-making, performance tracking, and strategic management.

Maintenance Phase: The final stage in the system lifecycle is the Maintenance Phase, which focuses on sustaining performance, resolving issues, and improving the system continuously after deployment. During this phase, the researchers provide ongoing technical support to address challenges encountered during regular real-world use. Software bugs, logical errors, calculation discrepancies, or interface issues reported by users or identified through monitoring are thoroughly investigated, documented, and resolved promptly to maintain operational quality. Continuous performance monitoring is conducted to ensure the platform remains stable, responsive, and efficient even as transaction volumes grow or usage patterns change over time. Critical security patches, library updates, and software upgrades are applied whenever necessary to protect system integrity, close vulnerabilities, and comply with evolving security standards. The researchers actively gather and review feedback from users regarding feature requests, workflow improvements, or usability enhancements that could add greater value to operations. Minor refinements, interface adjustments, and functional additions are implemented based on these insights and evolving business requirements. Compatibility testing is performed regularly to ensure consistent operation across updated web browsers, operating systems, mobile devices, and hosting environments used within the organization. Backup and recovery procedures are monitored, tested, and optimized to guarantee reliable protection against data loss or corruption. Maintenance activities ensure the platform remains fully functional, relevant, and capable of meeting business demands over an extended period. This phase supports the long-term sustainability of the BREWINSIGHT BI platform, ensuring it remains a dependable asset. Through continuous improvement and dedicated support, the system evolves alongside the organization and adapts to future operational and technological changes. Maintenance activities are therefore critical to preserving the effectiveness, accuracy, and usability of the developed system throughout its lifecycle.

Maintenance also presents valuable opportunities for expanding the platform's capabilities as Kape Tubong's operational needs grow and change. As new branches open, product lines expand, or reporting requirements become more detailed, the system undergoes enhancements to accommodate these developments effectively. New data fields, analytical dimensions, or reporting formats are introduced to capture and process additional information without disrupting existing functions. This adaptive approach ensures that BREWINSIGHT BI remains scalable, flexible, and sustainable for long-term organizational use. The maintenance phase includes periodic system audits and reviews to evaluate whether the platform continues to meet defined organizational objectives, performance expectations, and quality standards. Database optimization tasks such as indexing, query refinement, and storage management are performed regularly to maintain fast and efficient data retrieval even as historical records accumulate over time. The researchers analyze system logs, error reports, and usage statistics to identify recurring issues, bottlenecks, or opportunities for technical optimization. Preventive maintenance measures including system health checks, load testing, and security assessments are scheduled to reduce the likelihood of unexpected downtime or performance degradation. User training materials, help resources, and documentation are updated continuously to reflect new features, interface changes, or updated procedures.

This phase also supports the continuous alignment of the system with changing business strategies, goals, and operational workflows. As Kape Tubong introduces new marketing campaigns, pricing structures, or performance indicators, the platform is modified and configured to reflect these evolving requirements accurately. Long-term analysis of user feedback and operational trends helps identify emerging needs that were not evident or defined during initial development and deployment. The researchers evaluate opportunities for integrating advanced capabilities such as predictive analytics modules, automated forecasting tools, or machine learning models to deliver deeper insights and greater strategic value in future updates. Detailed records of all maintenance activities, revisions, configurations, and system updates are carefully documented and archived to ensure full traceability and to facilitate future technical support or development work. Through proactive maintenance, regular optimization, and iterative enhancement, the BREWINSIGHT BI platform remains adaptable, reliable, and strategically valuable to the organization over many years. This stage ensures the continued operational effectiveness, technical stability, and long-term success of the deployed business intelligence system.

3.3 System Architecture

The BREWINSIGHT BI: Revenue Pattern Analytics System is designed using a client-server architecture, which serves as the core framework for the system's web-based operation. This type of architecture separates the platform into client-side and server-side components to improve efficiency, organization, and scalability. The client side refers to the interface that authorized users access through web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge. Through these browsers, users can interact with the system by logging in, viewing dashboards, generating reports, and analyzing business performance data. The browser functions as the presentation layer of the platform, displaying visual analytics such as charts, graphs, tables, and summary metrics. Since the system is web-based, users are not required to install any software on their individual devices. This makes the platform more accessible and convenient for users across different locations and devices. The interface is also designed to be responsive so that it can adapt to various screen sizes and device types. As a result, Kape Tubong management can monitor business performance more flexibly and efficiently. This client-side accessibility improves convenience and supports timely managerial decision-making.In addition, the client-side design prioritizes usability and user experience to ensure that management can navigate the platform with minimal technical difficulty. The interface is structured to present information clearly and logically so that users can interpret business metrics without requiring specialized analytical training. Interactive dashboard controls such as filters, date selectors, and comparison tools are integrated to improve user engagement and analytical flexibility. These features enable management to customize data views according to specific reporting needs or operational concerns. As a result, the client layer functions not only as a display interface but also as an interactive decision-support environment.

On the server side, the platform handles all backend processing, business logic execution, and communication with the database. The system uses Apache Web Server through XAMPP as its local hosting and development environment. Apache is responsible for receiving requests from users through the browser and returning the appropriate data or interface response. Whenever a user requests information such as revenue reports, filtered analytics, or dashboard updates, the server processes the request before sending the results back to the client interface. Backend scripts execute the necessary logic for report generation, sales aggregation, and analytical calculations. The server also manages user authentication and ensures that only authorized personnel can access restricted system functions. By centralizing business logic on the server, the platform maintains consistent processing and improves overall performance. This setup also enhances security because sensitive processes and calculations remain hidden from the client side. In addition, centralized processing reduces the workload on user devices since computation is handled by the server. Overall, the server component serves as the processing engine of the BREWINSIGHT BI platform.

The backend architecture is also designed to support modular processing, allowing individual system functions such as reporting, analytics, and authentication to operate independently while remaining integrated. This modularity improves maintainability by enabling developers to modify or enhance one module without significantly affecting other components. It also facilitates future expansion should additional analytical features or integrations be required. The centralized backend model therefore contributes to both system efficiency and long-term scalability.

To manage data storage, the system utilizes MySQL as its relational database management system. MySQL functions as the centralized storage repository for all business and transactional data used by the platform. It stores records such as sales transactions, kiosk information, product details, timestamps, user accounts, and generated reports. The database is structured into organized relational tables to support efficient storage and retrieval of information. Through SQL queries, the system can process large volumes of data and generate analytical outputs needed for reporting and dashboard visualization. MySQL also supports data filtering, sorting, aggregation, and comparison functions required by the analytics features of the system. Centralized database management ensures that all sales information from Kape Tubong kiosks is stored in one unified source. This eliminates the inconsistencies and duplication commonly associated with decentralized spreadsheets or manual recordkeeping. Security controls are also applied to protect stored data and restrict unauthorized database access. The use of MySQL therefore provides a reliable and scalable data management solution for the system.

Database normalization techniques are also applied during schema design to reduce data redundancy and maintain data integrity across records. Relationships between tables are carefully structured to support efficient querying and consistent data linkage among transactions, products, branches, and user accounts. Backup procedures may also be implemented to preserve critical business records and ensure recoverability in the event of data loss. These database management practices improve reliability and support long-term operational sustainability.

The communication process within the system follows a structured request-response mechanism between the client, server, and database. When a user performs an action through the dashboard interface, the browser sends a request to the server for the required information or operation. The server then processes the request and retrieves relevant data from the MySQL database. Once the data is collected and processed, the server transforms it into a format suitable for visual presentation. The processed results are then returned to the client interface where they are displayed as charts, graphs, tables, or dashboard summaries. This workflow enables the system to provide dynamic and real-time updates based on user interaction. It also allows multiple users to access the system simultaneously while maintaining consistent and synchronized data. Because all requests are processed centrally, the system ensures data accuracy and uniformity across users. This request-response structure is essential for delivering responsive and interactive analytics features. It enables BREWINSIGHT BI to provide timely and accurate business insights to management.

This communication model also improves processing transparency by clearly separating data input, server-side computation, and presentation output into distinct operational layers. Such separation enhances debugging efficiency and simplifies troubleshooting when issues occur. It further ensures that data transformations and analytical calculations are consistently applied regardless of which user accesses the system. This contributes to analytical reliability and reporting standardization across the platform.

The client-server architecture also offers significant operational advantages for Kape Tubong’s multi-kiosk business setup. Since all sales data is processed and stored centrally, management can monitor performance across multiple kiosk locations from one unified platform. This removes the need for manually consolidating reports from different branches and reduces administrative workload. It also improves consistency in reporting because all users access the same centralized dataset. Remote access allows management to review business performance and analytics even when not physically present at the business location. The architecture further simplifies system maintenance because updates and enhancements can be deployed directly on the server without modifying individual client devices. This ensures that all users automatically access the latest version of the platform. The centralized design also supports future scalability, allowing the system to accommodate additional kiosks or expanded analytics features as the business grows. Security and backup processes are easier to manage in a centralized server environment. Overall, the client-server architecture provides a practical, scalable, and efficient technical foundation for the BREWINSIGHT BI platform.

Centralized architecture strengthens organizational control through uniform data processing rules and standardized reporting procedures applied across all kiosk locations. This approach maintains consistency in operational monitoring and analytical interpretation, regardless of differences in branch size or geographic location. It enables precise branch-to-branch performance comparisons, as every dataset undergoes evaluation using identical criteria and calculation methods. This level of uniformity serves as a fundamental requirement for fair, accurate, and reliable performance assessment throughout the business.

The client-server architecture functions as a suitable and effective framework for deploying the BREWINSIGHT BI system. It facilitates centralized processing, secure information management, responsive web access, and efficient support for multiple users simultaneously. This design fits the operational requirements of Kape Tubong perfectly, allowing management to monitor distributed kiosk locations from one central point. The web-based structure ensures broad accessibility, operational convenience, and real-time oversight of all business activities. Using Apache via XAMPP alongside MySQL creates a practical and economical environment for deployment and ongoing maintenance. The structure supports scalability and flexibility, enabling the platform to adjust as business needs change over time. Separating user interfaces, processing logic, and data storage into distinct layers improves overall organization and simplifies maintenance tasks. This design choice increases system stability and ensures analytical outputs remain consistent. The selected architecture delivers the technical structure required for the platform to operate successfully. It builds a solid and lasting foundation for providing meaningful business intelligence and revenue analytics to Kape Tubong.

This architectural setup also prepares the system for future enhancements and integration possibilities. As Kape Tubong expands operations or seeks more advanced analytical features, the modular and centralized design allows for efficient upgrades and expansion. New features or increased capacity can be added without disrupting current processes. This capability ensures the BREWINSIGHT BI platform stays relevant and capable of supporting the organization through growth and technological changes.

3.4 Tools and Technologies Used

- **Frontend:** HTML, CSS and JavaScript
- **Backend:** Python Flask
- **Database:** MySQL
- **Development Tools:** Visual Studio Code and XAMPP

3.5 Data Gathering Techniques

- Structured interviews with owner (Bryan Tacayon) and branch managers
- Observation of sales processes at kiosks and main shop
- Document analysis of sales records, receipts, and manual logs
- Surveys with staff on technology comfort and reporting preferences

3.6 Testing Procedures

To evaluate the quality, reliability, functionality, and usability of the BREWINSIGHT BI: Revenue Pattern Analytics System, the researchers implemented a comprehensive and structured testing process composed of Unit Testing, System Testing, and User Acceptance Testing (UAT). These testing procedures were conducted systematically throughout the development lifecycle and prior to the final deployment of the system. Every component underwent strict assessment to confirm it functions according to intended design and operational purpose. The testing process proved essential in validating whether the developed platform could effectively support the full spectrum of business intelligence and revenue analytics needs of Kape Tubong. Assessments ensured stability, accuracy, efficiency, and usability remained consistent even under the varying demands of actual business operations, including high transaction volumes and continuous daily usage.

Each testing level focused on distinct aspects of the system. Evaluation began with individual modules and progressed toward assessment of the fully integrated platform. This layered strategy enabled the researchers to identify, analyze, and correct errors at an early stage before problems could propagate and negatively affect other parts of the system. Testing occurred incrementally during development, establishing a continuous cycle of evaluation and improvement. This approach progressively strengthened reliability, security, and overall quality of the platform. Issues found during each phase were carefully documented, categorized by severity, and thoroughly investigated. Teams applied corrections through appropriate modifications and performed retesting to confirm solutions resolved detected matters without introducing new complications or disrupting existing features. This systematic process significantly reduced risks of operational failure, minimized software defects, and substantially improved platform readiness for real-world use.

Testing procedures confirmed the system directly addresses operational concerns identified during earlier study phases. These included previously encountered difficulties regarding real-time sales performance monitoring, generation of both standardized and customized reports, analysis of complex revenue trends, consolidation of data from multiple branches, and support for management decision-making through accurate information. Rigorous and repeated verification confirmed the developed solution is not only technically functional and compliant with specifications but also practical, dependable, and perfectly suited to the specific operational environment, workflow patterns, and business culture of Kape Tubong. The process provided objective, measurable evidence regarding system strengths, limitations, and overall performance prior to deployment. Teams completed necessary refinements, optimizations, and improvements well before introducing the platform into regular business activities.

Validation extended beyond technical features to assess accuracy, consistency, and reliability of analytical outputs generated by the system. Since the platform serves as the primary basis for business intelligence activities and data-driven decision-making, researchers confirmed reports, interactive dashboards, predictive forecasts, and detailed revenue analyses remain accurate, timely, complete, and meaningful. The team tested the platform using diverse datasets ranging from historical records to current transaction logs, inventory information, and sales reports covering different periods. This verified system correctly processes, organizes, calculates, and visualizes business information under varying data volumes and conditions, ensuring every insight derived reflects true operational status.

Researchers also designed tests around different workload scenarios to thoroughly examine performance during both peak and non-peak operational periods. Evaluations measured platform response when multiple users accessed the system at the same time, generated heavy reports, input transaction data, or performed complex filtering and analysis. These assessments determined whether performance remained stable and response times acceptable even during high activity or increased demand. Teams closely monitored processing speed, page loading duration, system responsiveness, database query efficiency, and overall operational efficiency. Results confirmed reliability and full functionality regardless of data volume or number of active users interacting with the system at any given time.

Procedures further evaluated how the system handles incomplete, inconsistent, or invalid data entries, occurrences common in daily operations. Researchers intentionally introduced incorrect, missing, or improperly formatted data to verify validation mechanisms, warning messages, and error-handling procedures function effectively and prevent incorrect information from entering the database. This approach assessed platform resilience, reliability, and capability to avoid data-related errors that might negatively impact reports or decision-making. Teams also confirmed the system provides clear, understandable notifications and helpful guidance whenever users encounter invalid inputs, duplicate records, or operational mistakes. Such features reduce confusion and support correct usage habits.

Testing included comprehensive security and access-control assessments to verify sensitive business information such as daily sales figures, profit margins, and customer details remain strictly protected from unauthorized access or modification. User authentication procedures, account permissions, login mechanisms, password policies, and data access restrictions underwent thorough checks. Teams confirmed only authorized personnel access confidential records and analytical reports relevant to their roles. These evaluations protected organizational data integrity and maintained required confidentiality levels. Researchers also verified proper recording of user activities, login attempts, and transaction history to support continuous monitoring, operational accountability, and audit-related requirements.

Multiple testing levels allowed evaluation from both technical and user-centered perspectives. Technical assessments focused on program code functionality, database connectivity, data processing accuracy, integration between different modules, operational stability, and overall performance efficiency. Every component—including data input forms, dashboard interfaces, report generators, analytics modules, and database management functions—was tested individually and collectively. Teams verified performance aligns with specified requirements and design objectives established during the analysis phase, ensuring smooth interaction between all technical elements.

Technical evaluation examined integration effectiveness with existing hardware and software infrastructure currently used by Kape Tubong. Compatibility tests confirmed proper operation across different devices, operating systems, web browsers, and network conditions commonly applied in the main office and various branches. Teams verified consistent and reliable function without software conflicts, display errors, or operational interruptions. These assessments ensured platform implementation proceeds smoothly and efficiently without major adjustments or expensive upgrades to current technological environments.

User-centered testing focused on usability, accessibility, and practicality from the perspective of actual end-users interacting with the system daily. This phase involved selected staff members, sales personnel, branch managers, and administrative users performing regular duties using the platform. Participants provided detailed feedback regarding experiences navigating and operating the system. Researchers observed movement through interfaces, information entry, generation of different report types, interpretation of visual analytics, and access to various functions. Special attention focused on identifying areas causing confusion, delays, or difficulties regarding specific features or presented information.

Structured interviews and survey questionnaires supported further usability evaluation. Participants provided feedback on interface clarity, navigation ease, system responsiveness, readability of dashboards and reports, and overall satisfaction levels. Additional input covered relevance of system features, usefulness of analytical outputs, accessibility of stored information, and convenience of daily operational task completion using the new platform compared to previous methods. This feedback guided adjustments to ensure the system matches user expectations and operational habits.

Researchers assessed levels of technical knowledge and training required for employees to effectively operate, manage, and maintain the system. Evaluations identified needs for additional training materials, simplified user manuals, or ongoing technical support mechanisms during implementation to ensure smooth adoption. Understanding learning needs, preferences, and existing skill levels allowed researchers to recommend strategies for faster adoption and effective long-term utilization within the organization. This preparation helped reduce resistance to change and encouraged confidence among users.

User Acceptance Testing (UAT) stands as one of the most critical phases in the process. Actual users evaluated the system based on real operational scenarios and business activities common within Kape Tubong. Tests utilized authentic sales transactions, inventory records, customer order details, and operational reports. Participants determined whether the platform meets expectations and fully satisfies operational requirements. The primary objective confirmed the developed system is practical, functional, and capable of supporting daily activities and decision-making processes in real-world environments.

Feedback gathered during UAT helped identify final adjustments and improvements needed before full deployment. Suggestions regarding interface enhancements, report formats, workflow adjustments, terminology changes, and feature refinements underwent careful review and feasibility analysis. Teams incorporated recommendations whenever possible to align the platform closer with user expectations. This collaborative process ensured the final version matches user preferences, work habits, and operational requirements closely, increasing chances of successful integration.

The comprehensive testing framework significantly improved confidence regarding readiness, quality, and effectiveness of the BREWINSIGHT BI: Revenue Pattern Analytics System. Detailed evaluation and continuous refinement confirmed capability to deliver accurate analytics, stable performance, secure data management, and user-friendly functionality suited to Kape Tubong’s unique operational environment. The process established a reliable basis for future maintenance, system updates, scalability expansion, and further enhancements as the organization grows and adapts to changing business conditions and market demands.

Findings obtained from testing procedures created important documentation regarding system performance and behavior under different operational scenarios, data volumes, and user loads. Records serve as valuable references for future troubleshooting, routine maintenance activities, performance monitoring, and expansion projects. Results also highlighted opportunities for future feature enhancements and improvements that may further increase platform value, capability, and effectiveness over time.

Rigorous and systematic implementation of Unit Testing, System Testing, and User Acceptance Testing confirmed the BREWINSIGHT BI: Revenue Pattern Analytics System is not only technically sound and built according to standards but also operationally practical, highly reliable, and perfectly suited for real-world business application. Procedures validated capability to effectively support revenue analysis, business monitoring, operational efficiency, and data-driven decision-making within Kape Tubong. Thorough evaluation of every component, feature, and interaction ensures the final system delivers long-term value, improves organizational performance, and supports sustainable growth and continued business success well into the future.

Unit Testing - Unit testing was conducted to examine individual system components separately and confirm that each module functioned correctly and efficiently on its own before being integrated into the complete system. This testing stage focused on validating the smallest functional parts of the BREWINSIGHT BI: Revenue Pattern Analytics System to ensure that every module performed exactly according to its intended design and technical specifications. By isolating individual components during evaluation, researchers identified errors, inconsistencies, and functional issues with greater accuracy and efficiency. This approach allowed problems to be detected and corrected at the earliest possible stage of development, preventing defects from affecting other parts of the platform later on.

During this phase, specific system modules underwent independent assessment. These included the login and authentication module, database connection handlers, dashboard visualization components, chart display functions, sales filtering tools, report generation logic, user account management features, data input forms, and analytics processing units. Each module received different sets of input data to verify whether expected outputs, calculations, and system responses were produced correctly and completely. Researchers carefully evaluated whether each function executed properly under normal operating conditions and whether the module could consistently maintain accurate performance even during repeated operations.

The primary objective of unit testing involved detecting programming errors, logical inconsistencies, calculation mistakes, database communication failures, and unexpected behavior before individual components were combined into the fully integrated platform. Conducting tests at the module level made it easier to isolate defects, trace the exact source of problems, and implement corrective actions without disrupting or altering other parts of the system. This method significantly reduced the complexity of troubleshooting and minimized the risk of unresolved errors propagating into later development stages where they would become more difficult and costly to fix.

Input validation testing was performed extensively throughout this phase to ensure the system processed both valid and invalid user inputs correctly. Researchers examined whether the platform properly accepted authorized and correctly formatted data while rejecting incomplete, duplicate, inconsistent, or invalid entries. Various test cases were applied, including incorrect usernames and passwords, missing required fields, unsupported characters, invalid numerical values, and improperly formatted dates. These tests ensured that the system maintained high levels of data accuracy and integrity by preventing erroneous or misleading information from being stored or processed within the database.

Error messages, warning prompts, and exception-handling mechanisms were likewise evaluated to determine whether the platform responded appropriately whenever operational issues occurred. Researchers verified that the system displayed clear, understandable, and specific messages to users whenever errors were encountered. These notifications helped prevent confusion and guided users toward appropriate corrective actions. Exception handling routines were tested to confirm that unexpected failures, connection interruptions, or invalid operations did not cause the system to crash, freeze, or lose stored data. This process strengthened the reliability and resilience of the platform by ensuring errors could be managed gracefully without disrupting overall functionality.

Additional testing assessed the responsiveness and stability of each module under varying operational conditions. Researchers evaluated whether components continued functioning correctly when subjected to repeated requests, rapid user interactions, or simultaneous processing operations. This included measuring how quickly charts loaded, how efficiently reports were generated, and how accurately filters processed and displayed requested information. Database queries and data retrieval functions were also tested to confirm that records could be accessed, processed, and displayed without unnecessary delays or inconsistencies in results.

Researchers performed compatibility checks to ensure smooth interaction between each module and supporting technologies such as database servers, web interfaces, and integrated software libraries. These tests confirmed that each component could communicate effectively with related system resources and external services required for full operation. Particular attention was given to verifying that data transferred between modules remained accurate, synchronized, and entirely free from corruption or alteration during processing.

Edge cases and uncommon input conditions were included in the testing process to assess the robustness and dependability of individual modules. Boundary values, empty input fields, invalid credentials, unsupported file formats, incorrect filter combinations, duplicate records, and unusual transaction conditions were intentionally evaluated. These tests determined whether the system could continue operating correctly even in non-standard or unexpected situations. This approach helped identify hidden defects and weaknesses that might not appear during normal use but could potentially cause operational issues in real-world environments.

Performance consistency formed another important focus during unit testing. Researchers repeatedly executed functions and operations to verify that modules consistently produced the same accurate results over time. Retesting procedures were conducted whenever corrections, modifications, or updates were applied to any component. This process, known as regression testing, ensured that newly implemented fixes successfully resolved identified issues without introducing additional errors or negatively affecting previously functioning features.

Detailed documentation of testing results was maintained throughout the unit testing phase. Researchers recorded detected issues, test conditions applied, corrective actions taken, and retesting outcomes. These records created a clear reference for future debugging, maintenance, and system enhancement activities. The information gathered provided valuable insights regarding the behavior and stability of each module and served as supporting evidence that all components had been thoroughly validated prior to integration.

Through unit testing, researchers verified that each individual component of the BREWINSIGHT BI: Revenue Pattern Analytics System functioned properly, accurately, and consistently before progressing to integration and system-level testing phases. This stage established a strong and reliable foundation for the entire platform by ensuring that all core functions operated correctly on their own. As a result, unit testing significantly contributed to improving the overall quality, stability, reliability, and maintainability of the developed system. It also reduced the likelihood of major technical issues arising during later stages of development and final implementation.

System Testing: Once all individual modules successfully passed the Unit Testing phase, System Testing was conducted to evaluate the performance, functionality, reliability, and stability of the fully integrated BREWINSIGHT BI: Revenue Pattern Analytics System. This stage focused on determining whether all system components worked together correctly as a complete and unified business intelligence platform capable of supporting the operational and analytical requirements of Kape Tubong. Unlike Unit Testing, which examined individual modules independently, System Testing evaluated how the various modules interacted with one another when combined into a single operational environment.

The researchers tested the interaction between the user interface, backend processing logic, database management system, reporting engine, analytics modules, and dashboard visualization components to ensure smooth and accurate communication among all parts of the platform. This phase aimed to verify that information entered into the system flowed properly through each process without loss, corruption, duplication, or inconsistency. By examining the integrated behavior of the platform, the researchers were able to confirm whether the system performed according to the functional and technical specifications defined during the planning and design stages of the project.

During System Testing, common operational tasks and user activities were performed repeatedly under controlled testing conditions. These activities included user login and authentication, account access verification, sales transaction retrieval, report filtering, dashboard visualization, analytics generation, inventory monitoring, revenue trend analysis, and summary report generation. The researchers carefully observed whether the system processed data accurately and displayed correct outputs throughout the platform. Particular attention was given to ensuring that reports, charts, tables, and dashboard indicators reflected updated and synchronized information retrieved from the central database.

The researchers also verified that all modules exchanged data correctly and maintained consistency across the entire platform. For example, updates made within transaction records were checked to determine whether changes were immediately reflected in dashboard analytics, summary reports, and filtering tools. This process ensured that the system maintained real-time or near real-time synchronization between operational data and business intelligence outputs. Any issues involving delayed processing, incomplete updates, mismatched information, duplicate records, or synchronization failures were identified, documented, corrected, and retested to confirm that the implemented solutions resolved the problem successfully.

In addition to functional verification, the researchers evaluated the stability and reliability of the integrated system under regular operational conditions. Repeated testing procedures were conducted to determine whether the platform could maintain consistent behavior during prolonged usage and continuous interaction. The researchers observed whether the system remained stable while performing multiple tasks simultaneously, such as generating reports while processing data queries or updating dashboard visualizations while users navigated through different sections of the platform.

Basic performance testing was also performed during this phase to examine system responsiveness, processing efficiency, and operational speed under normal workload conditions. The researchers monitored loading times, dashboard responsiveness, report generation speed, and database query performance to ensure that the system delivered information within acceptable response times. These evaluations helped determine whether users could interact with the system efficiently without experiencing unnecessary delays or interruptions during daily operations.

To further evaluate performance, the researchers simulated scenarios involving multiple users accessing the platform simultaneously. This allowed them to assess how the system handled concurrent activities such as simultaneous dashboard requests, report filtering, data retrieval, and account access operations. The testing process examined whether increased user activity negatively affected system responsiveness, data accuracy, or operational stability. Through these simulations, the researchers verified that the platform could support regular business operations without significant performance degradation.

System Testing also included compatibility testing to verify that the platform functioned properly across different supported web browsers, operating systems, screen sizes, and hardware devices commonly used within the organization. The researchers tested the system using various browser environments to ensure that interface elements, dashboard components, navigation menus, charts, and reporting tools displayed correctly and behaved consistently regardless of the user's device or browser preference. Compatibility checks were important in ensuring that all employees and managers could access and operate the system without encountering technical limitations caused by hardware or software differences.

The researchers additionally evaluated the responsiveness and adaptability of the user interface across different display resolutions and device types. This ensured that the platform maintained usability and readability when accessed through desktop computers, laptops, or other supported devices. Interface elements such as charts, buttons, tables, filters, and navigation panels were examined to confirm that they adjusted properly to varying screen dimensions without affecting usability or visual clarity.

Security-related functions were also assessed during System Testing. The researchers examined whether user authentication, access restrictions, account permissions, and session management features operated correctly within the integrated environment. Tests were conducted to verify that unauthorized users could not access restricted information or administrative functions. Logout procedures, session expiration controls, and password validation mechanisms were also evaluated to ensure that the platform maintained appropriate security standards and protected sensitive business data from unauthorized access.

Error handling and recovery mechanisms were another important aspect of the testing process. The researchers intentionally introduced invalid operations, incomplete data entries, interrupted connections, and unexpected user actions to determine how the integrated system responded to unusual conditions. The platform was evaluated to ensure that errors were handled appropriately without causing crashes, data corruption, or operational instability. Clear notifications and guidance messages were also verified to ensure that users received understandable instructions whenever problems occurred.

Data integrity testing was likewise conducted to confirm that information remained accurate and consistent throughout all system processes. The researchers checked whether stored records matched displayed outputs, whether calculations produced correct analytical results, and whether database transactions were completed successfully without missing or altered information. This was particularly important because the BREWINSIGHT BI platform relies heavily on accurate business intelligence outputs to support managerial decision-making and operational analysis.

Documentation of identified issues, corrective actions, testing procedures, and retesting results was maintained throughout the System Testing phase. These records provided a detailed reference regarding the behavior, performance, and stability of the platform under different operational conditions. Maintaining comprehensive documentation also supported future maintenance activities, troubleshooting procedures, and system enhancement efforts after deployment.

Through System Testing, the researchers confirmed that the entire BREWINSIGHT BI: Revenue Pattern Analytics System operated as a fully integrated, stable, and functional business intelligence platform. The testing process verified that all modules worked together efficiently, data was processed accurately, and system features performed according to established functional requirements. Furthermore, System Testing ensured that the integrated platform met the operational, technical, and usability standards necessary for practical deployment within Kape Tubong’s business environment.

The successful completion of this phase provided greater confidence in the overall readiness of the platform for real-world use. By thoroughly evaluating functionality, integration, compatibility, security, responsiveness, and data consistency, the researchers ensured that the system was capable of supporting the organization’s operational activities, delivering accurate business insights, and assisting management in making informed decisions. Ultimately, System Testing played a critical role in validating the reliability, effectiveness, and deployment readiness of the BREWINSIGHT BI platform before proceeding to the final stages of implementation and User Acceptance Testing.

User Acceptance Testing (UAT): was carried out to determine whether the developed BREWINSIGHT BI: Revenue Pattern Analytics System met the expectations, operational requirements, and practical needs of its intended users. This testing phase focused on evaluating the system in a realistic business environment to confirm that the platform was suitable for actual day-to-day operations within Kape Tubong. Unlike technical testing phases that primarily assessed system functionality and integration, User Acceptance Testing emphasized the experiences, satisfaction, and overall usability perceptions of the people who would directly utilize the system in their daily work activities.

During this phase, the business owner, Bryan Tacayon, together with selected staff members, branch personnel, and administrative users of Kape Tubong, participated in the testing activities. The participants were instructed to use the BREWINSIGHT BI platform in practical and realistic business-related scenarios that closely reflected actual operational conditions. These activities included accessing dashboards, retrieving reports, filtering sales data, monitoring branch performance, reviewing revenue trends, analyzing product sales, generating summaries, and navigating through various analytical features of the platform. By performing real operational tasks, the participants were able to evaluate whether the system effectively supported their responsibilities and business processes.

The primary purpose of User Acceptance Testing was to assess the system from the perspective of actual end-users and determine whether the platform was practical, understandable, efficient, and beneficial in supporting operational and managerial activities. The researchers observed how users interacted with the system and evaluated whether the platform's features, navigation structure, and dashboard presentations aligned with the users' expectations, work habits, and decision-making needs. Particular attention was given to evaluating whether users could easily access the information they needed and whether the analytical outputs generated by the platform were meaningful and useful for monitoring business performance.

Users assessed several important aspects of the platform during testing, including ease of use, clarity of dashboard presentation, responsiveness of the interface, accessibility of reports, speed of data retrieval, accuracy of displayed information, and overall convenience in performing daily operational tasks. The participants also evaluated whether charts, tables, summaries, and analytical visualizations were presented in a clear and understandable manner. Since many of the intended users were not technical specialists, it was important to ensure that the system communicated information in a simple, organized, and user-friendly format that could easily support managerial interpretation and business analysis.

The researchers also examined how quickly users were able to learn and navigate the platform without extensive technical guidance. This helped determine whether the system interface was intuitive enough for employees with varying levels of technological familiarity and experience. Tasks involving dashboard filtering, report generation, navigation between modules, and data interpretation were carefully observed to identify any difficulties or confusion experienced by participants while interacting with the system. The testing process also evaluated whether the terminology, menu labels, and interface controls used within the platform were understandable and appropriate for the organization's operational context.

Feedback was gathered from participants through direct observation, interviews, informal discussions, and structured evaluation forms. Users were encouraged to share their experiences, opinions, suggestions, concerns, and recommendations regarding the system's functionality, appearance, and usability. The researchers documented all feedback related to navigation issues, interface layout, report readability, dashboard organization, processing speed, accessibility of features, and clarity of analytical outputs. Participants also identified areas where additional improvements or refinements could further enhance the usefulness and convenience of the platform.

User Acceptance Testing also provided an important opportunity to evaluate whether the analytical outputs generated by the BREWINSIGHT BI platform were understandable, relevant, and meaningful to non-technical users. This aspect was particularly significant because the primary users of the platform rely on the system for business monitoring, operational analysis, and managerial decision-making rather than advanced technical analysis. The researchers assessed whether users could easily interpret charts, sales trends, revenue summaries, branch comparisons, and product performance reports without requiring extensive technical explanation or assistance.

The feedback obtained during this phase helped refine several aspects of the platform, including dashboard layouts, chart arrangements, report formatting, navigation flow, and interface organization. In response to user suggestions, the researchers implemented necessary revisions and adjustments to improve readability, simplify navigation, and enhance the overall user experience. Modifications were also made to ensure that important information could be accessed more quickly and displayed more clearly to users. These improvements contributed to making the platform more intuitive, accessible, and practical for everyday business use.

The researchers additionally evaluated the responsiveness of the system during actual user interaction. This included observing how quickly dashboards loaded, how efficiently reports were generated, and how smoothly filtering operations were processed while users navigated through the platform. The testing process ensured that users could interact with the system efficiently without experiencing delays or interruptions that could negatively affect productivity or operational workflows.

Another important objective of UAT was to determine whether the system effectively supported the operational workflows already established within Kape Tubong. The researchers examined whether the platform integrated naturally into existing business procedures and whether it simplified tasks related to sales monitoring, report preparation, and performance evaluation. This ensured that the implementation of the BREWINSIGHT BI system would improve operational efficiency rather than introduce unnecessary complexity or disruption to the organization's daily activities.

User Acceptance Testing also allowed the researchers to identify the level of user training and support required for successful implementation. By observing how participants interacted with the platform, the researchers were able to determine whether additional user manuals, tutorials, orientation sessions, or technical support materials would be necessary to facilitate smooth adoption of the system. This information became valuable for planning future training programs and implementation strategies within the organization.

All usability concerns, recommendations, operational difficulties, and enhancement requests identified during UAT were carefully reviewed and analyzed by the researchers. Necessary revisions were implemented, and the affected system components were retested to ensure that the modifications successfully addressed user concerns without negatively affecting other functionalities of the platform. This iterative improvement process ensured that the final version of the system was refined according to actual user experiences and operational requirements.

The successful completion of User Acceptance Testing confirmed that the BREWINSIGHT BI: Revenue Pattern Analytics System was suitable for deployment and aligned with the practical needs of Kape Tubong. The testing process verified that the platform was not only technically functional but also user-friendly, accessible, understandable, and effective in supporting business monitoring and decision-making activities. Furthermore, UAT strengthened user confidence and acceptance of the system by actively involving future users in the evaluation and refinement process.

Ultimately, User Acceptance Testing played a critical role in validating the overall effectiveness, practicality, and usability of the BREWINSIGHT BI platform. Through direct participation from actual users, the researchers ensured that the system was capable of delivering meaningful analytical insights, improving operational efficiency, and supporting informed managerial decisions within the organization. The feedback gathered during this phase contributed significantly to the optimization of the platform, ensuring that the final system was well-suited to the operational environment, business objectives, and long-term needs of Kape Tubong.

3.7 Evaluation Criteria

The developed BREWINSIGHT BI: Revenue Pattern Analytics System will be evaluated using a set of criteria designed to measure its effectiveness, technical quality, and suitability for practical deployment in Kape Tubong's operational environment. These criteria provide a structured basis for determining whether the system meets its intended objectives and performs according to the requirements established during the planning and design phases. Each criterion focuses on a specific aspect of software quality relevant to business intelligence systems, particularly those intended for real-world organizational use, addressing core factors that influence how well the system supports decision-making, integrates with existing workflows, and delivers tangible value to the business. They take into account the specific nature of Kape Tubong's operations, including daily transaction processes, inventory management practices, sales tracking methods, and reporting requirements that define how the organization functions and makes strategic choices. They also consider the unique characteristics of the coffee shop industry, such as seasonal demand fluctuations, product variety, customer behavior patterns, and market trends that directly impact revenue generation and business performance. By focusing on these specific elements, the evaluation framework ensures that every aspect of the system's functionality is measured against the actual context and requirements of the organization, rather than relying on generic standards that may not apply directly to its operations.

Through evaluation based on these standards, the researchers can determine the strengths and limitations of the platform and identify areas requiring improvement prior to full implementation. This assessment phase is critical to ensuring that the final product is not just a theoretical concept, but a functional tool that can be adopted smoothly and used consistently by the organization's staff. It serves as a bridge between the development stage and actual deployment, allowing adjustments and refinements to be made while the system is still in the testing phase, which reduces risks associated with full-scale implementation and ensures that resources are used efficiently. The evaluation process also ensures that the developed system is not only technically operational but also practical, dependable, secure, and beneficial for its intended users, taking into account factors such as ease of use, reliability under varying workloads, and alignment with the daily operations and long-term goals of the organization. It considers how the system interacts with other tools and processes already in place, whether it can adapt to changes in business needs over time, and how well it addresses specific challenges or gaps that the organization has identified in its current data management practices. It also examines how the system handles different types of data, from basic transaction records to more complex analytical datasets, and whether it can process information efficiently even as the volume of data grows over time. Furthermore, it looks at how the system supports different levels of users within the organization, from frontline staff who input data to management personnel who analyze results and make decisions, ensuring that each group can access the information they need in a format that is appropriate and useful for their specific roles and responsibilities.

By assessing the platform across multiple dimensions, the study can provide a more comprehensive validation of system quality and readiness, moving beyond basic functionality to evaluate how well the system fits within the unique context and operational framework of Kape Tubong. This multi-faceted approach allows for a thorough understanding of not only what the system can do, but also how well it performs those tasks in real-world conditions, how it is perceived by those who will use it regularly, and what impact it is likely to have on overall business performance. It also helps in distinguishing between features that are essential to the system's purpose and those that may be useful but not necessary, ensuring that resources are focused on delivering the most value to the organization. This balanced assessment prevents over-engineering and ensures that the system remains user-friendly and cost-effective, while still providing all the capabilities required to meet the organization's analytical and operational needs. It also enables a fair comparison between the developed system and existing methods or tools currently used by the organization, highlighting improvements in efficiency, accuracy, and decision-making support that the new platform can offer.

The criteria selected for evaluation are aligned with common software quality standards and reflect the practical needs of Kape Tubong as the intended beneficiary of the platform, balancing technical benchmarks with real-world usability and business relevance. They are derived from established guidelines for software development and system evaluation, as well as direct consultation with the organization's management and staff to ensure that they capture all elements that matter most to the success of the system. These criteria collectively help determine whether BREWINSIGHT BI successfully fulfills its role as a business intelligence and revenue analytics solution, verifying that it can accurately process data, generate meaningful insights, and support the strategic and operational needs of the business effectively. They cover everything from the accuracy and reliability of data outputs to the clarity and accessibility of information presented to users, ensuring that the system serves as a reliable source of information for planning, analysis, and decision-making at all levels of the organization. They also include considerations related to system performance, such as processing speed, response time, and ability to handle concurrent users, as well as factors related to maintenance, support, and long-term sustainability. Additionally, they address aspects of data integrity, confidentiality, and compliance, ensuring that sensitive business information is protected and that the system operates within legal and ethical standards applicable to business operations and data management.

The evaluation results will serve as evidence of the platform's effectiveness and suitability for supporting business operations, providing clear documentation of performance levels, areas of excellence, and opportunities for further refinement. This information will be valuable not only for the researchers and developers who created the system, but also for Kape Tubong as they consider integrating the platform into their regular operations. It will help the organization understand what to expect from the system, how to maximize its benefits, and what steps may be needed to ensure its long-term success and continued usefulness as business conditions evolve over time. The findings will also provide insights that can guide future updates and enhancements, allowing the system to grow and adapt alongside the organization's changing needs and market dynamics. Furthermore, the evaluation outcomes can be used to demonstrate the value of the investment made in developing and implementing the system, providing justification for continued support and expansion of its use within the organization. By establishing clear measures of success and performance, the evaluation process creates a foundation for monitoring the system's impact over time and ensuring that it continues to deliver value long after it has been deployed. The following criteria will be used in assessing the developed system. The system will be evaluated based on the following criteria:

Functionality- Functionality refers to the ability of the system to perform its intended tasks accurately, completely, and strictly according to specified requirements and business rules. This criterion evaluates whether the BREWINSIGHT BI platform successfully delivers all features and services required by Kape Tubong to support daily operations, monitoring, and strategic analysis. Core capabilities under assessment include interactive dashboard visualization, detailed historical and real-time sales tracking, side-by-side branch performance comparison, long-term revenue trend analysis, individual product performance monitoring, and flexible report generation. Researchers verify that these features operate properly, remain stable, and consistently produce correct outputs under various usage conditions and data volumes. Evaluation confirms that every required module is fully implemented, active, and accessible only to authorized personnel based on their roles. Missing features, calculation errors, incorrect data presentation, or malfunctioning components indicate limitations or gaps in system performance. This criterion also checks if the platform responds appropriately to all types of user commands and processes data exactly as defined during the design phase. Feedback gathered during testing and user evaluation helps determine whether the system meets or exceeds defined functional expectations. High functionality demonstrates that the platform effectively fulfills its purpose as a reliable and comprehensive business intelligence solution. This measure proves essential in confirming that the developed system achieves the primary goals and objectives outlined in this study.

Assessment of functionality determines whether analytical outputs generated by the platform accurately and faithfully reflect the underlying sales records, inventory logs, and transactional details stored in the central database. The correctness of revenue computations, profit margin calculations, branch comparisons, and product performance summaries undergoes careful examination to ensure absolute analytical integrity and mathematical precision. The system's ability to handle multi-layered filtering options, custom date-range selection, comparative reporting, and data grouping without generating inaccurate, misleading, or incomplete results also receives thorough review.

Functional completeness gets verified by confirming that all business requirements identified during the analysis phase translate successfully into fully working features within the system. This close alignment ensures the platform matches operational expectations set by Kape Tubong management and supports actual workflows effectively. Furthermore, the platform supports simultaneous execution of multiple functions—such as report generation while data synchronization occurs—without conflict, lag, or performance degradation. Strong functionality confirms the system serves as a comprehensive and dependable operational support tool. This criterion validates whether BREWINSIGHT BI delivers intended business intelligence capabilities both effectively and accurately.

Evaluation examines the consistency of system operations across different user interactions, usage patterns, and operational scenarios. The platform maintains accurate, fast, and predictable performance regardless of the volume of transactions processed or the number of users accessing the system at the same time. Features related to data retrieval, automated report generation, dynamic filtering, and real-time dashboard updating operate efficiently without delays, missing records, duplicate entries, or processing errors. The system integrates, merges, and organizes information coming from multiple branches or kiosks into one centralized database while preserving data accuracy, logical relationships, and consistency throughout. This integration capability proves critical in ensuring management receives reliable, unified, and comprehensive business insights covering all operational locations in a single view.

Another key aspect involves assessing the effectiveness and reliability of automated processes built into the platform. Functions such as automated daily sales consolidation, real-time data synchronization across locations, dynamic dashboard refreshing, and scheduled report generation must function correctly to minimize manual intervention and reduce operational inefficiencies. The system correctly categorizes products by type or group, summarizes complex transaction histories, and calculates key performance indicators such as average sales, best-selling items, and peak hours without requiring extensive user adjustments or corrections. These automated features allow the system to significantly improve speed, reduce human error, and increase accuracy during business analysis activities. Evaluation determines if these processes contribute positively to overall operational productivity and enhance decision-making efficiency for management and staff.

This criterion also examines the reliability, strictness, and clarity of user access controls and role-based permissions implemented throughout the platform. Authorized administrators, managers, and staff members access only features, menus, and information strictly relevant to their designated responsibilities and job functions. This structured separation ensures sensitive sales figures, financial summaries, and confidential business details remain fully protected from unauthorized viewing or unintended modification. Proper functioning of login authentication, session management, account creation, password recovery, and permission settings forms a central part of the assessment process. Effective access control mechanisms contribute directly to both system security and clear operational organization within the business.

Assessment includes measuring the system's capability to generate meaningful, relevant, and visually understandable reports suitable for different levels of decision-making. Dashboards and analytical summaries present clear, interpretable information that supports managers in evaluating performance, identifying trends, and spotting opportunities or issues. Charts, graphs, tables, and statistical summaries display accurate data representations that users can analyze, compare, and interpret easily without requiring advanced technical knowledge or specialized training. The platform also supports highly customizable reporting options allowing management to examine specific business periods, individual branches, selected product categories, or custom-defined groups based on current needs and inquiries. This flexibility increases the usefulness of the business intelligence platform and enables it to address diverse operational concerns effectively.

Evaluation determines whether the platform effectively supports future operational growth and long-term scalability requirements. As Kape Tubong expands operations, introduces new products, opens additional branches, or serves a growing customer base, the platform accommodates increasing volumes of transactional data, higher user traffic, and rising analytical demands without loss of performance or stability. Functional scalability ensures the system continues operating efficiently, accurately, and responsively even as organizational requirements evolve and grow over months or years. This adaptability remains important in maintaining long-term relevance and ensuring the platform stays a useful and valuable asset within the changing business environment.

Functionality testing involves identifying, recording, and documenting any system limitations, processing delays, logic errors, or feature inconsistencies encountered during actual usage. Errors or gaps discovered during assessment provide valuable insights necessary for system refinement, debugging, and improvement before final deployment. User feedback regarding ease of feature execution, report accuracy, calculation correctness, and operational reliability contributes significantly to the overall evaluation of functional quality. Continuous testing, validation, and refinement of system functions ensure the final platform operates strictly according to original design specifications and intended business objectives.

Functionality serves as a comprehensive measure of the system's capability to meet all operational, analytical, and managerial requirements established by the organization. It verifies whether BREWINSIGHT BI successfully performs all intended tasks while delivering accurate, reliable, and efficient business intelligence services to Kape Tubong. Confirming that every feature works correctly, securely, and supports business processes effectively allows this study to validate the practical value, completeness, and overall effectiveness of the developed solution.

Usability- Usability refers to the overall ease, convenience, and comfort experienced by users while learning, understanding, navigating, and operating the system, and this evaluation criterion specifically assesses whether the BREWINSIGHT BI platform offers an interface that is clear, accessible, intuitive, and genuinely user-friendly for all personnel within Kape Tubong, regardless of their prior technical background or skill level. During this assessment, the research team carefully examines how easily and quickly users can navigate through different sections of the platform, locate specific features or tools they require, interpret complex data presented on dashboards, and successfully complete all necessary operational tasks without unnecessary difficulty or confusion. Every element related to interface design—including menu placement, overall dashboard layout, the clarity of labels and headings, the design and readability of charts and graphs, and the functionality of filtering controls and navigation buttons—is thoroughly reviewed and tested to ensure maximum clarity, logical flow, and effectiveness in supporting user actions. A system that achieves high usability significantly reduces mental effort and confusion for the user, allowing individuals to complete their intended tasks with minimal external assistance, limited training, and very little trial and error; furthermore, strong usability greatly enhances the overall quality of the user experience by making the entire platform feel natural, intuitive, and comfortable to interact with, which helps users feel more confident and productive while working. Direct feedback collected from end-users during actual testing sessions provides deep and valuable insights into how practical, understandable, and helpful the system is when used in real operational scenarios, offering essential information about what works well and what areas may still need improvement. The importance of usability cannot be overstated; even the most technically advanced, feature-rich, and powerful systems may remain underutilized, ignored, or abandoned entirely if users find them complicated, difficult to navigate, or hard to understand. A platform that is designed to be easy to use naturally encourages wider adoption, consistent daily use, and deeper engagement, ensuring that the full potential of the business intelligence system is realized within daily operations and that it becomes a trusted tool rather than a source of frustration. This criterion serves as a fundamental measure to determine whether the developed system is not only technically sound but also practically applicable and truly centered around the needs and capabilities of the people who will use it most.

Additional evaluation efforts focus extensively on how effectively the system's dashboard communicates complex analytical information through its visual design, layout structure, and presentation style, recognizing that since the platform is built to present large volumes of business data through charts, graphs, tables, and visual summaries, these elements must remain clear, meaningful, and understandable even to users who do not possess advanced analytical skills or technical backgrounds. Researchers carefully review the clarity and consistency of all labels, legends, axis descriptions, color coding systems, and the logical grouping of related data points to ensure that information can be interpreted quickly, accurately, and without confusion or guesswork. Navigation efficiency is also measured objectively by determining exactly how many steps, clicks, or actions are required for users to complete common and essential tasks—such as generating a specific report, filtering data by date or branch, or comparing performance metrics—to ensure that workflows are streamlined and free from unnecessary complexity. The responsiveness and adaptability of the interface across different hardware environments, including desktop computers, laptops, tablets, and mobile devices, forms a critical part of this evaluation, ensuring that the system remains fully functional, visually consistent, and equally intuitive regardless of the screen size or device being used. A system that maintains high usability and consistent quality across multiple device types significantly improves accessibility and provides greater operational flexibility, allowing staff members to access important information and perform necessary tasks from various locations and workstations within the business. Furthermore, the researchers assess whether the platform actively works to minimize user errors through the use of clear guidance, helpful prompts, real-time validation checks, and interactive assistance, ensuring that users are supported at every step and that mistakes are avoided or corrected before they cause issues. This comprehensive usability evaluation ultimately confirms whether the BREWINSIGHT BI platform can be effectively adopted, comfortably learned, and consistently utilized by all personnel within Kape Tubong to support their routine operations and decision-making responsibilities.

The assessment also thoroughly examines the learning curve experienced by first-time users or individuals who have not interacted with the system previously, aiming to determine how quickly and easily users can familiarize themselves with available features, understand the purpose of different tools, and successfully complete required tasks without needing extensive technical support, formal training, or constant reference to manuals or documentation. Every element designed to guide the user—including written instructions, icons, symbols, navigational cues, and help messages—is evaluated to ensure they remain simple, direct, self-explanatory, and logically positioned, effectively reducing dependency on external assistance and making the system accessible to everyone. A platform that features a short and gentle learning curve enables employees to become fully productive and confident in using the system much more quickly, which significantly reduces the amount of time and resources spent on training sessions and allows the business to begin benefiting from the new system sooner. This aspect of usability holds particular value for small and medium-sized business environments such as Kape Tubong, where staff members may possess varying levels of technical expertise, experience with technology, and educational backgrounds, and where complex or difficult-to-learn systems can create significant barriers to effective use. By ensuring that the platform is easy to learn and straightforward to master, the design supports a smoother and faster operational implementation process, minimizes resistance to change, and contributes directly to higher levels of user acceptance and satisfaction across the organization.

This evaluation criterion also places strong emphasis on assessing the consistency of interface behavior, design elements, and interaction patterns throughout every section and module of the platform. Uniformity in menu structures, button placements, typography, color schemes, and navigation styles helps users develop a clear mental model of how the system works, allowing them to predict how different elements will behave and reducing the need to relearn how to perform similar actions in different parts of the application. When interface components behave in a similar and predictable manner across all modules and functions, users are able to navigate more efficiently, work with greater confidence, and complete tasks more quickly without encountering unexpected changes or confusing variations in design or operation. The evaluation process specifically identifies whether similar actions produce uniform responses, whether terminology remains consistent, and whether system workflows follow standardized logic across different features and functions. Consistency in design and behavior contributes significantly to user comfort and familiarity, reduces cognitive load, and greatly improves the overall quality and fluidity of the interaction between the user and the system, making the entire experience feel cohesive and professional.

Another important area of assessment involves the readability and accessibility of all information displayed within the platform, ensuring that every element of content—from text and numbers to charts and tables—can be easily read, understood, and interpreted by users under various working conditions and over extended periods of time. Researchers examine factors such as text size, line spacing, color contrast, font style, and overall visual presentation to confirm that content remains clear, legible, and comfortable to view, even during long working sessions or under different lighting environments. Charts and tables are evaluated to ensure they avoid excessive complexity, clutter, or unnecessary detail, instead presenting information in a concise, organized, and logical manner that highlights the most important insights and supports quick decision-making. The platform is also assessed to ensure that key performance indicators, critical values, and essential details appear prominently and are visually emphasized so that users can immediately identify what matters most without having to search through large amounts of data. Improving readability and accessibility directly reduces user fatigue, minimizes eye strain, and enhances overall operational efficiency during daily use, ensuring that users can work effectively and accurately without unnecessary physical or mental discomfort.

Usability testing also thoroughly evaluates the effectiveness of system feedback and response mechanisms, which are essential for helping users understand exactly what is happening during their interaction with the platform and ensuring they feel informed and in control at all times. The researchers verify that users receive clear, timely, and unambiguous confirmations whenever actions such as data filtering, report generation, record updates, or data saving processes are completed successfully, so there is never uncertainty about whether a task was done correctly. Additionally, the platform is assessed to ensure it provides understandable, helpful, and non-technical error messages and guidance whenever invalid inputs are entered, mistakes are made, or operational issues occur, turning potential frustrations into learning opportunities and helping users correct problems quickly and independently. Effective feedback mechanisms help users understand the system's status, reduce anxiety or confusion during interaction, and prevent misunderstandings or incorrect assumptions about how the system works. This leads to a more reliable, transparent, and satisfying user experience while minimizing operational mistakes and ensuring that users always feel supported and guided throughout their work.

The researchers also examine how well the system supports and enhances user productivity and task efficiency, focusing on whether the platform enables users to complete their analytical, reporting, and management tasks quickly, smoothly, and without unnecessary steps, delays, or repetitive procedures. The evaluation looks closely at features such as quick-access menus, automated filtering and sorting options, searchable records and databases, pre-built templates, and streamlined workflows that are designed to reduce manual effort and save time. The importance of time efficiency is recognized as critical in business environments where managers and staff members must process information, generate insights, and make decisions promptly to respond to market changes, monitor performance, and guide daily operations. A highly usable system that is optimized for speed and efficiency directly improves user satisfaction by making work easier and faster, while also contributing significantly to overall organizational productivity and operational effectiveness, as more work can be accomplished in less time with fewer resources.

Evaluation efforts also consider the adaptability and flexibility of the platform to accommodate different user roles, responsibilities, and operational needs within Kape Tubong, recognizing that managers, administrators, and regular staff members each interact with distinct features and require different types of information based on their specific duties. The researchers assess whether the interface provides appropriate levels of access, relevant tools, and simplified workflows that are specifically tailored to each type of user, ensuring that individuals only see what they need and are not overwhelmed by features or data that are irrelevant to their work. Customized dashboards, role-specific displays, and personalized settings are reviewed to confirm they reduce unnecessary complexity and improve the relevance of the information presented to each user group. This high level of adaptability enhances user convenience and satisfaction by making the system feel customized to their needs, while also ensuring that the platform effectively supports diverse operational functions and responsibilities across the entire organization.

The system's visual appeal and overall aesthetic quality also form an important part of the usability assessment, as the way a platform looks and feels has a direct impact on how users perceive its value, quality, and ease of use. An organized, visually balanced, and professionally designed interface significantly improves user engagement, creates a positive first impression, and encourages continued and consistent system use over time. Researchers evaluate the appropriate use of colors, spacing, icons, typography, and graphical elements to ensure the platform is attractive and visually pleasing while maintaining high standards of professionalism, clarity, and functionality. While functionality and ease of use remain the primary concerns, visual design plays a powerful role in influencing user perception, trust, and comfort during interaction; a visually appealing system contributes positively to user confidence, makes data more engaging, and helps create a more enjoyable and motivating working environment for all users.

Furthermore, the researchers assess whether the platform supports efficient recovery from user mistakes, operational errors, or incorrect inputs, ensuring that the system is forgiving and allows users to correct problems easily without negative consequences or frustration. Features such as undo functions, editable inputs, confirmation prompts before executing important actions, and clear recovery options are evaluated to confirm they prevent permanent mistakes, reduce stress, and help users maintain smooth workflow continuity even when things do not go exactly as planned. Error tolerance is recognized as a key component of usability in real-world business environments, where mistakes are natural and inevitable, and where system interruptions or data loss can be costly or disruptive. By making it easy to correct errors and go back to previous states, the platform becomes more user-friendly and reliable, ensuring that users feel safe to explore, learn, and work efficiently without fear of causing irreversible damage or losing important work.

Usability serves as a critical and comprehensive measure of how effectively the BREWINSIGHT BI platform supports user interaction, operational convenience, and overall system accessibility, ultimately determining whether the platform can be used comfortably, confidently, and efficiently by all personnel within Kape Tubong, regardless of their technical background, experience level, or role within the organization. Through detailed and comprehensive evaluation, researchers confirm whether the developed system successfully promotes ease of use, efficient task completion, and a consistently positive user experience in every aspect of its operation. A highly usable platform strengthens user acceptance, encourages regular and deep engagement, and ensures that the powerful capabilities of the business intelligence system are fully understood, trusted, and integrated seamlessly into daily operational activities, delivering maximum value and benefit to the entire business.

Reliability- Reliability refers to the consistency and dependability of the system while operating under defined conditions over a specific period. This criterion evaluates whether the BREWINSIGHT BI platform functions continuously without frequent failures, crashes, or processing errors. It also determines if the system produces accurate, stable, and identical outputs even when specific tasks are performed repeatedly over time. Core functionalities such as dashboard loading speed, automated report generation, and data filtering options undergo multiple rounds of testing to verify that performance levels remain uniform and consistent across different usage instances. Beyond basic functionality, reliability encompasses the system's capability to handle unexpected situations, including invalid inputs, incomplete data entries, or minor operational mistakes, without causing a complete halt in service or disrupting overall performance. The platform maintains high levels of data integrity and stays responsive to user commands even when encountering common operational anomalies or routine technical issues. A system demonstrating high reliability builds user confidence, ensuring stakeholders that the information presented is accurate, valid, and dependable enough to serve as the foundation for critical business decision-making. Frequent interruptions, calculation errors, or inconsistent outputs reduce trust and may discourage users from relying on the platform for daily operations. Since this system supports important strategic and operational decisions, reliability serves as a primary measure of its practical value and professional quality. This evaluation criterion ensures the platform serves as a dependable, robust, and consistent analytical tool within the daily workflow of the organization.

Evaluation of reliability focuses further on the system's capacity to sustain stable operation during extended periods of continuous or heavy usage. Researchers observe the platform's behavior during prolonged testing sessions to determine if repeated access requests, heavy report generation tasks, or simultaneous user activity lead to any decline in performance quality or output consistency. Recovery mechanisms following minor errors, connection interruptions, or unexpected shutdowns are assessed to understand how well the platform restores normal function and resumes correct operation after a disruption occurs. A reliable system continues functioning in a predictable manner without significant degradation in speed, accuracy, or stability over extended periods. The platform's capability to preserve stored data accurately—ensuring no corruption, loss, or unintended modification occurs during processing or storage—receives thorough examination as a core component of reliability. Internal backup procedures and data restoration workflows undergo review to confirm the system's preparedness against accidental data loss or technical failure. This aspect of reliability is especially critical because management decisions based on inaccurate, incomplete, or unstable reports can lead to incorrect strategies and negatively impact daily operations or long-term planning. The system demonstrates consistent and dependable performance under realistic operating conditions that mirror the actual environment of Kape Tubong. High reliability confirms BREWINSIGHT BI as a stable, durable, and consistent source of business intelligence over time.

Assessment of reliability examines the platform's capability to maintain uninterrupted and smooth performance during varying levels of operational workload. The system remains stable and responsive regardless of whether it processes a small batch of daily transactions or handles large volumes of historical sales data consolidated from multiple branches and outlets. Sudden increases in user activity, such as during end-of-day reporting or peak business hours, or spikes in data requests never cause the platform to become unresponsive, laggy, or produce incomplete or distorted outputs. Maintaining consistent responsiveness even during high-demand periods ensures management and staff access necessary business information exactly when needed most. The ability of the system to scale effectively and maintain operational continuity under changing workload conditions directly reflects the robustness of its design, coding efficiency, and overall implementation quality. This characteristic ensures the platform remains useful and effective regardless of how much data undergoes processing or how many users interact with it at any given moment.

A key aspect of reliability involves evaluating the consistency of analytical computations and final results generated through repeated executions of the same tasks. Revenue summaries, branch-to-branch comparisons, and detailed product performance analyses produce identical, precise, and accurate results every time, provided that the input data and applied filters remain unchanged. Researchers systematically verify that all calculations, aggregations, and interpretations remain precise and error-free even after multiple cycles of report generation, data sorting, or filtering operations. This level of consistency is essential because inconsistent analytical outputs or fluctuating data interpretations can lead to confusion, mistrust, and poor judgment during managerial decision-making processes. Reliable analytical processing ensures users confidently depend on the information displayed by the platform, knowing the data never changes arbitrarily or produces different insights based on the same underlying facts. This predictability allows management to track trends accurately and measure performance changes with certainty.

The reliability criterion assesses the effectiveness and sophistication of the system's internal error-handling mechanisms. The platform automatically detects invalid inputs, incomplete records, format mismatches, or operational conflicts without crashing, freezing, or causing permanent damage to stored data. Whenever issues appear, the system provides clear, understandable notifications and guides users through proper recovery procedures or correction steps. Effective error handling minimizes operational disruptions and helps maintain stable service delivery even during unexpected situations or when users make mistakes during data entry. The capability to recover gracefully from minor faults, invalid actions, or system warnings contributes significantly to the overall dependability and user-friendliness of the system in real-world operational environments. Small issues never escalate into major problems or result in data loss, thereby protecting the integrity of business information at all times.

Researchers thoroughly evaluate the platform's capability to maintain database integrity during all transactions, updates, and data modification activities. Every record of sales, inventory levels, customer transactions, and analytical history remains complete, accurate, logically structured, and fully synchronized across the entire system. The platform enforces strict controls to prevent data duplication, accidental deletion, unauthorized modification, or inconsistent updates that could compromise the quality of stored information. Reliability in data management is critical because inaccurate, missing, or conflicting information immediately compromises the quality of business analysis, reporting, and decision support. By preserving database consistency and ensuring every entry remains valid and correct, the system guarantees all analytical outputs and reports generated are trustworthy, valid, and useful for management review and strategic planning.

System availability likewise forms an important and measurable part of the reliability evaluation process. Researchers assess whether the platform remains fully operational, accessible, and responsive throughout regular business hours with minimal downtime or service interruptions. Unexpected outages, long loading times, or prolonged periods of inaccessibility severely interrupt daily operations and gradually reduce user confidence in the platform's value. A dependable system provides continuous access to dashboards, real-time reports, and analytical tools exactly when required by management and operational staff.

High system availability ensures the platform effectively supports ongoing activities, allows for timely monitoring, and enables decision-making processes to continue without unnecessary delays or technical obstacles.

Reliability testing includes specific evaluations regarding the platform's capability to support multiple concurrent users without creating operational conflicts or causing performance failures. Several users accessing dashboards, generating reports, filtering datasets, or entering data simultaneously never negatively affect system stability, processing speed, or the accuracy of results for any user. The platform maintains consistent speed, responsiveness, and precision regardless of the number of active users or the volume of requests undergoing processing at the same time. This capability is especially important for organizations with multiple branches, departments, and several managerial personnel who may need to access the system at overlapping times. Reliable multi-user support enhances operational coordination, improves workflow efficiency, and strengthens the overall dependability of the platform as a shared business tool.

Researchers assess the reliability of data synchronization processes and real-time updating mechanisms integrated within the platform architecture. Information entered or updated from different branches, kiosks, or terminals undergoes capture, processing, and accurate reflection within the centralized dashboard without noticeable delays, lags, or inconsistencies. Real-time synchronization ensures managers always view the most current, up-to-date, and reliable operational data when making critical business decisions. Delayed updates or mismatched information between local terminals and the central database reduce the usefulness of the platform and lead to decisions based on outdated figures. The consistency, speed, and timeliness of all data synchronization procedures receive careful measurement and validation throughout the evaluation phase.

The platform's backup and recovery capabilities undergo examination in detail as a key component of the reliability assessment. The system preserves critical business information through automated or manual backup procedures and secure storage mechanisms designed to prevent data loss. In the event of accidental deletion, software failure, hardware malfunction, or technical error, the platform allows for efficient, complete, and accurate restoration of any lost or damaged data. Reliable recovery procedures help minimize operational disruptions, reduce downtime, and protect important organizational records from permanent loss or corruption. These features contribute directly to the long-term stability, security, and sustainability of the business intelligence platform within the organization.

Another important consideration involves the platform's ability to maintain consistent reliability and performance standards after software updates, feature enhancements, or system modifications are applied. New updates or configuration changes never negatively affect existing functions, alter stored data, or introduce new operational inconsistencies or errors. Researchers verify and document whether the platform continues to operate correctly, accurately, and efficiently immediately following maintenance activities, version upgrades, or setting adjustments. Stable performance after changes reflects effective system architecture, high-quality coding, and proper software management practices. Maintaining reliability even during periods of system evolution or improvement ensures continuous and uninterrupted operational support for the needs of Kape Tubong.

Reliability serves as a critical indicator of the platform's overall stability, consistency, and trustworthiness when supporting daily business operations and strategic planning. It confirms whether BREWINSIGHT BI continuously delivers accurate analytical outputs, preserves the highest levels of data integrity, and maintains consistent operational performance under both normal and demanding conditions. Comprehensive reliability evaluation determines whether the system offers enough dependability, resilience, and quality to effectively support managerial analysis, monitoring, and decision-making processes. A highly reliable platform strengthens organizational confidence, encourages regular usage, and ensures the system functions as a stable, long-term, and valuable business intelligence solution tailored specifically for Kape Tubong.

Efficiency - Efficiency refers to the speed and resource effectiveness of the system while performing its intended functions. This criterion measures how quickly the BREWINSIGHT BI platform processes user requests, retrieves stored data, and displays meaningful results to those accessing the system. Researchers evaluate response times when loading dashboards, generating detailed reports, and applying specific filters to analytical datasets. Fast and smooth performance ensures users access important business insights without unnecessary delay or waiting periods. Efficiency also covers how well the system utilizes available computing power, memory, and server resources during daily operation. A system completing tasks efficiently supports improved productivity and delivers a better overall experience for every user. Slow processing speeds hinder usability and reduce the practical value of the platform within active operational settings. Since business decisions often require timely access to updated information, efficiency holds great importance for analytical systems. A highly efficient platform allows users to retrieve and analyze data rapidly, thereby enabling more responsive and well-informed management decisions. This factor stands as a key determinant regarding the practicality and success of the developed system.

Assessment of efficiency includes evaluating the optimization of database queries and backend processing logic. These elements ensure the system handles large transaction datasets without excessive delay or lag. As Kape Tubong expands its operations and accumulates more sales records over time, the platform must continue processing increasing data volumes effectively. Evaluation procedures verify whether performance remains acceptable even as stored information grows in size and complexity. Efficient data handling directly improves scalability and supports long-term usability of the platform. Researchers also measure loading speeds for visual analytics and check if dashboard components render smoothly across various devices and differing internet connection strengths. High performance reduces user waiting time and increases satisfaction during regular use. Optimization efforts additionally lower strain on server infrastructure and improve sustainability regarding hosting and maintenance. This criterion confirms that BREWINSIGHT BI delivers business intelligence outputs promptly and consistently.

Evaluation of efficiency further examines processing speed and update frequency for transactional information stored within the centralized database. Sales records entered from multiple branches should appear in the dashboard within a reasonable timeframe. This ensures management receives updated insights quickly and acts based on current data. Delays in data synchronization reduce the effectiveness of real-time monitoring and may impact the accuracy of operational analysis. The system maintains fast performance even when handling simultaneous data submissions coming from different locations or terminals. Rapid and accurate updating improves responsiveness and supports timely decision-making processes across the organization.

A key aspect of efficiency involves assessing the system’s capability to run complex analytical operations without excessive consumption of computing resources. Revenue analysis, comparative reporting, trend visualization, and filtering functions complete quickly while keeping performance stable. Researchers verify that calculations and data aggregations occur efficiently without causing lag, freezing, or delays in dashboard updates. Efficient analytical processing allows continuous interaction with the platform without interruptions or technical slowdowns. This characteristic proves essential for managers requiring immediate access to reports and insights during daily activities or decision-making sessions.

This criterion also assesses responsiveness within the user interface during navigation and interaction. Menus, charts, filters, and reporting tools react immediately to commands entered by users without noticeable delay. Smooth transitions between dashboard pages and fast rendering of graphical elements increase user satisfaction and operational convenience. The platform minimizes unnecessary waiting periods so users complete their intended tasks faster and with less effort. A responsive interface contributes significantly to the practicality and usability of the system in real-world business environments.

Researchers also evaluate how efficiently the platform manages concurrent user activity. Multiple individuals accessing the system at the same time should not reduce speed or negatively affect operational performance. The platform maintains acceptable response times even when managers, administrators, and staff use different modules or perform different tasks simultaneously. Efficient handling of concurrent operations supports organizational coordination and prevents workflow interruptions or bottlenecks. This capability remains particularly valuable for businesses with multiple branches relying on centralized access to shared analytical resources.

Efficiency testing involves measuring startup and loading performance during initial access to the platform. The time taken for user authentication, dashboard initialization, and retrieval of relevant analytical data undergoes careful observation and recording. A system that initializes quickly improves accessibility and reduces delays during operational use. Long startup times discourage frequent usage and lower overall user productivity. Efficient initialization ensures users access business intelligence tools whenever required without unnecessary inconvenience or wasted time.

Assessment procedures also review optimization applied to data storage and retrieval mechanisms within the platform. Efficient indexing, query processing methods, and logical database organization significantly increase speed when searching for information or generating reports. The platform locates and processes specific records rapidly even as the database grows larger over months or years of operation. Proper database optimization supports performance efficiency and contributes to overall system scalability and long-term sustainability. Effective storage management reduces risks of performance degradation as business operations expand and datasets multiply.

This criterion further examines the platform’s ability to perform automated functions within minimal processing time. Automated report generation, sales data consolidation, and dashboard refreshing occur smoothly without disrupting ongoing operations or user activities. Efficient automation reduces manual workload and enhances operational productivity. Repetitive processes complete quickly and accurately, freeing staff to focus on other important tasks. Researchers determine if these automated features positively impact workflow efficiency and time management within Kape Tubong operations.

Evaluation also considers how well the platform adapts under varying internet connectivity conditions. Since users may access the system from different locations and devices, the platform maintains reasonable responsiveness even during periods of moderate network limitations or slower connections. Efficient data transmission protocols and optimized dashboard rendering ensure smooth performance across diverse operational environments. This flexibility improves accessibility and supports uninterrupted use of the platform regardless of connection quality or location.Energy and resource consumption also form part of the efficiency evaluation process. Efficient systems operate effectively while using fewer server resources and hardware capacity. Lower consumption reduces operational costs related to hosting, maintenance, and system management activities. Optimization of resource utilization makes the platform more sustainable and cost-effective for long-term deployment and support. Effective resource management additionally contributes to improved system stability and reliability over extended periods.

Researchers verify whether the platform maintains high efficiency even after prolonged periods of continuous use. Systems that gradually slow down during extended operation negatively impact productivity and user experience. The platform sustains stable processing speed and responsiveness even after running for long operational hours or during heavy usage periods. Consistent efficiency over time reflects effective system architecture and proper software optimization practices. This consistency ensures the platform remains dependable during routine daily business activities and long analysis sessions.

Efficiency serves as a critical measure of the platform's speed, responsiveness, and effective use of resources while supporting business operations. It confirms whether BREWINSIGHT BI processes, analyzes, and presents information promptly while keeping performance quality high. Comprehensive evaluation verifies if the system delivers timely analytical outputs that support fast and informed management decisions. A highly efficient platform enhances productivity, improves user satisfaction, and strengthens the overall operational value of the business intelligence system developed for Kape Tubong.

Security - Security refers to the system's ability to protect stored data, regulate user access, and safeguard internal processes against unauthorized entry, improper manipulation, or potential breaches. This criterion evaluates whether BREWINSIGHT BI includes adequate protective measures designed to secure Kape Tubong's confidential sales history and sensitive operational records. Key mechanisms such as user authentication protocols, password validation rules, role-based access control, and restricted system permissions undergo thorough examination during assessment. These features ensure only authorized personnel can access sensitive reports or perform administrative functions. Security also covers measures preventing unauthorized modification, accidental deletion, or corruption of critical information. Session management controls and secure database interaction methods receive careful review as well. Since the system holds financial and sales-related details, strong protection stands essential for maintaining confidentiality and building trust among users. Weak safeguards expose the business to operational risks and unauthorized data access. Evaluation of security measures proves necessary before full deployment. This criterion confirms the system operates safely and remains trustworthy for regular business use.

Further evaluation is conducted to rigorously verify that access restrictions are properly and consistently enforced in accordance with defined user roles, levels, and responsibilities. Researchers carefully confirm that individuals assigned to lower-level permissions are strictly prevented from viewing, modifying, or managing administrative features, system configurations, or confidential datasets that fall beyond their designated authority boundaries. Additionally, session timeout settings and automatic logout controls are thoroughly reviewed and tested to reduce potential risks associated with unattended devices, forgotten active sessions, or prolonged idle periods that could leave the system exposed. Input sanitization and comprehensive validation procedures are also examined to confirm they effectively defend against common security vulnerabilities, including injection attacks, malicious code submissions, or improperly formatted data entries that could compromise system stability or data integrity. Backup procedures and disaster recovery protocols form a vital part of this overall protection strategy, ensuring that information remains safe and recoverable under all circumstances. Implementing strong security controls significantly increases confidence regarding system integrity, trustworthiness, and reliability during full-scale operational deployment. Protection of sensitive information holds particular importance in analytical platforms like this one, which centralizes financial records, sales history, and operational data collected from multiple locations and branches. Through this comprehensive security evaluation, the team ensures that BREWINSIGHT BI meets or exceeds accepted industry standards and best practices for safeguarding the organization’s critical digital assets.

The assessment of security also involves detailed measurement of the effectiveness of authentication mechanisms designed to accurately verify user identity before granting entry or access to the system. Login procedures are tested extensively to ensure they strictly prevent unauthorized individuals from accessing confidential data or system functionalities through rigorous credential verification and properly validated authentication steps. Specific aspects such as password policies—including minimum length rules, complexity requirements, expiration periods, and secure hashing and storage methods—are reviewed in detail to confirm they align with established security benchmarks. These measures collectively reduce risks related to compromised accounts, stolen credentials, or unauthorized entry attempts while strengthening the overall defensive posture of the platform. Reliable and consistent authentication practices remain essential for maintaining individual accountability, establishing clear user identities, and protecting organizational records from misuse, alteration, or unauthorized distribution.

A key focus of the security assessment involves evaluating how information is protected while being transmitted between the user interface and the central database server infrastructure. Sensitive content such as detailed sales transactions, user login credentials, financial summaries, and performance data is required to move through fully encrypted and secure communication channels to minimize exposure to interception, eavesdropping, or unauthorized viewing during transfer. Researchers closely examine the implementation and configuration of secure communication protocols to confirm that data exchanges remain protected at all times, especially during queries, updates, and report generation activities. Protection during transmission holds special importance in web-based platforms like BREWINSIGHT BI, where data frequently travels across public networks or internet connections that may be vulnerable to monitoring or interference. Robust secure communication practices contribute significantly to preserving both data integrity and confidentiality throughout daily operations and information exchange.

This evaluation criterion also examines the platform's capability to maintain highly accurate activity logs and comprehensive audit trails for all actions performed within the system. Logging features are assessed to ensure they reliably record important user actions, including login attempts (both successful and failed), access to sensitive reports, modifications made to datasets, administrative adjustments, and changes to configuration settings. These detailed records allow system administrators and management to monitor usage patterns, track user behavior, and identify suspicious activities, policy violations, or unauthorized actions whenever necessary. The presence of thorough audit trails greatly improves organizational accountability by enabling management to trace exactly who performed specific actions, when they occurred, and what changes were made within the system. Effective monitoring capabilities strengthen organizational control, support compliance requirements, and provide essential evidence and context to support investigations should security incidents, discrepancies, or errors occur.

Researchers conduct specific assessments to measure the system's resistance against common cybersecurity threats, external attacks, and operational vulnerabilities that could be exploited. The platform is validated to ensure it effectively prevents unauthorized database modifications, malicious data entry attempts, privilege escalation efforts, or attempts to expand access permissions beyond assigned limits. This is achieved through strict input validation routines, adherence to secure coding standards, and carefully controlled database access policies that limit exposure to risks and potential exploits. Evaluation confirms that the system maintains stable operation and secure performance even when it receives potentially harmful, unexpected, or invalid inputs designed to test its defenses. The presence of strong defenses against known security weaknesses improves the overall resilience, robustness, and long-term reliability of the platform in real-world usage scenarios.

Evaluation procedures also include a detailed review of the protection applied to all information stored within the centralized database structure. Researchers verify that all business records, sales details, transaction logs, inventory data, and generated analytical reports remain fully protected from accidental loss, unauthorized modification, deletion, or corruption over time. The platform implements strict database management controls, access governance, and integrity checks designed to preserve the accuracy, consistency, and reliability of stored information throughout its lifecycle. Backup strategies, retention policies, and recovery mechanisms are tested to confirm that restoration of important records is possible, complete, and accurate following technical failures, system errors, or accidental removal events. Secure storage practices, combined with regular integrity checks, prove critical for the long-term preservation and trustworthiness of valuable organizational information assets.

Researchers measure how effectively the platform separates user privileges, defines distinct operational responsibilities, and enforces role-based access control across all modules. The system is evaluated to ensure that administrators, managers, supervisors, and regular staff members are only able to access features, data sets, and functions strictly relevant to their assigned roles, duties, and authority levels. Limiting access rights and clearly defining permissions significantly reduces the chances of accidental misuse, unintentional errors, or exposure of highly confidential data to unauthorized personnel. Proper role segregation supports organizational discipline, establishes clear boundaries, and strengthens operational accountability by ensuring that users only interact with what is necessary for their work. Careful management and regular review of access rights ensure stronger, layered protection for sensitive business data across all levels of system usage.

The assessment also considers the platform's capability to sustain consistent, reliable, and effective security measures during prolonged periods of continuous use and evolving operational conditions. Researchers verify that the system maintains secure session handling, reliable authentication states, and strict access controls continuously over time without degradation or the introduction of new vulnerabilities. Protective features remain fully functional and effective even after software updates, maintenance work, configuration changes, or extended operational periods under heavy load. The ability to maintain consistent security standards reflects careful design, thorough testing, and high-quality development practices integrated from the earliest stages of the project. This consistency ensures the platform stays protected, compliant, and dependable throughout its entire operational lifecycle and as the business grows.

This criterion further examines the safeguards and error-prevention mechanisms built specifically against accidental operational errors that might compromise data integrity or reliability. Features such as confirmation prompts for critical actions, restricted deletion rights, controlled editing capabilities, and data validation rules are designed to minimize unintended modifications, accidental removal, or data loss caused by user mistakes. Such protections prove highly valuable because human error remains a common risk that could negatively impact reporting accuracy, historical records, and the quality of analysis produced by the system. Reducing the likelihood and impact of accidental errors helps preserve the correctness, completeness, and reliability of stored information, ensuring it remains trustworthy and useful for future analysis and decision-making.

Researchers also evaluate the user guidance, security education, and awareness features integrated directly within the platform's interface and workflows. Clear notifications regarding failed login attempts, password strength requirements, session expiration warnings, and secure handling instructions are included to help users understand, remember, and consistently follow recommended security procedures.

Security-related prompts, reminders, and inline feedback improve user awareness and encourage responsible interaction with the system, reducing risks caused by negligence or lack of knowledge. This indirect education through interface guidance and user assistance strengthens the overall security environment and culture within the organization, helping users become active participants in maintaining data safety.

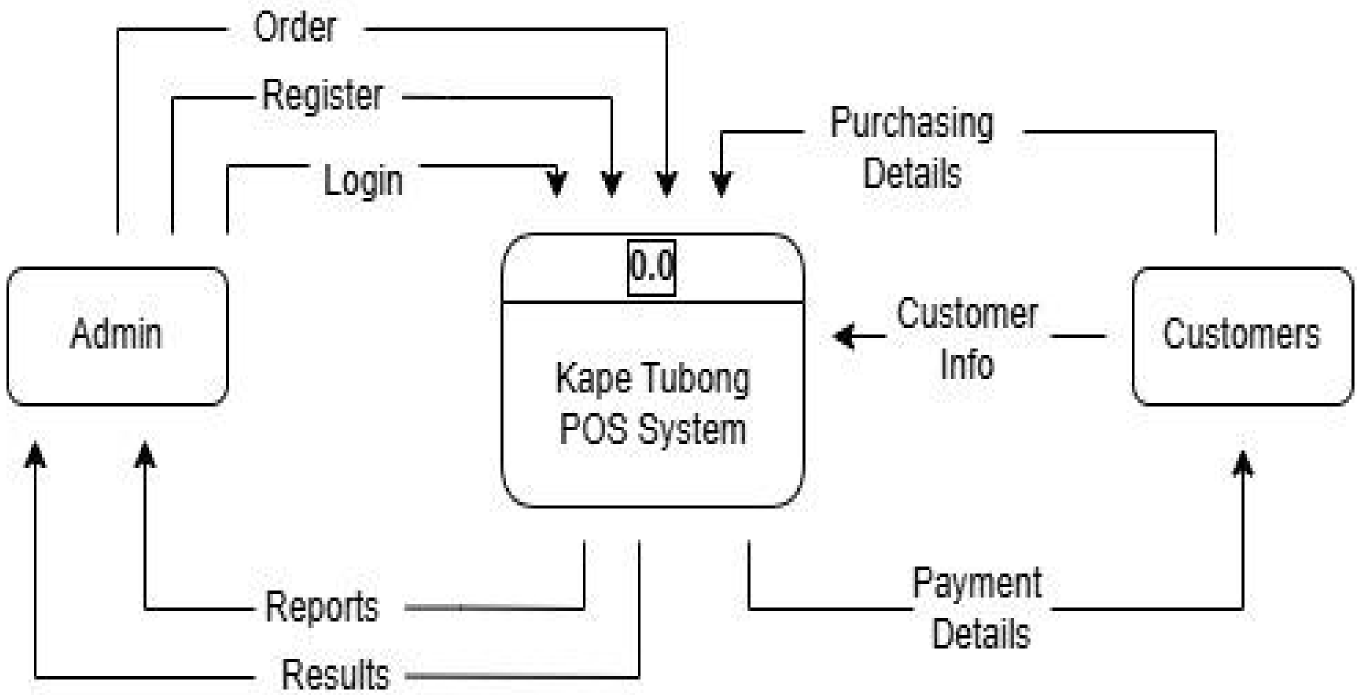
The platform's ability to maintain full security protections during periods of simultaneous user access and high concurrency also undergoes rigorous testing. Researchers simulate scenarios where multiple individuals are viewing reports, modifying records, using dashboard features, or performing administrative tasks at the same time to ensure these activities never create conflicts that compromise data consistency, accuracy, or protection measures. Secure transaction handling, database locking mechanisms, and carefully controlled data access prevent overlapping changes, unauthorized modifications, deadlocks, or operational inconsistencies during peak usage. This capability remains especially important in centralized business intelligence platforms like BREWINSIGHT BI, which serve multiple branches, departments, and user groups accessing the same datasets concurrently.

Security is recognized as a critical measure of the platform's overall capability to protect vital business information, maintain strict confidentiality standards, prevent unauthorized access or misuse, and ensure data privacy. It determines whether BREWINSIGHT BI successfully creates a safe, controlled, and compliant environment for managing Kape Tubong's financial, operational, and strategic records. Comprehensive evaluation and continuous verification confirm adherence to acceptable standards regarding digital asset protection, data integrity preservation, privacy controls, and secure operational processes. A highly secure platform strengthens organizational trust, significantly minimizes operational risks, protects against financial or reputational loss, and ensures the system functions confidently as a reliable, protected, and professional business intelligence solution.

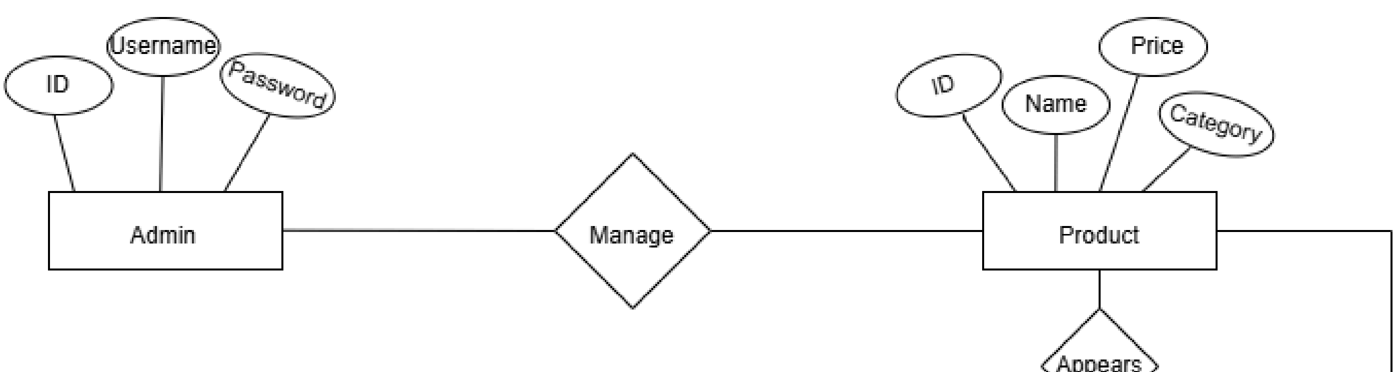
SUPPORTING DIAGRAMS (Included within Chapter 3)

Data Flow Diagram (DFD – Level 0)

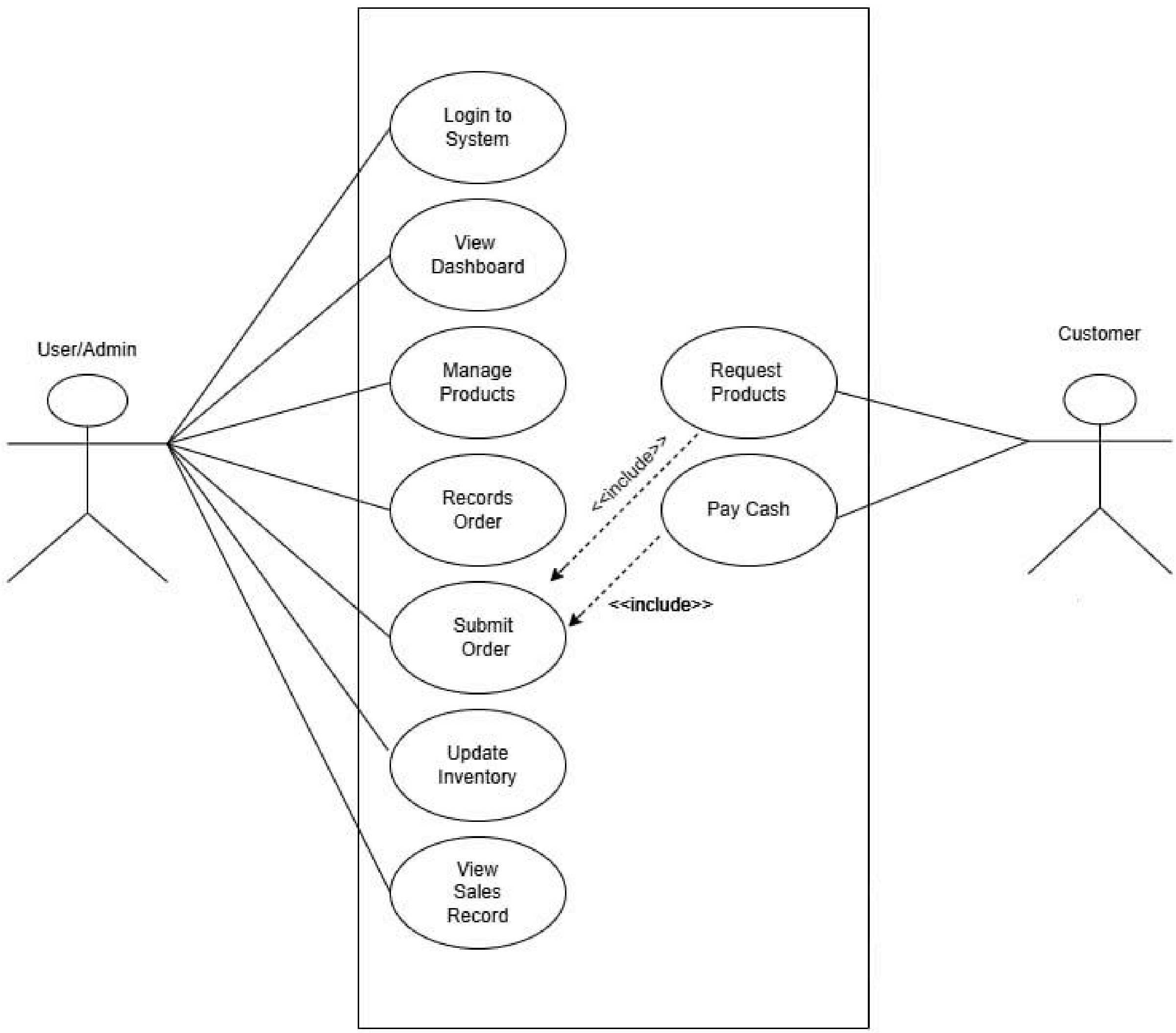
Admin → Pos System→ Customers



Entity Relationship Diagram (ERD)



Use Case Diagram


User/Admin
Login to System
View Dashboard
Manage Products
Records Order
Submit Order
Update Inventory
View Sales Record
Request Products
Pay Cash
<<include>>
<<include>>
Customer

End of Proposal

Note to the instructors handling this subject

CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in application development, database management, systems analysis and design, and emerging technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical considerations, user safety, and data privacy.